Technical training.
Product information.

F10H Complete Vehicle

Edited for the U.S. market by:
BMW Group University
Technical Training
ST1203 5/1/2012

BMW Service

General information

Symbols used

The following symbol is used in this document to facilitate better comprehension or to draw attention to very important information:

⚠

Contains important safety information and information that needs to be observed strictly in order to guarantee the smooth operation of the system.

Information status and national-market versions

BMW Group vehicles meet the requirements of the highest safety and quality standards. Changes in requirements for environmental protection, customer benefits and design render necessary continuous development of systems and components. Consequently, there may be discrepancies between the contents of this document and the vehicles available in the training course.

This document basically relates to the European version of left-hand drive vehicles. Some operating elements or components are arranged differently in right-hand drive vehicles than shown in the graphics in this document. Further differences may arise as a result of the equipment specification in specific markets or countries.

Additional sources of information

Further information on the individual topics can be found in the following:

- Owner's Handbook
- Integrated Service Technical Application.

Contact: conceptinfo@bmw.de

The information contained in this document forms an integral part of the technical training of the BMW Group and is intended for the trainer and participants in the seminar. Refer to the latest relevant information systems of the BMW Group for any changes/additions to the technical data.

Information status: **October 2011**
VH-23/International Technical Training

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1. Introduction

1.1. Positioning

Within the BMW EfficientDynamics framework, the BMW Group introduced the second generation of hybrid vehicles in the spring 2012. The BMW ActiveHybrid 5 is the third series vehicle with hybrid technology. The BMW ActiveHybrid 5 sets new standards in sporty driving pleasure and sustainability in this vehicle segment by combining for the first time a BMW six-cylinder in-line engine with an electric drive. The BMW ActiveHybrid 5 (whose development code is F10H) is based on the BMW 5 Series (F10). It embodies the consistent further development of the drive technology used in the series models BMW ActiveHybrid X6 and BMW ActiveHybrid 7: The ActiveHybrid 5 is a full hybrid vehicle with a lithium-ion high-voltage battery.

The powertrain of the BMW ActiveHybrid 5 comprises a 6-cylinder in-line engine with TwinPower turbo technology (N55B30M0), an 8-speed automatic gearbox (GA8P70HZ) and an electrical machine. The BMW 535i sedan (which is already commended for the efficiency of its combustion engine) experiences a further reduction in fuel consumption and emissions levels of over 10% thanks to the integration of BMW active hybrid technology. Its powertrain generates a total system power output of 250 kW/335 HP. It accelerates the BMW ActiveHybrid 5 in 5.7 seconds from zero to 60 mph and restricts the fuel consumption to average values between 6.4 and 7.0 l for every 100 kilometers/60 ml, as well as CO_2 emissions to 149 to 163 g/km (values based on EU test cycle, dependent on tire type selected).

The electric motor of the BMW ActiveHybrid 5 enables emission-free fully electric driving at speeds of up to 60 km/h/37 mph. At an average speed of 35 km/h/22 mph, the high-voltage battery provides enough energy to enable fully electric driving across a distance of up to four kilometers (2.5 miles). Furthermore, a engine Start/Stop function specific to a hybrid enhances the overall efficiency by switching off the combustion engine each time the car is stopped for example at traffic lights or in traffic.

Another unique feature of the BMW ActiveHybrid 5 is the advanced energy management which is capable of adapting the operating strategy not only to the current driving situation, but also to an imminent driving situation, thus enabling a more efficient journey.

Also in the BMW Active Hybrid 5, the driver can choose ECO PRO mode, in addition to the settings "Sport+", "Sport", "Comfort" and "Comfort+", via the standard driving experience switch. ECO PRO mode supports a particularly relaxed and highly economical driving style thereby preferring a fully electric driving style.

The high-voltage batteries for the ActiveHybrid X6 and ActiveHybrid 7 still originate from the cooperation ventures with other automobile manufacturers, whereas the high-voltage battery used in the ActiveHybrid 5 is "Made by BMW".

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1. Introduction

Generations of the ActiveHybrid vehicles to date

Index	Explanation
1	The E72 was launched on the market at the end of 2009 as the first BMW hybrid vehicle. The technology used (so-called generation 1.0) was a product created of the cooperation venture between General Motors, Daimler Chrysler and BMW. A nickel metal hydride battery was used as an electric energy storage unit. .
2	The second hybrid vehicle from BMW the ActiveHybrid 7 came to the market in 2010. It is a mild hybrid vehicle with 1.5 generation technology. This technology was developed together with Mercedes Benz. The highly efficient lithium ion battery was used in the high-voltage electrical system.
3	The ActiveHybrid 5, the third BMW hybrid vehicle, with a SOP from the end of 2011. A lithium-ion battery is also used here as an electrical energy storage unit.
4	Further BMW hybrid vehicles with 2.0 generation technology will be launched on the market from 2012.

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1. Introduction

1.2. Identifying features

1.2.1. Exterior

The BMW ActiveHybrid 5 distinguishes itself from the conventional 5-Series through a range of special features. These includes the unique exterior color "Liquid Blue Metallic" which represents the innovative BMW active hybrid technology. In the 5-Series this exterior color is reserved exclusively for the active hybrid vehicles. However, the customer can also choose other colors for his active hybrid vehicle. The "ActiveHybrid 5" badge on the entry sills, on two C-pillars, as well as on the trunk lid indicate that it is a hybrid car. In addition, the "ActiveHybrid" wording was positioned on the acoustic cover of the combustion engine. The dark, chrome-plated exhaust tailpipes of the exhaust system are another special feature of the BMW ActiveHybrid 5. To increase the range further, new sporty 18" aero wheel rims are offered in a five-spoke design, which reduce the drag of the vehicle. Another trademark of the new hybrid is the galvanically chrome-plated fins of the BMW radiator grille which can be seen in the design template for the BMW 550i.

External view of the ActiveHybrid 5

Index	Explanation
1	"Liquid Blue Metallic" paint color
2	"ActiveHybrid 5" badge
3	Exhaust tailpipe of the exhaust system (dark chrome)
4	"ActiveHybrid 5" badge
5	Entry sills with "ActiveHybrid 5" badge
6	18" aero wheel rims
7	BMW radiator grille galvanically chrome-plated finish
8	Acoustic engine cover with "ActiveHybrid" badge

1.2.2. Interior

TH11-1368

Interior of the ActiveHybrid 5

The interior equipment of the BMW ActiveHybrid 5 also distinguishes itself from other F10 by some special features.

A plate with "ActiveHybrid 5" badge on the flap above the cup holders indicates it is a hybrid vehicle.

The hybrid-specific operating conditions and the state of charge of the high-voltage battery are displayed in the instrument cluster and if desired in the Central Information Display. Both the display in the CID and in the instrument cluster are activated upon start-up.

The plate with the "ActiveHybrid Power Unit" writing in the luggage compartment refers to the high-voltage battery unit.

1.3. Energy management

Just like in the BMW ActiveHybrid X6 and BMW ActiveHybrid 7, energy management plays an important role in the BMW ActiveHybrid 5 with regard to efficiency and driving dynamics in various driving situations.

1.4. Driving situations

The ActiveHybrid 5 has a range of hybrid-specific driving situations just like the ActiveHybrid X6 and ActiveHybrid 7.

These include the following:

- Automatic engine start/stop function
- Driving-off and driving (fully electric or combustion engine)
- Acceleration (boost function)
- Brake energy regeneration
- Coasting (Rolling without energy consumption).

The "Coasting" driving situation is the only new function.

1.4.1. Coasting

This innovation (which has been introduced for the first time in the BMW ActiveHybrid 5) ensures that a consistent increase in efficiency is possible by switching off the combustion engine not only when the vehicle is stationary and in city traffic, but also at highway speeds.

If the combustion engine is not required for the powertrain, it can be switched off by the hybrid system also during the journey. In this regard, the vehicle must be in ECO PRO mode and the driver cannot press the accelerator pedal. The combustion engine is then switched off up to a speed of 160 km/h/ 100 mph and disconnected from the rest of the drivetrain by a decoupler.

The BMW ActiveHybrid 5 rolls silently and emissions-free on the roadway without the influence of a engine drag torque. Although, to guarantee the unlimited operation of all safety and convenience functions, a small part of the kinetic energy is converted to electrical energy by energy recovery.

1.5. Technical data

	Unit	BMW 535i	ActiveHybrid 5
Engine and transmission			
Design		in-line 6	in-line 6
Number of valves per cylinder		4	4
Displacement	[cm^3]	2979	2979
Transmission		GA8HP70Z	GA8P70HZ
Powertrain		Rear-wheel	Rear-wheel
Maximum power, combustion engine	[kW (HP)] [rpm]	225 (300) 5800–6400	225 (300) 5800–6400
Maximum torque, combustion engine	[Nm (lb-ft)] [rpm]	400 (300) 1200–5000	400 (300) 1200–5000
Complete system power	[kW (HP)] [rpm]		250 (335) 5800–6400
Complete system torque	[Nm (lb-ft)] [rpm]		400 (300) 1000–5000
High-voltage battery type		-	Lithium-ion

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	Unit	BMW 535i	ActiveHybrid 5
Power of electrical machine	[kW / bhp] [rpm]	-	40/55 from 1800
Maximum torque, electrical machine	[Nm (lb-ft)] [rpm]	-	210 (155) from 1300 rpm
Vehicle performance			
acceleration 0 – 60 mph	[s]	5.7	5.7
Maximum speed (limited)	[km/h (mph)]	250 (150)	250 (150)
Consumption and emissions			
Fuel consumption (EU cycle, urban driving)	[l/100 km]	11.9	5.7 - 6.2 *
Consumption EU cycle, non-urban driving	[l/100 km]	6.4	6.7 - 7.4 *
Consumption EU cycle, total	[l/100 km]	8.4	6.4 - 7.0 *
CO_2 emissions	[g/km]	195	149 - 163 *
Dimensions and weights			
Length/width/height	[mm] [in]	4905 / 1860 / 1464 193.1 / 73.2 / 57.6	4905 / 1860 / 1464 193.1 / 73.2 / 57.6
Wheelbase	[mm/in]	2968 / 116,9	2968 / 116.9
Turning circle	[m/ft]	11.95 / 39.2	11.95 / 39.2
Track width front/rear	[mm] [in]	1600 / 1627 63 / 64.1	1600 / 1627 63 / 64.1
Curb weight	[kg/lbs]	1855 / 4090	1980 / 4365
Payload in accordance with DIN	[kg/lbs]	480 / 1080	390 / 860
Fuel tank capacity	[liters/gals]	70 / 18.5	67 / 17.7
Luggage compartment volume	[m³ / ft³]	0.520 / 18.4	0.375 / 13.2

*** Values of EU test cycle, dependent on tire format selected. The lower value with standard tires: Wheel rims 8J x 17, tires 225/55 R17 (SA 2K1) not available in the US.**

1.6. Equipment

The F10H and F10 differ not only in the technical data but also in the range of optional equipment offered. The most important optional equipment which is not offered in the F10H is briefly summarized below:

- All-wheel drive system xDrive (installation location of the front axle differential is partly taken up by the electrical machine electronics)
- Integral Active Steering
- Dynamic Drive
- Ski bag and through-loading facility (due to lithium-ion high-voltage battery)

The following equipment is part of the standard equipment:

- 4-zone A/C system
- Navigation system

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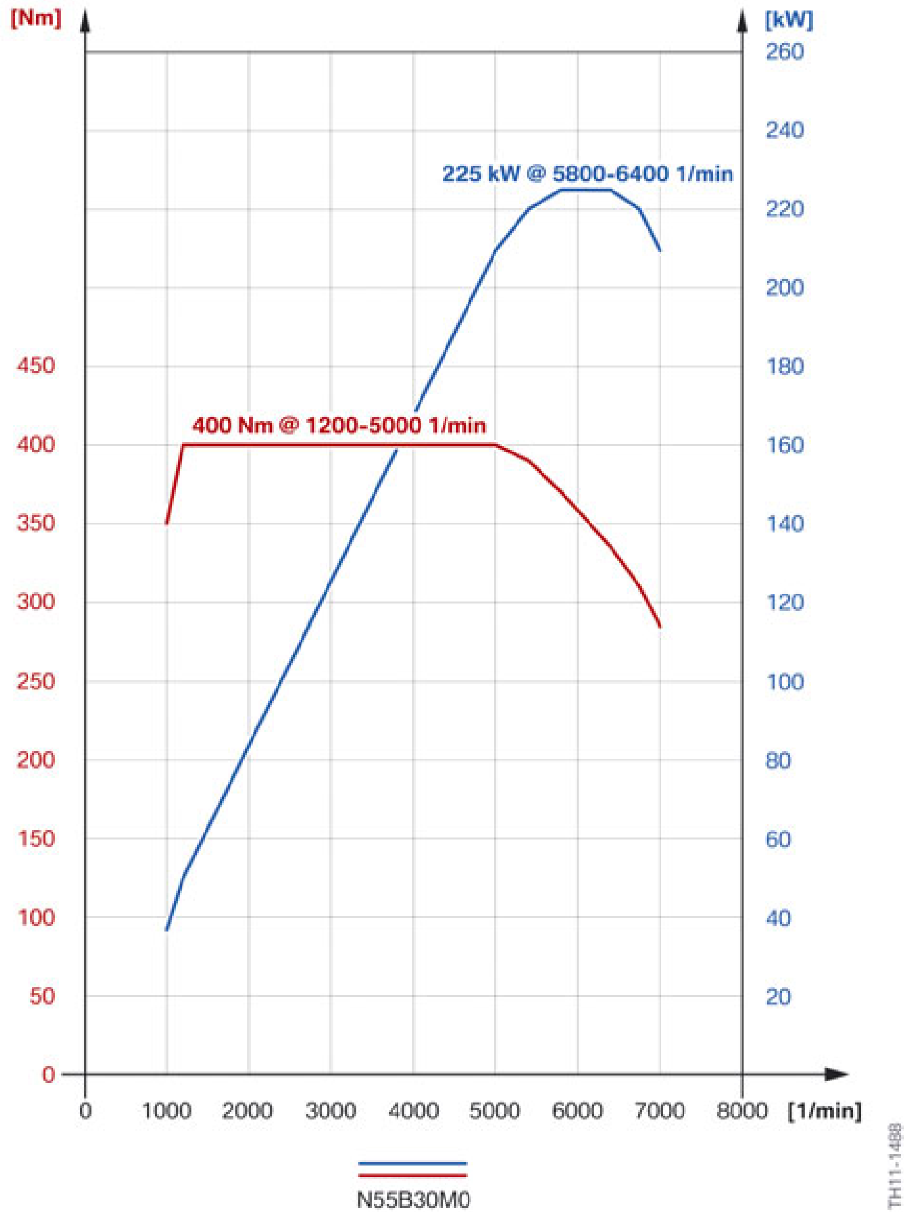
2. Powertrain Components

2.1. Introduction

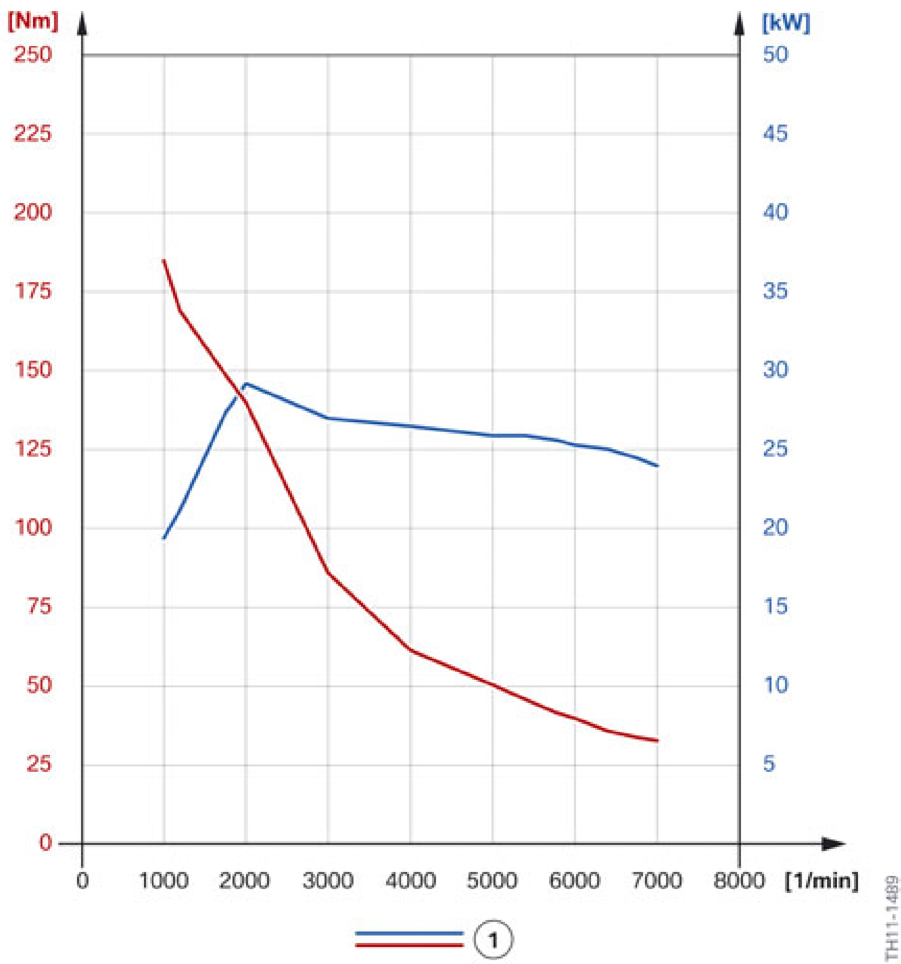
For the first time a 6-cylinder in-line engine is part of a BMW ActiveHybrid powertrain. The 3.0 l gasoline engine, which generates 300 HP and a maximum torque of 300 lb-ft, is a guarantee of enhanced driving pleasure and efficiency. The BMW TwinPower Turbo technology of the 6-cylinder N55 engine, which has already been awarded the international "Engine of the Year Award" twice in succession, combines a single twin-scroll exhaust turbocharger with direct fuel injection and the fully variable valve control Valvetronic.

2.1.1. Performance data

The F10H is driven by both a combustion engine and an electrical machine. The vehicle can be accelerated using fully electrical means, a combustion engine and a combination of the two. There is a different full load diagram for each of these situations.



Full load diagram for N55B30M0 engine in the F10H

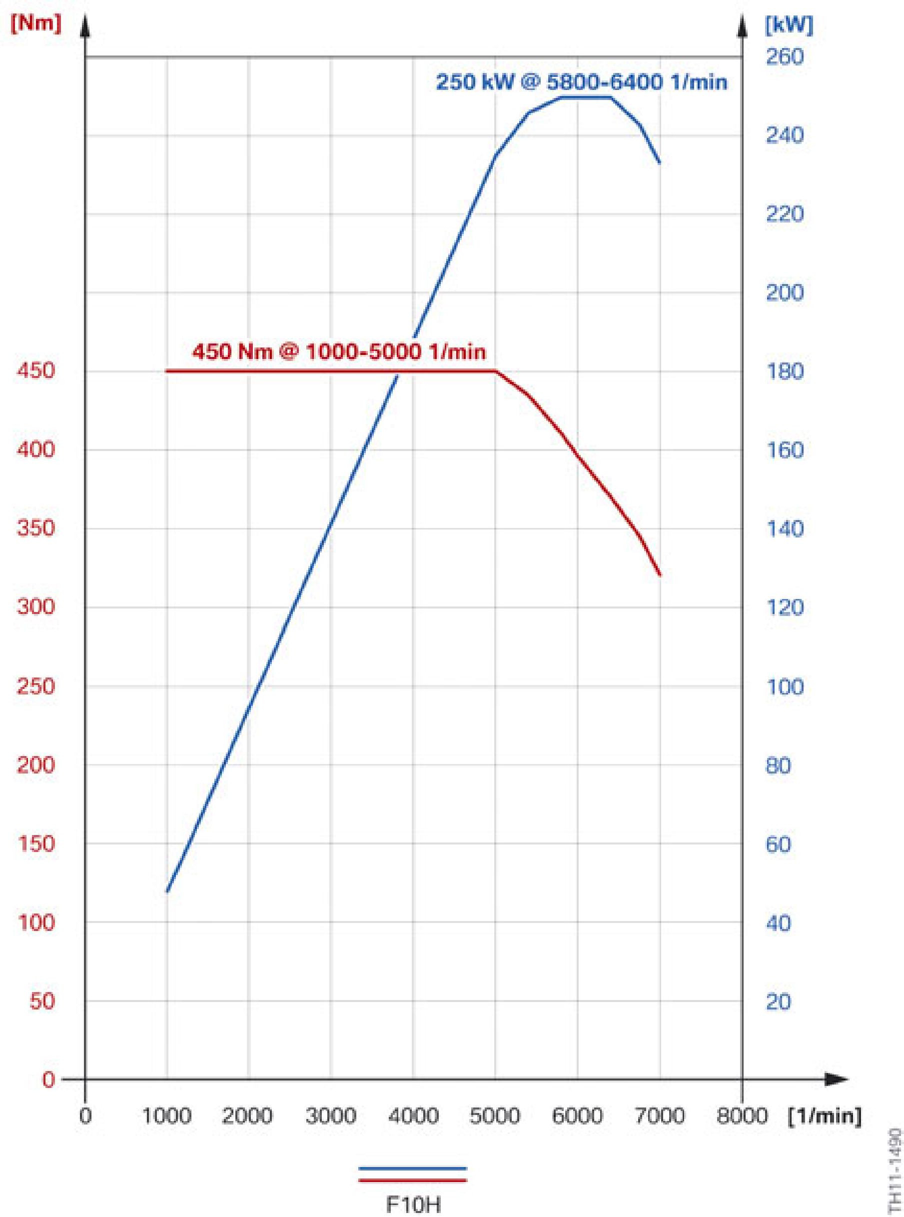


Full load diagram for electrical machine in the F10H

Index	Explanation
1	Electrical machine in the GA8P70HZ transmission of the F10H

2. Powertrain Components

In the following you see the full load diagram of the F10H with a combined use of the combustion engine and electrical machine. The system power is restricted to a maximum torque of 450 Nm/ 332 lb-ft and a maximum power of 250 kW/335 HP. Due to the direct response characteristics of the electrical machine and the slightly delayed response of the combustion engine, acceleration in the F10H is much more spontaneous than in the F10 with a N55B30M0 engine.



Full load diagram for system power of the F10H

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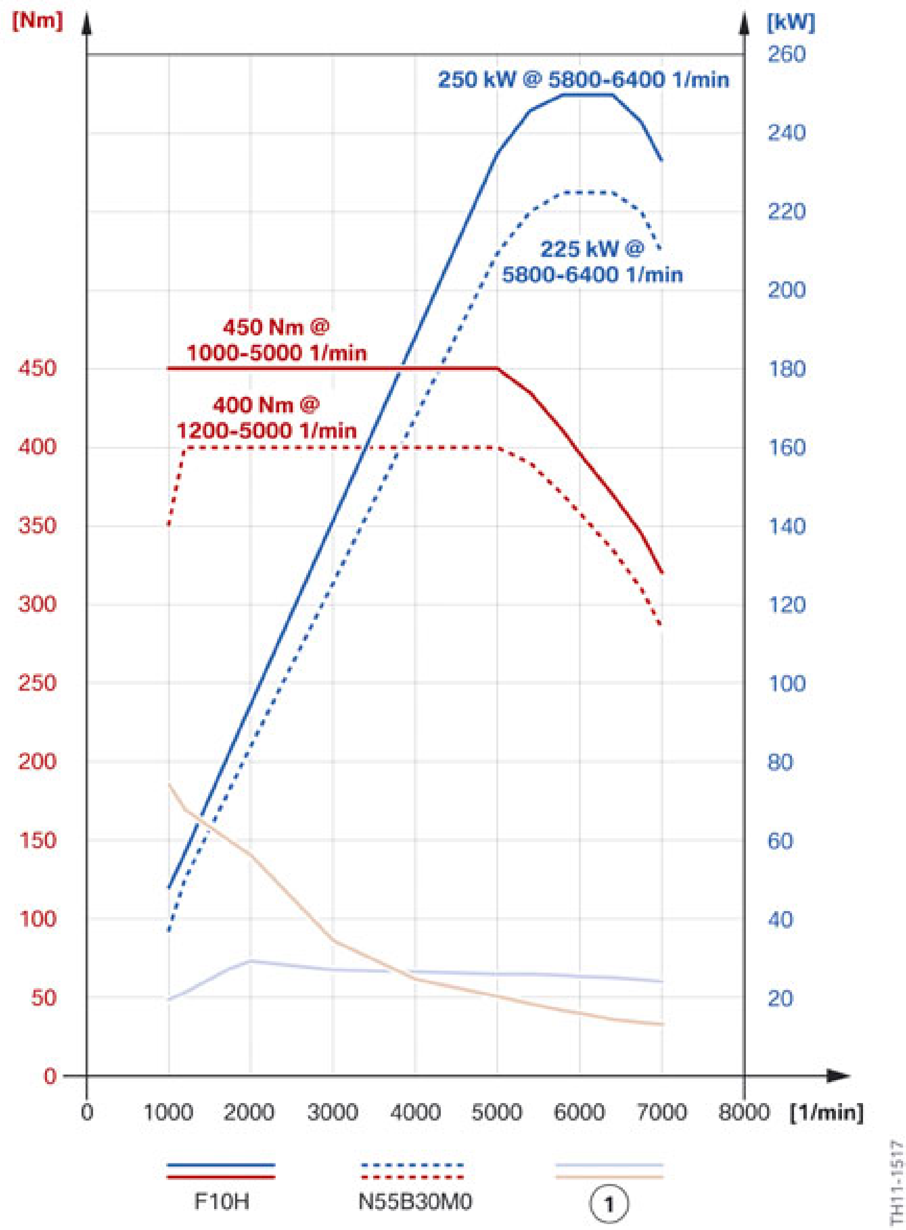
2. Powertrain Components

The following full load diagram shows all relevant components and the system power of the F10H in a joint diagram.



F10H full load diagram of the drive components and system power

Index	Explanation
1	Electrical machine in the GA8P70HZ transmission of the F10H

2.2. N55 engine Modifications

2.2.1. Introduction

BMW introduced the N55 engine in the BMW 535i Gran Turismo at the end of 2009. The N55 engine sees the combines an exhaust turbocharger, Valvetronic and direct fuel injection for the first time. BMW calls this combination Turbo Valvetronic Direct Injection, TVDI for short.

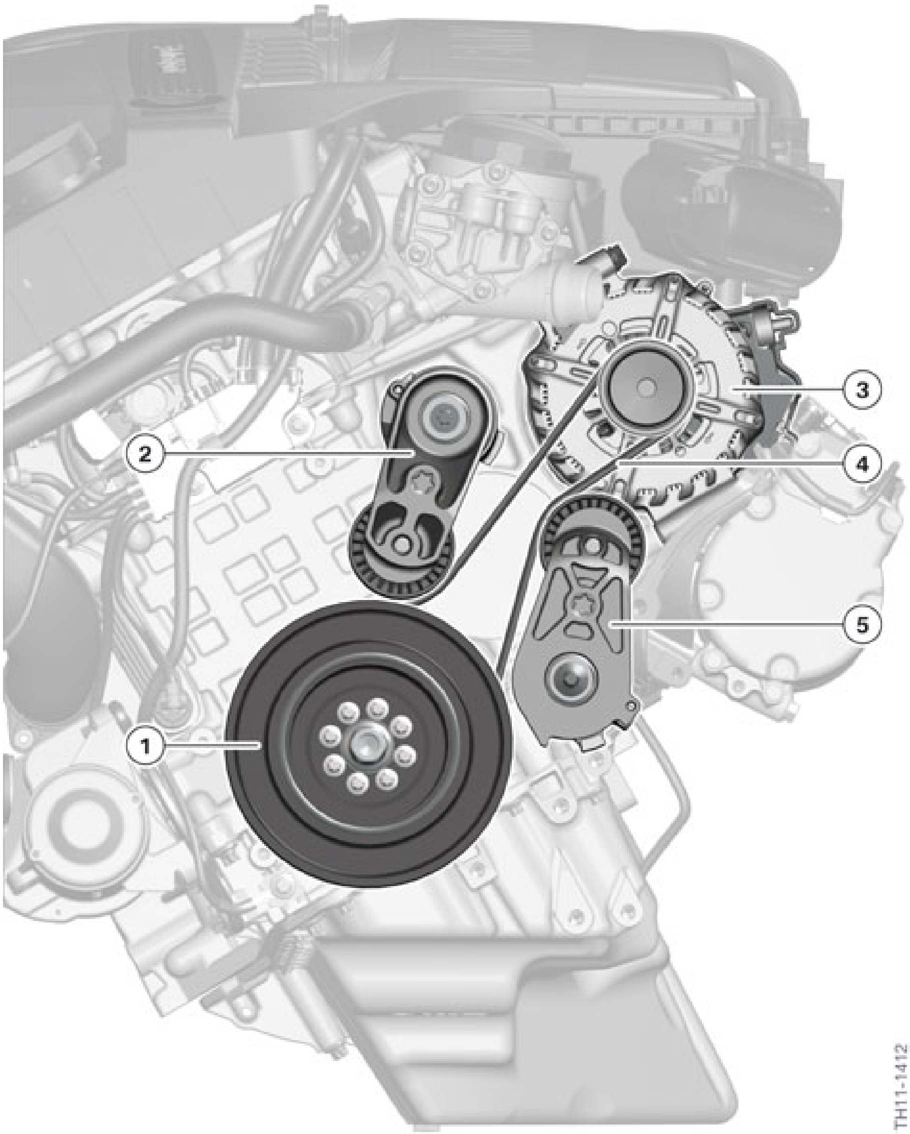
The N55 engine and its peripherals have been modified for use in the ActiveHybrid 5. Details of the individual modifications are provided in the following section.

2.2.2. Belt drive

The F10H is a full hybrid vehicle, i.e. its capable of purely electric driving. In this case, functions such as hydraulic steering, heating and air-conditioning should also still be available. As the combustion engine is off in this operating condition, it cannot activate a power steering pump or an A/C compressor. For this reason these two systems are operated electrically and discontinued from the belt drive.

In place of the alternator, a starter motor generator is used for the first time in a BMW hybrid. The generator can start the combustion engine in start/stop phases or during fully electric driving. The belt drive of the N55 engine had to be adapted for the integration of the starter motor generator and the modified loading.

2. Powertrain Components

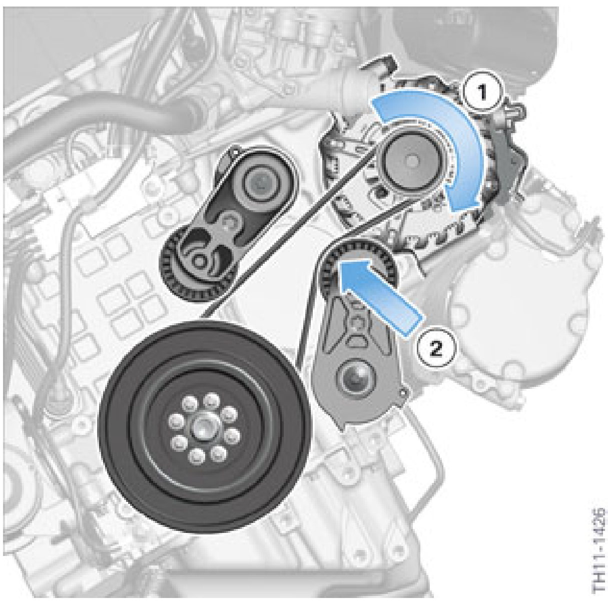


Belt drive F10H

Index	Explanation
1	Vibration damper with belt pulley
2	Top belt tensioner
3	Starter generator
4	Belt
5	Lower belt tensioner

Start-up

BMW engines are right-turning engines. If one looks at the engine from the front, the crankshaft rotates clockwise. To start the engine after a stop phase or during an electric journey, the starter motor generator has to rotate the engine. The top part of the belt is tensioned and the bottom is slackened. To prevent the belt slipping, it is held taut by the bottom belt tensioner.

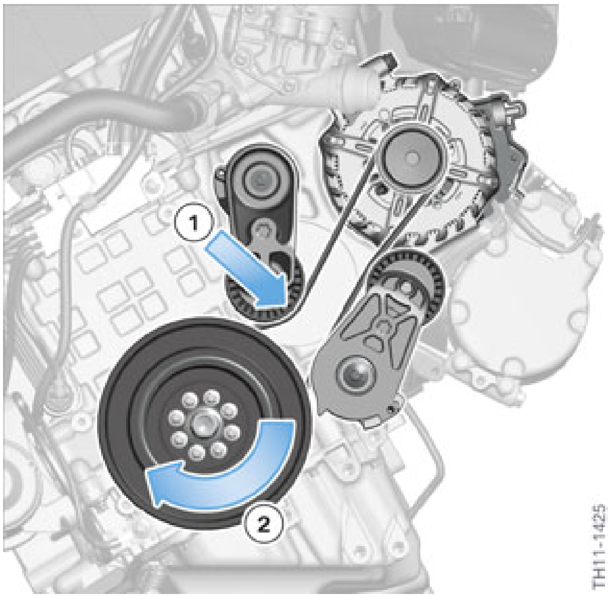


Belt drive in starting mode of the starter motor generator

Index	Explanation
1	Direction of force when starter motor generator powers the combustion engine
2	Direction of force of lower belt tensioner

Charging mode

If the starter motor generator is operated as an alternator, it takes energy from the combustion engine. The combustion engine powers the starter motor generator. The lower belt is thus tensioned; the top belt is slackened. To also prevent the belt slipping in charging mode, the belt is held taut by the top belt tensioner.



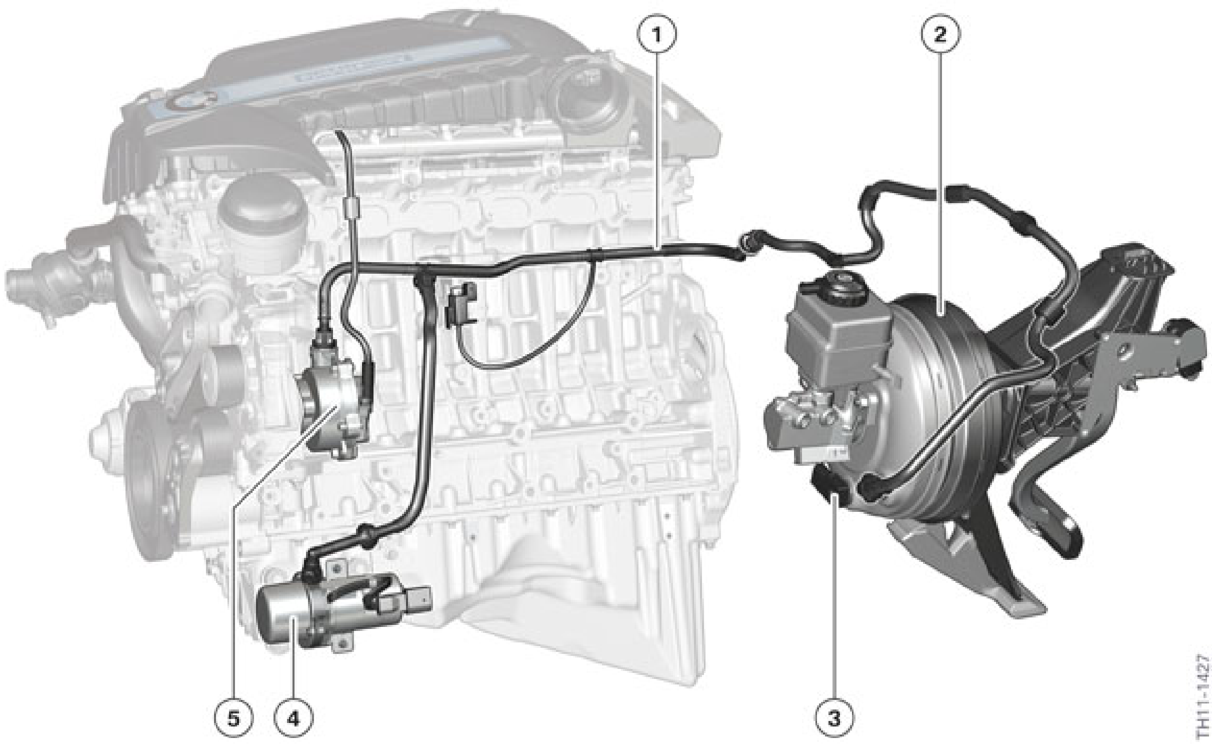
Belt drive in charging mode of the starter motor generator

Index	Explanation
1	Direction of force of the top belt tensioner
2	Direction of force when combustion engine powers the starter motor generator

2.2.3. Vacuum system

The N55 engine has a mechanical vacuum pump for generating the necessary vacuum to supply various components. As the vacuum supply must also be guaranteed in phases in which the N55 engine is switched off, the vacuum system has been enhanced with an electric vacuum pump. As soon as the vacuum in the system drops below a certain threshold, the electric vacuum pump is activated. The vacuum is recorded by a pressure sensor in the brake booster, which is already known from vehicles with an automatic start/stop system.

The following graphic provides an overview of the respective components.



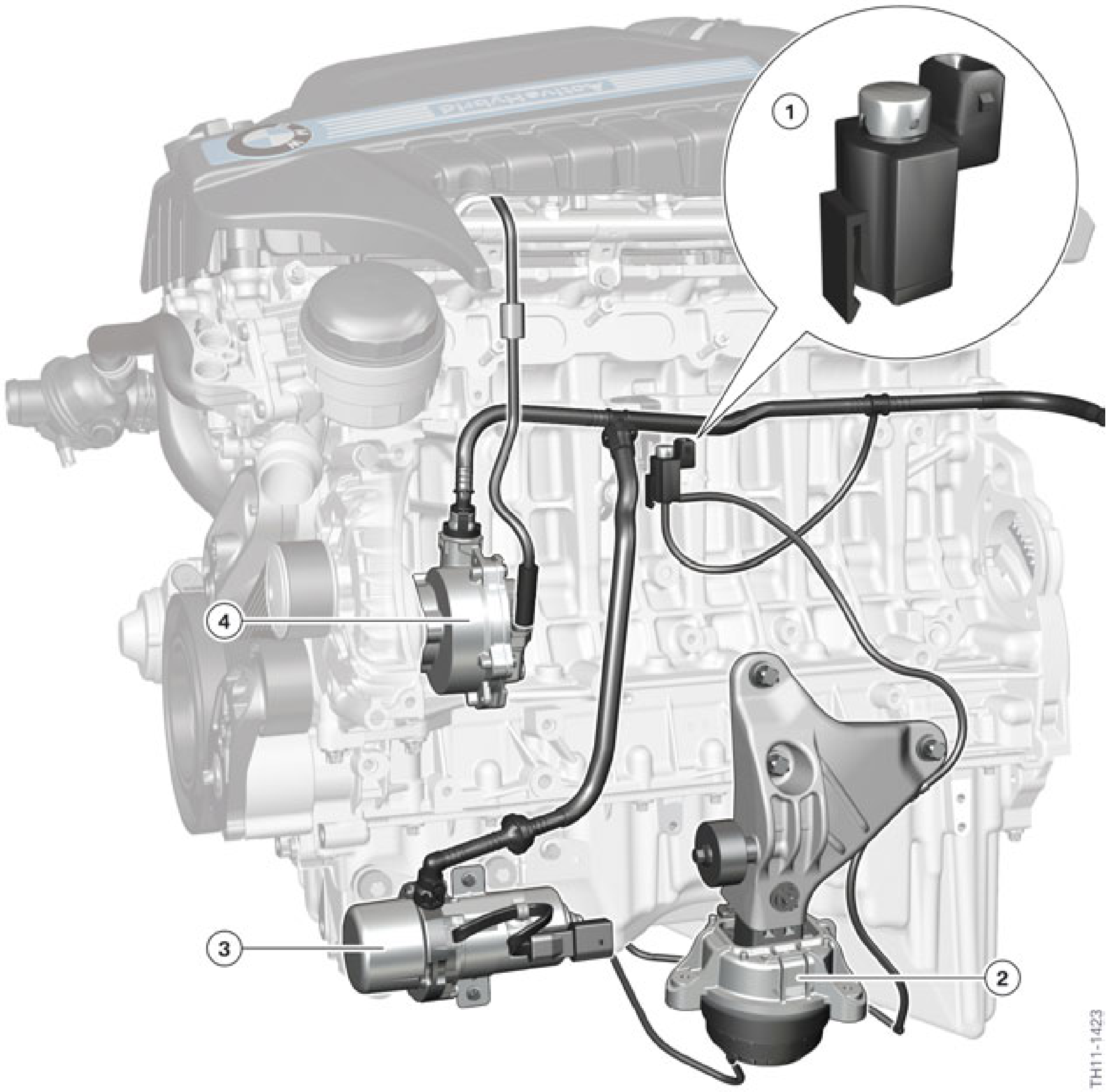
F10H Vacuum system

Index	Explanation
1	Vacuum line
2	Brake booster
3	Pressure sensor
4	Electric vacuum pump
5	Mechanical vacuum pump

2.2.4. Engine mounts

In the F10H the engine is mounted using the controlled variable-damping motor mounts already known from vehicles with diesel engines. These are adjusted to hard or soft by means of a vacuum in order to guarantee a comfortable start and idle state of the combustion engine. The valve for activating the controlled variable-damping mounts of the combustion engine is activated by the Digital Engine Electronics. The behavior of the controlled variable-damping mounts of the F10H is similar to that in vehicles with diesel engines. If a vacuum is applied to these, they are soft. This corresponds to the setting for idle state and start-up of the combustion engine and ensures comfortable damping action. As soon as the vacuum is no longer applied to the mounts and an ambient pressure is established, the mounts become hard. This setting is activated while driving off in the F10H.

The vacuum to operate the controlled variable-damping mounts is supplied by the vacuum system (described above).



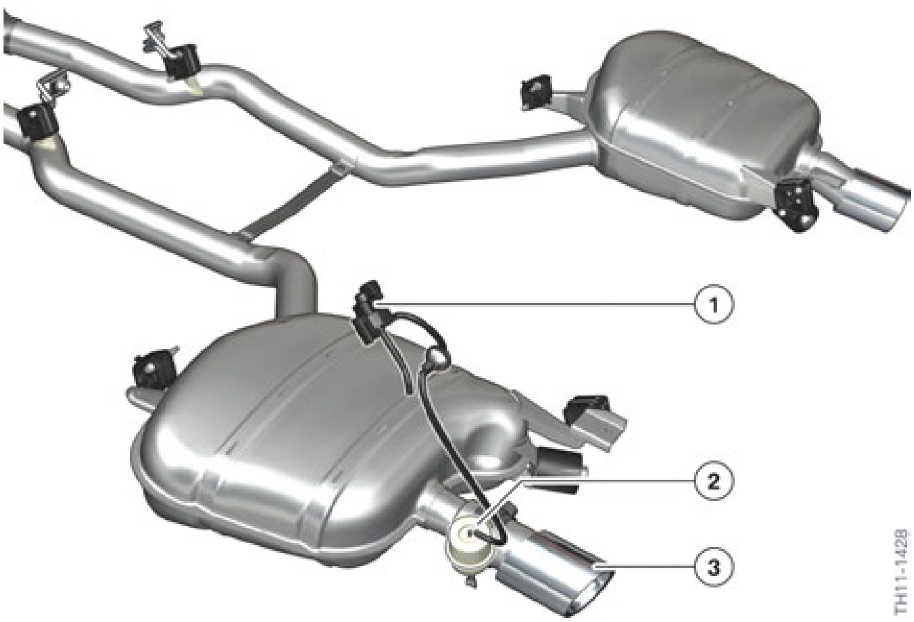
F10H engine mounts

Index	Explanation
1	Valve for activating the controlled variable-damping mount
2	Controlled variable-damping mount, left
3	Electric vacuum pump
4	Mechanical vacuum pump

2.2.5. Exhaust system

Due to the higher requirements regarding the acoustics of the F10H, the exhaust system has been equipped with a exhaust flap. It is located on the left exhaust tailpipe and is activated by the DME. For this purpose, an electrical changeover valve is activated which applies a vacuum to an actuator. The vacuum is supplied from the vacuum system in the engine compartment.

The exhaust tailpipes of the exhaust system are also equipped with matt, chrome-plated covers.



F10H Exhaust system

Index	Explanation
1	Changeover valve, exhaust flap
2	Actuator, exhaust flap
3	Matt, chrome-plated cover

2.2.6. Other adaptations

The mounts of the N55 engine are designed for increased stress due to the frequent starting and stopping of the engine.

The acoustics cover of the N55 engine in the F10H has "ActiveHybrid" lettering.

F10H front end

Index	Explanation
1	Acoustics cover in the F10H

2.3. Fuel supply systems

The fuel supply system of the F10H is based on that of the non–hybrid F10.

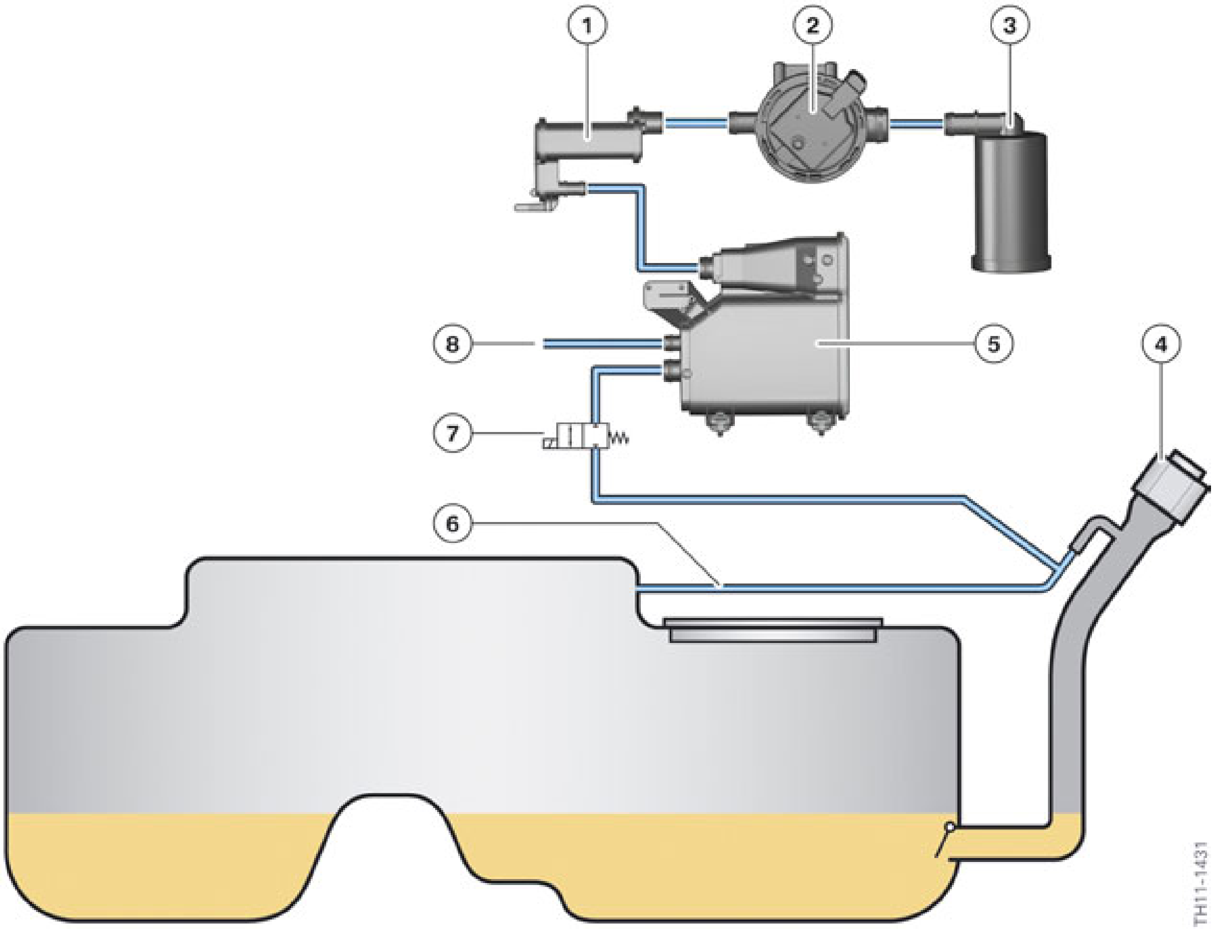
The fuel tank capacity has been reduced by three liters to 67 liters (0.79 gal) to make room for laying the high-voltage cables on the vehicle underbody of the F10H.

Note: The fuel tank ventilation in US-version vehicles had to be adapted due to legal requirements.

2.3.1. Fuel tank ventilation

As with all vehicles in the US, the fuel tank ventilation in the F10H is realized via a carbon canister which purifies the hydrocarbon (HC) gases arising in the fuel tank. The hydrocarbon collected in the carbon canister is fed to the combustion engine and burned. This process is described as purging or flushing the carbon canister. As purging is only carried out when the combustion engine is running and as there are considerably more phases without a running combustion engine in the operation of the F10H, the fuel tank ventilation of the F10H had to be adapted.

For this purpose, a shutoff valve (7) and an additional carbon canister (1) were added to the fuel tank ventilation.

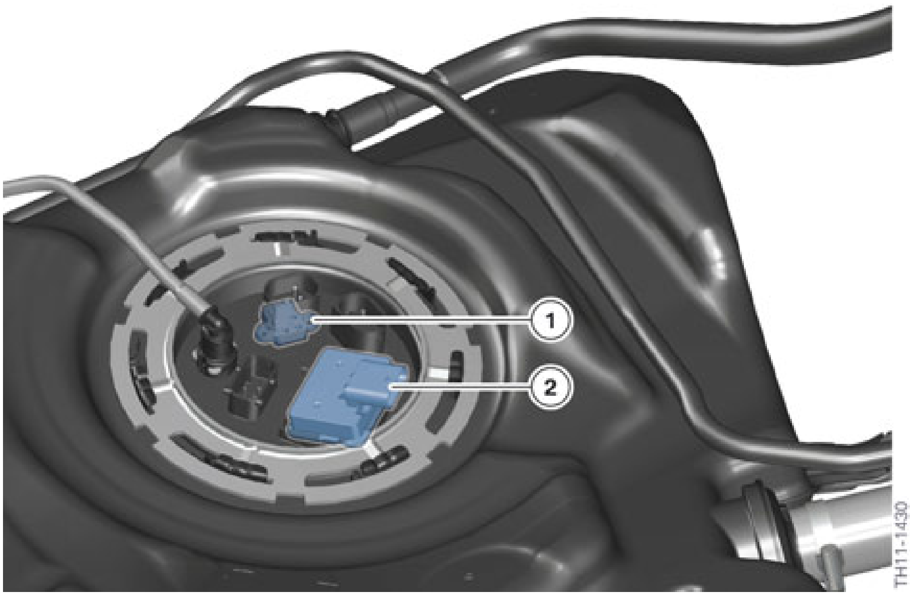


Fuel tank ventilation for US-version vehicles

Index	Explanation
1	Additional carbon canister (activated carbon pressed in honeycomb shape)
2	Tank leak diagnosis (Natural Vacuum Leak Detection, NVLD)
3	Air cleaner (spinning strainer)
4	Fuel filler neck
5	Carbon canister
6	Fuel tank ventilation line
7	Shutoff valve, tank ventilation system
8	Purge air line to the combustion engine

The shutoff valve for the fuel tank ventilation is closed in certain operating phases of the F10H. The purging process of the carbon canister is accelerated with the closing of the shutoff valve. In addition, no other gases pass through the carbon canister from the fuel tank.

A certain pressure and temperature range of the fuel tank is part of the operating phases in which the shutoff valve is closed. The pressure range goes from a slight vacuum to slight excess pressure in the fuel tank; the temperature range goes from slightly above freezing point to operating-temperature state. If the pressure or temperature range are exceeded, the shutoff value is opened. The pressure and temperature of the fuel tank are determined by a sensor installed on the service cover of the fuel tank.

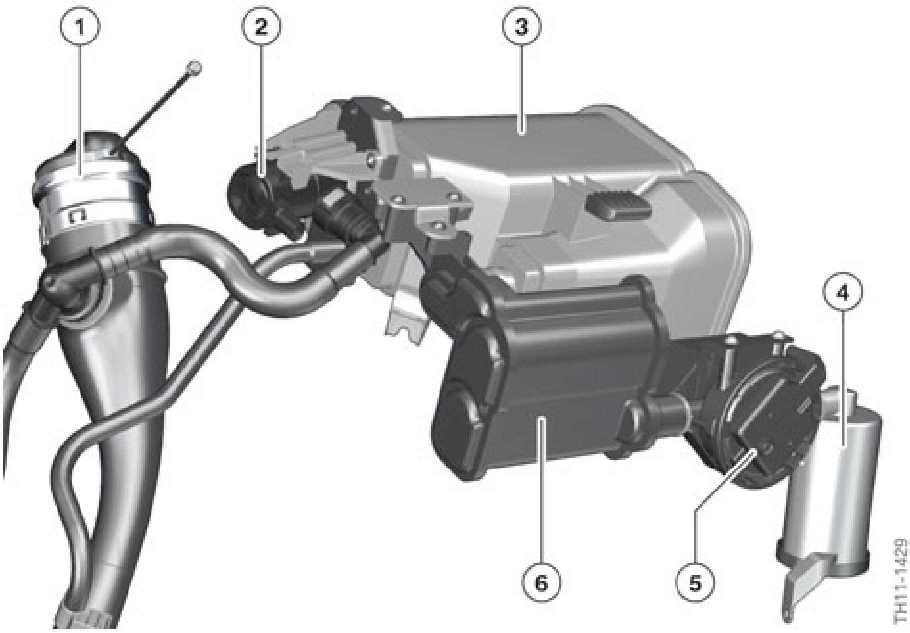


Service cap of the fuel tank for US-version vehicles

Index	Explanation
1	Pressure and temperature sensor
2	Tank leak diagnosis electronics (Natural Vacuum Leak Detection, NVLD)

If the drive readiness has been deactivated or the vehicle is exited, the shutoff valve is always opened. This way, for example, the excess pressure arising during refuelling the vehicle can escape via the fuel tank ventilation.

Another new component in the tank ventilation system of the F10H is the additional carbon canister. This is downstream of the standard carbon canister and contains activated carbon which is pressed in a honeycomb shape. The activated carbon in a honeycomb shape filters out the hydrocarbon (HC) particularly well from the slow flowing gases and was installed to comply with legal requirements.



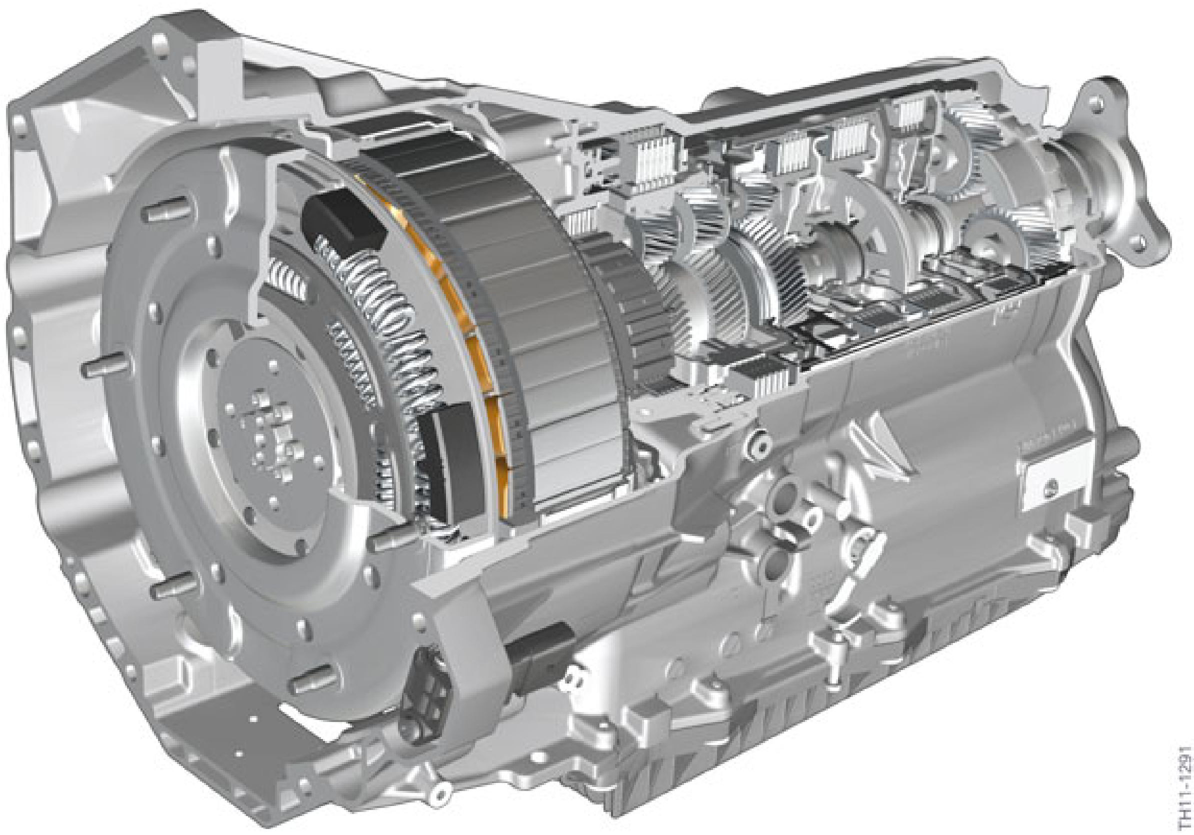
Components of the fuel tank ventilation in US-version vehicles

Index	Explanation
1	Fuel filler neck
2	Shutoff valve for fuel tank ventilation
3	Carbon canister
4	Air cleaner (spinning strainer)
5	Tank leak diagnosis (Natural Vacuum Leak Detection, NVLD)
6	Additional carbon canister (activated carbon pressed in honeycomb shape)

2.4. Automatic transmission

2.4.1. Introduction

The automatic transmission GA8P70HZ of the F10H is based on the 8 speed transmission introduced in the F07 at the end of 2009 (GA8HP70Z) which is also manufactured by ZF.



GA8P70HZ transmission

The electrical machine, a decoupler for the combustion engine, a torsional vibration damper and an auxiliary electric oil pump, are integrated in the 8-speed automatic transmission (Within the same installation space of the non hybrid model). In addition, various components have been adapted for operation in the vehicle.

2.4.2. Structure and function

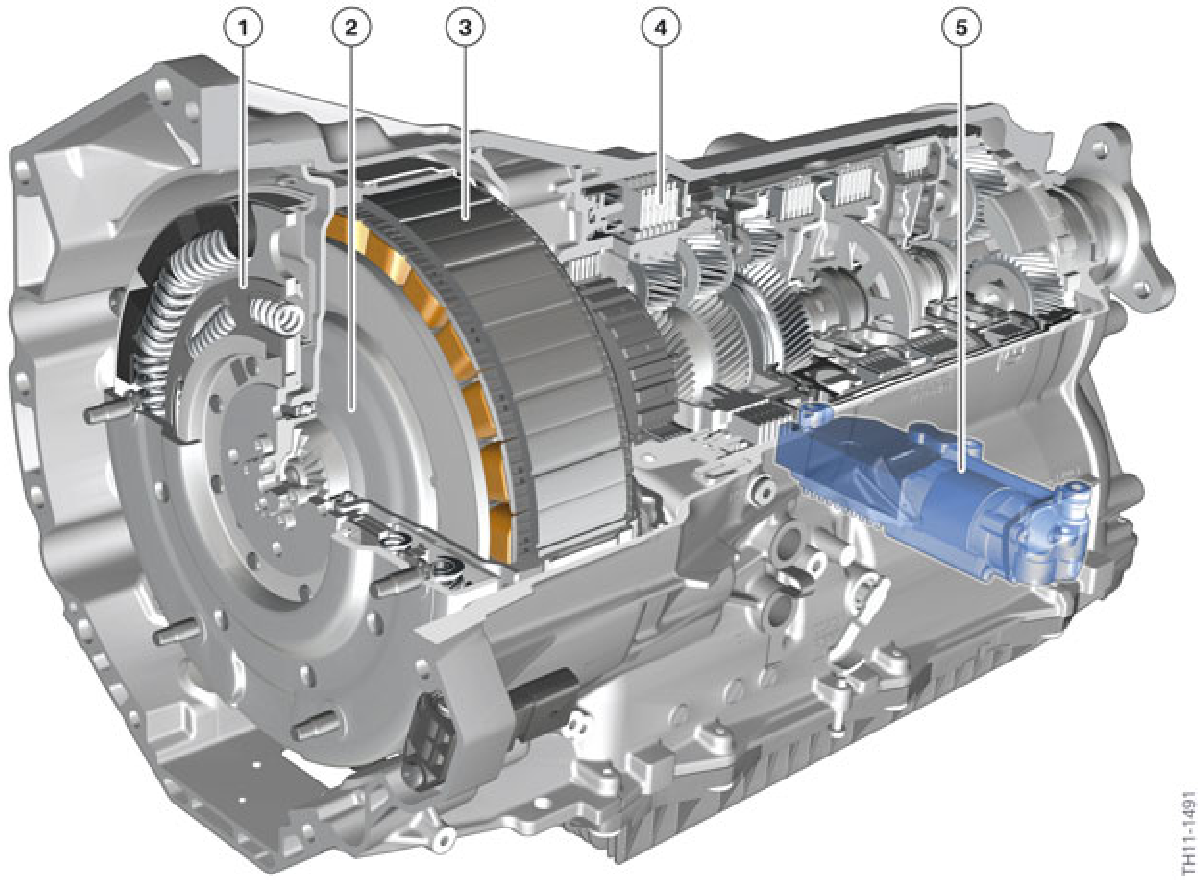
Overview

The F10H automatic transmission has been adapted for the particular requirements of hybrid operation. Although some of the existing components have been modified or replaced, the housing of the GA8P70HZ transmission has the same dimensions as the housing of the GA8HP70Z transmission.

The GA8P70HZ transmission has four new components:

- Torsional vibration damper
- Electrical machine
- Engine decoupler
- Auxiliary electric oil pump (for supplying transmission oil pressure when transmission input shaft is not engaged).

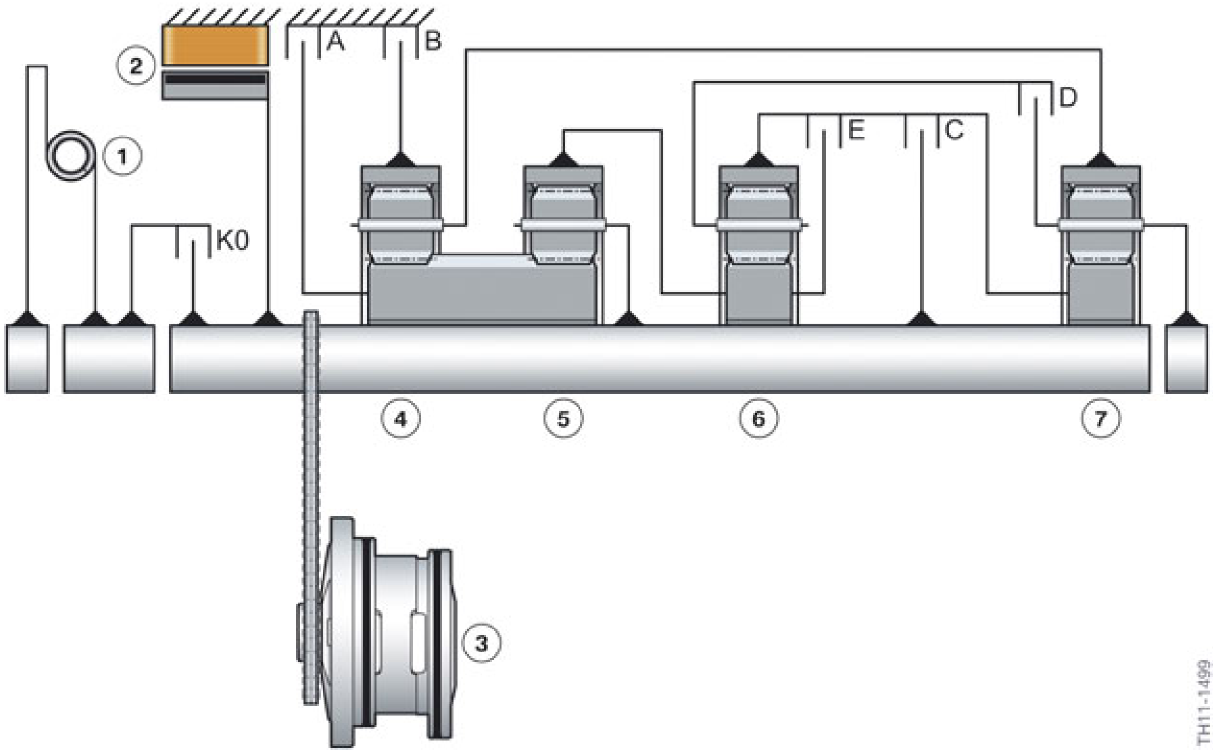
Due to the absence of the torque converter, the multi-disc brake B of the automatic transmission has been modified. In addition to its function as a shift element, in the F10H it enables the vehicle to start-up and crawl.



GA8P70HZ

Index	Explanation
1	Torsional vibration damper
2	Decoupler
3	Electrical machine
4	Multi-disc brake B
5	Auxiliary electric oil pump

The following transmission schematic diagram of the GA8P70HZ transmission shows how the new components have been integrated in the automatic transmission.



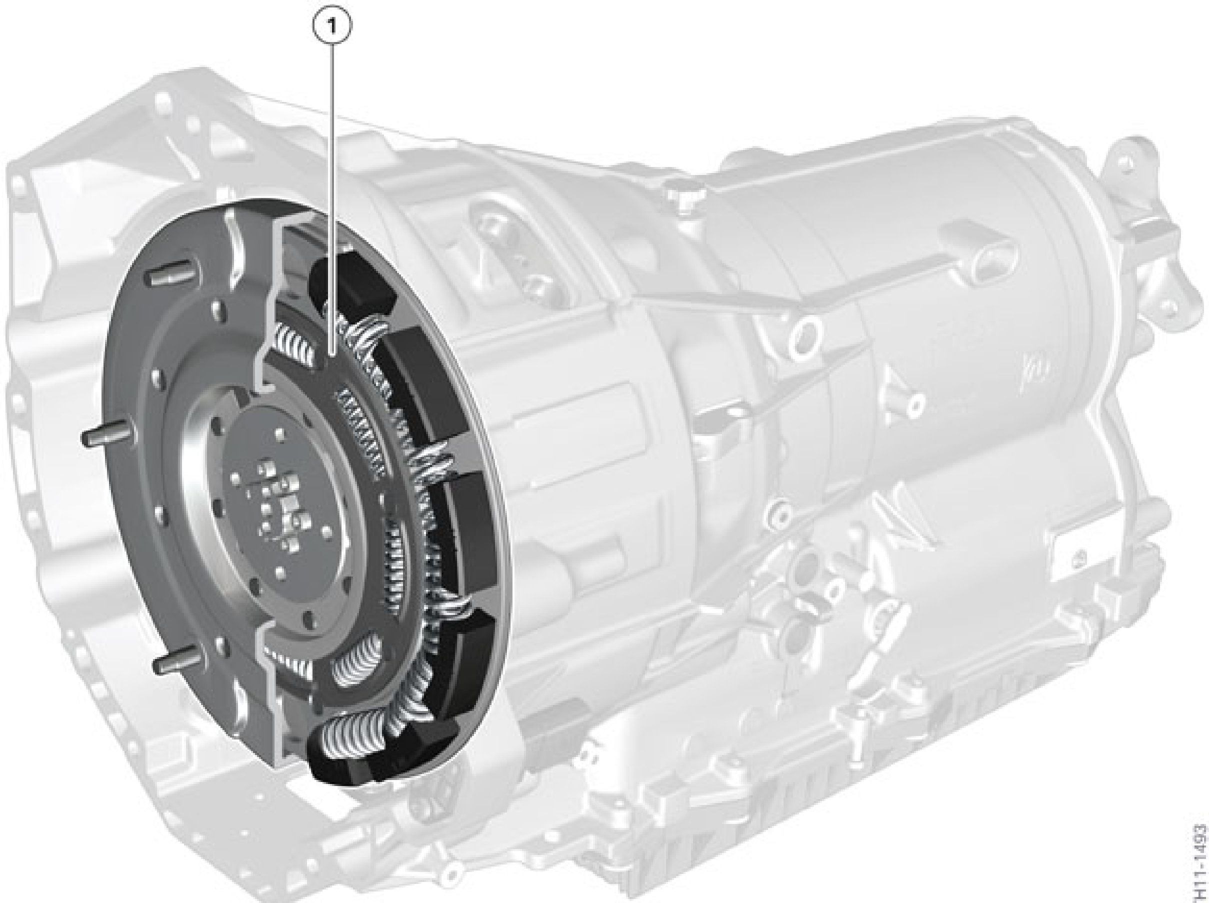
Schematic structure of the GA8P70HZ transmission

Index	Explanation
1	Torsional vibration damper
2	Electrical machine
3	Mechanical oil pump
4	Gear set 1
5	Gear set 2
6	Gear set 3
7	Gear set 4
A	Multi-disc brake A
B	Multi-disc brake B
C	Multi-disc clutch C
D	Multi-disc clutch D
E	Multi-disc clutch E
K0	Decoupler

Torsional vibration damper

The unequal running and the resulting torsional vibrations of combustion engines may cause strong droning or rattling noises at certain speeds and operating conditions. To isolate these torsional vibrations a torsional vibration damper is used in the automatic transmission of the F10H. The torsional vibration damper establishes the mechanical connection between the flywheel of the combustion engine and decoupler of the automatic transmission.

The torsional vibration damper can be replaced in the event of a fault.



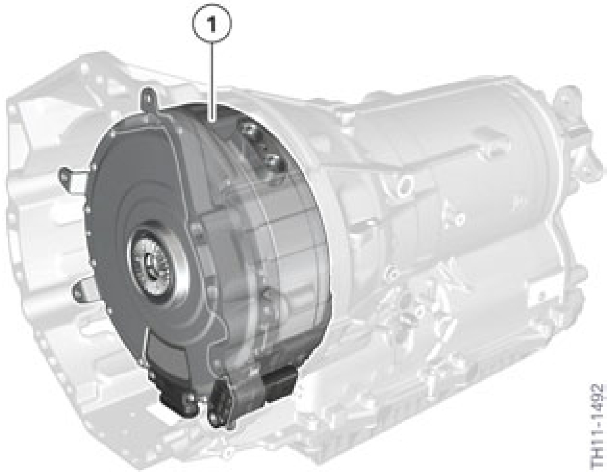
Torsional vibration damper of the GA8P70HZ transmission

Index	Explanation
1	Torsional vibration damper

Electrical machine

Another new feature of the GA8P70HZ transmission is the integration of the electrical machine and a decoupler. These are located behind the torsional vibration damper. Together with the torsional vibration damper the electrical machine and the decoupler occupy the space of the conventional torque converter.

Further information on the electrical machine is available in the chapter entitled "Electrical machine".



Electrical machine in the GA8P70HZ transmission

Index	Explanation
1	Electrical machine

Decoupler

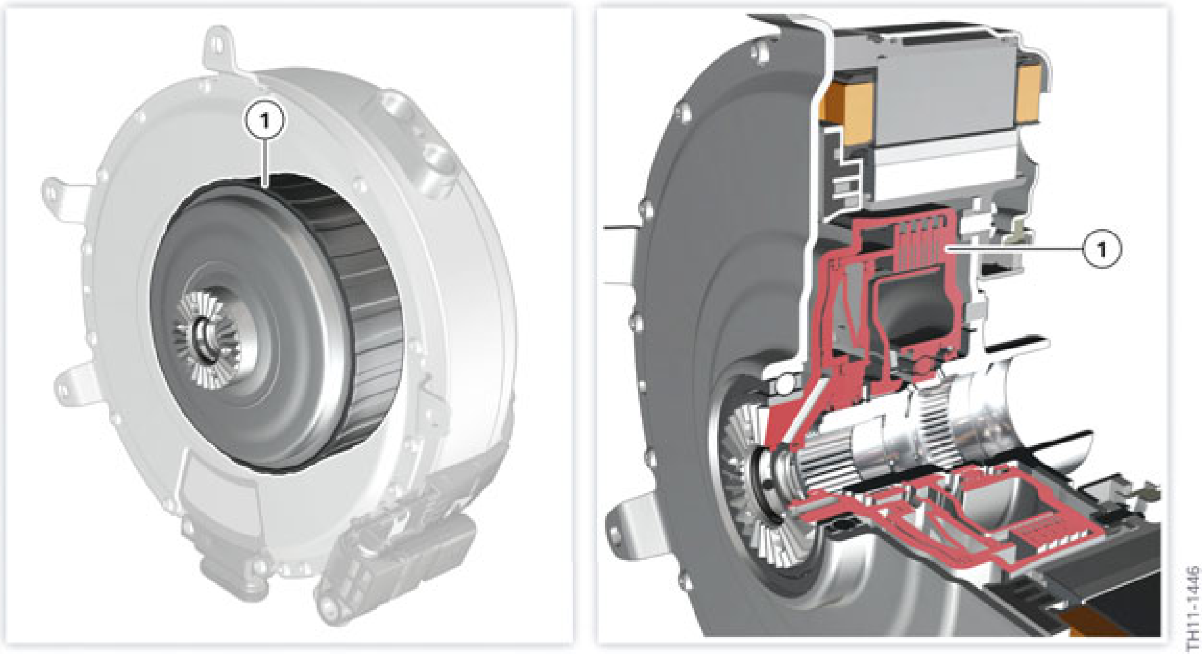
The decoupler and the electrical machine are located within a common housing.

The decoupler is designed as a wet multi-disc clutch and is used to disconnect the combustion engine from the electrical machine and the rest of the drivetrain in certain operating conditions. This is carried out for example while fully electric driving and when "Coasting".

The extreme accuracy of the decoupler makes the connection and disconnection of the combustion engine virtually imperceptible. As soon as the decoupler is closed, the electrical machine, transmission input shaft and the combustion engine rotate at the same speed (except for a slip control in the decoupler to some extent from an acoustic viewpoint).

The decoupler is (like all clutches and multi-disc brakes of the automatic transmission) operated by the mechatronics module. It is opened when in a depressurized state. Transmission oil pressure is thus required to close the clutch. This is supplied either by the auxiliary electric oil pump or the mechanical oil pump.

Because the mechanical oil pump is powered by the electrical machine when the decoupler is open, it may not be possible to close the decoupler in the event of a malfunction with the electrical machine and at a transmission oil temperature below 0° C/ 32° F and thus no starting process may take place.



Decoupler of the GA8P70HZ transmission

Index	Explanation
1	Decoupler/clutch

Similar to the torque converter in the non hybrid (GA8HP70Z) transmission, the decoupler is able to eliminate rotational oscillations of the engine/drivetrain through microslip control. This results in significantly improved acoustic comfort in the vehicle especially at very low engine speeds.

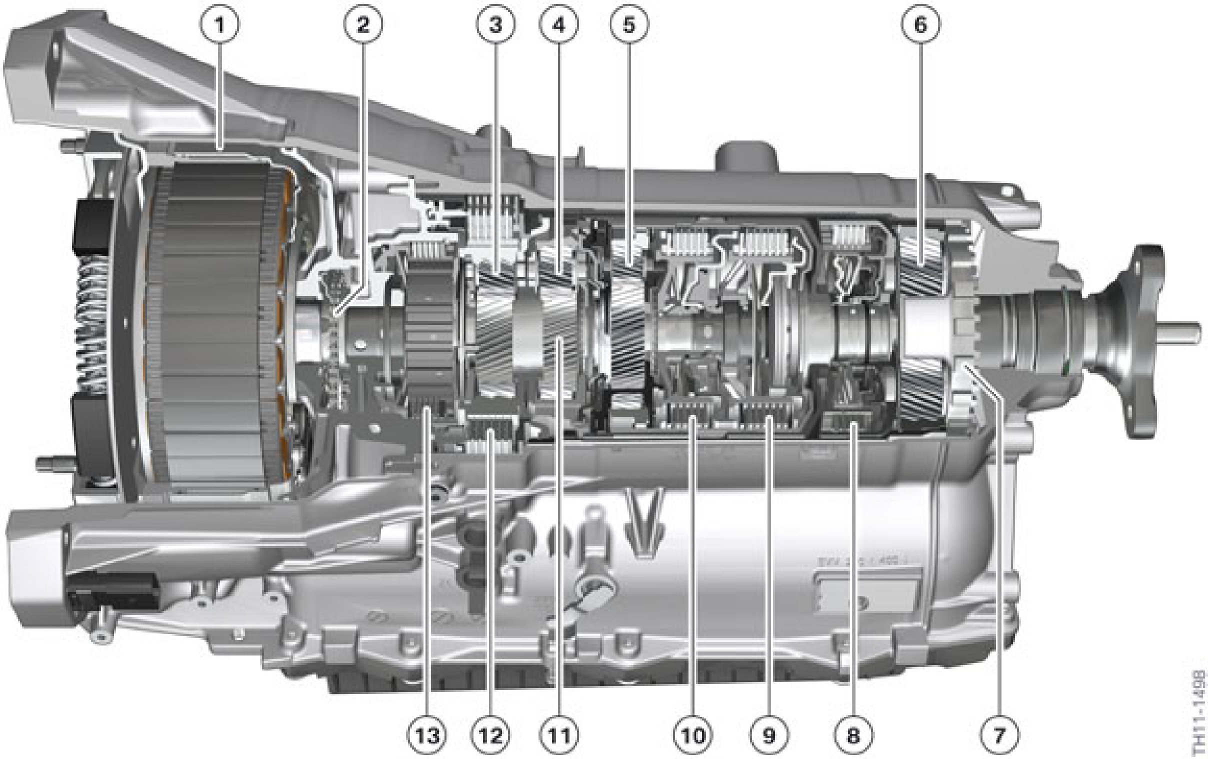
Shift elements

Brakes and clutches are described as shift elements which make possible the shifting and changing of all gears. Like in the GA8HP70Z transmission, the following shift elements are also used in the GA8P70HZ transmission:

- Two fixed multi-disc brakes (brake A and B)
- Three rotary multi-disc clutches (clutch C, D and E).

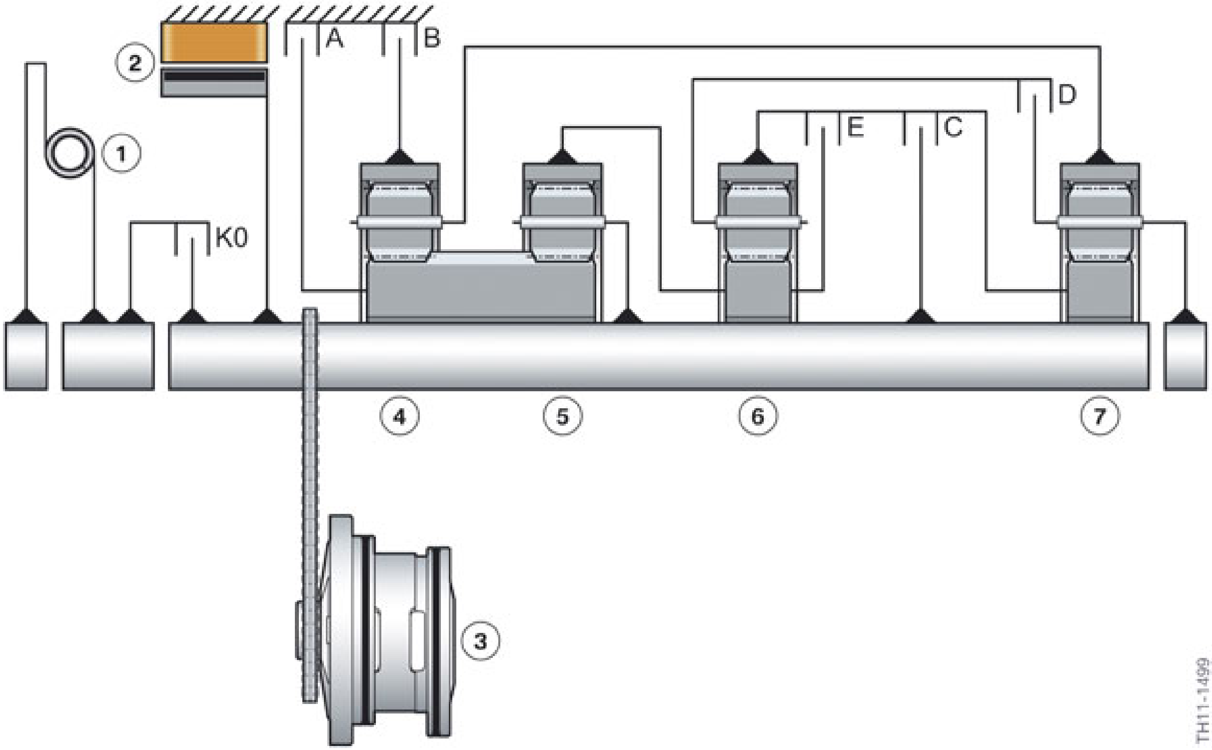
The multi-disc clutches (C, D and E) feed the drive torque to the planetary gear. The multi-disc brakes (A and B) support the torque against the transmission housing.

The clutches and brakes are closed hydraulically. For this oil pressure is applied to a piston enabling it to press the disc sets together.



Overview of the GA8P70HZ transmission

Index	Explanation
1	Electrical machine
2	Drive chain of the mechanical oil pump
3	Gear set 1
4	Gear set 2
5	Gear set 3
6	Gear set 4
7	Parking lock
8	Multi-disc clutch D
9	Multi-disc clutch C
10	Multi-disc clutch E
11	Common sun gear for gear set 1 and 2
12	Multi-disc brake B
13	Multi-disc brake A



Transmission schematic diagram of the GA8P70HZ transmission

Index	Explanation
1	Torsional vibration damper
2	Electrical machine
3	Mechanical oil pump
4	Gear set 1
5	Gear set 2
6	Gear set 3
7	Gear set 4
A	Multi-disc brake A
B	Multi-disc brake B
C	Multi-disc clutch C
D	Multi-disc clutch D
E	Multi-disc clutch E
K0	Decoupler

The shift elements of the GA8HP70Z transmission correspond in number and arrangement to those of the GA8P70HZ transmission. The eight gears are realized in the same way.

The following table shows what shift element is closed in what gear.

Gear	Brake A	Brake B	Clutch C	Clutch D	Clutch E
1	●	●	●		
2	●	●			●
3		●	●		●
4		●		●	●
5		●	●	●	
6			●	●	●
7	●		●	●	
8	●			●	●
R	●	●		●	

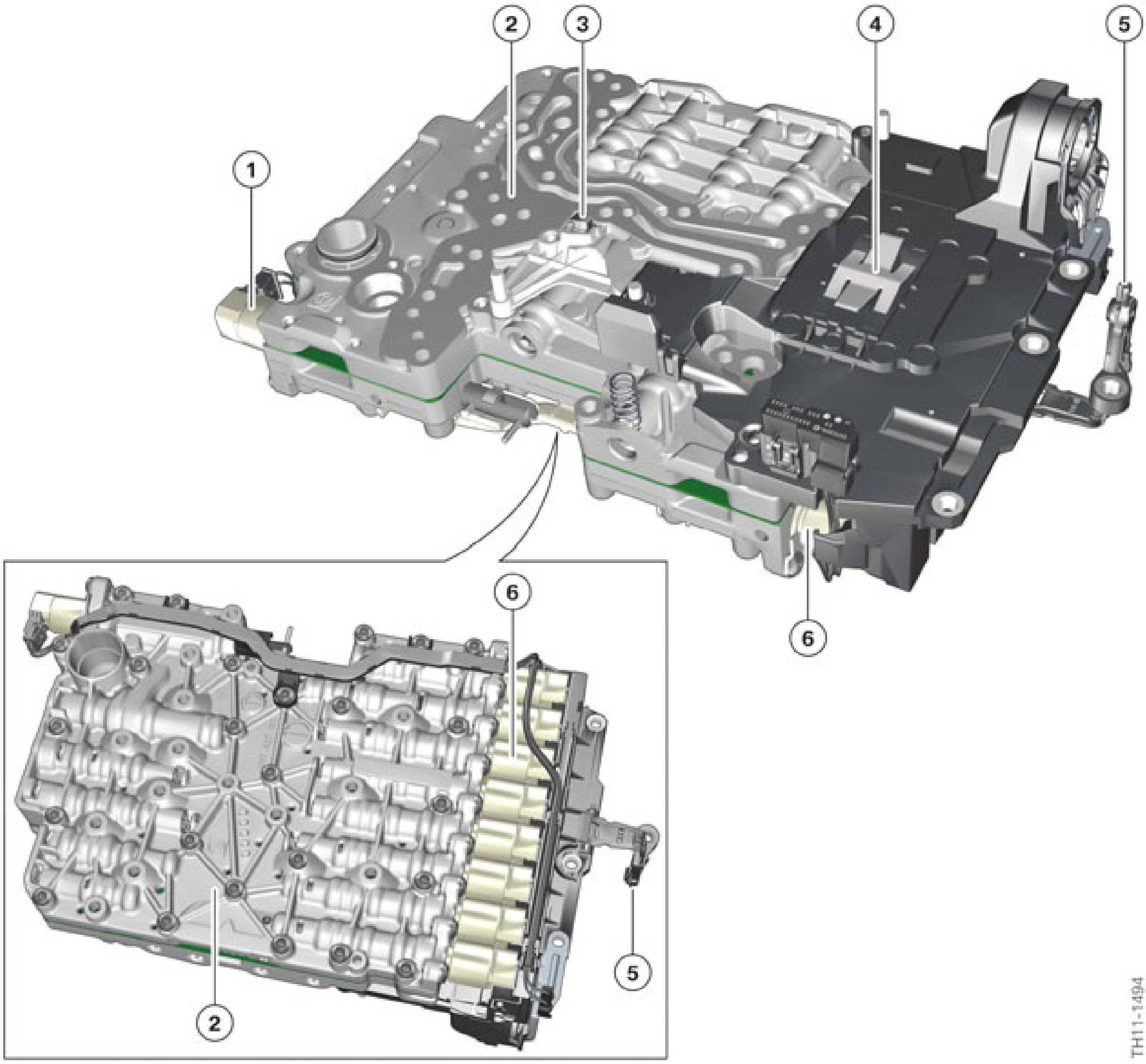
As previously mentioned, due to the discontinuation of the torque converter, the multi-disc brake B of the automatic transmission has been modified. Starting-up and crawling are realized via the multi-disc brake B. For this the number of discs is increased and their diameter enlarged. Here the multi-disc brake B is surrounded by transmission oil depending on the application purpose in order to guarantee cooling.

Mechatronics module

The mechatronics module is a combination of the hydraulic shift unit and the electronic control unit. It is installed in the transmission oil sump. The hydraulic shift unit contains the mechanical components of the transmission control unit such as valves, dampers and actuators.

The electronic control unit contains the complete electronic control unit of the transmission. The electronic control unit is welded and oil-tight.

The mechatronics module had to be adapted for operation in the GA8P70HZ transmission. Thus for example the speed sensor (which records the transmission input speed in the GA8HP70Z) in the GA8P70HZ is used for recording the speed of the planet spider of the first gear set. For this purpose, the multi-pole sensor gear of the speed sensor was connected to the planet spider of the first gear set. This is necessary in order to determine the slip of the multi-disc clutch B during start-up (drive off). Therefore the transmission input speed is determined in the GA8P70HZ transmission with help of the rotor position sensor of the electrical machine.



Mechatronics module of the GA8P70HZ transmission

Index	Explanation
1	Parking lock magnet
2	Hydraulic shift unit
3	Speed sensor, planet spider, gear set 1
4	Electronic control unit
5	Output speed sensor
6	Electronic pressure control valves and solenoid valves

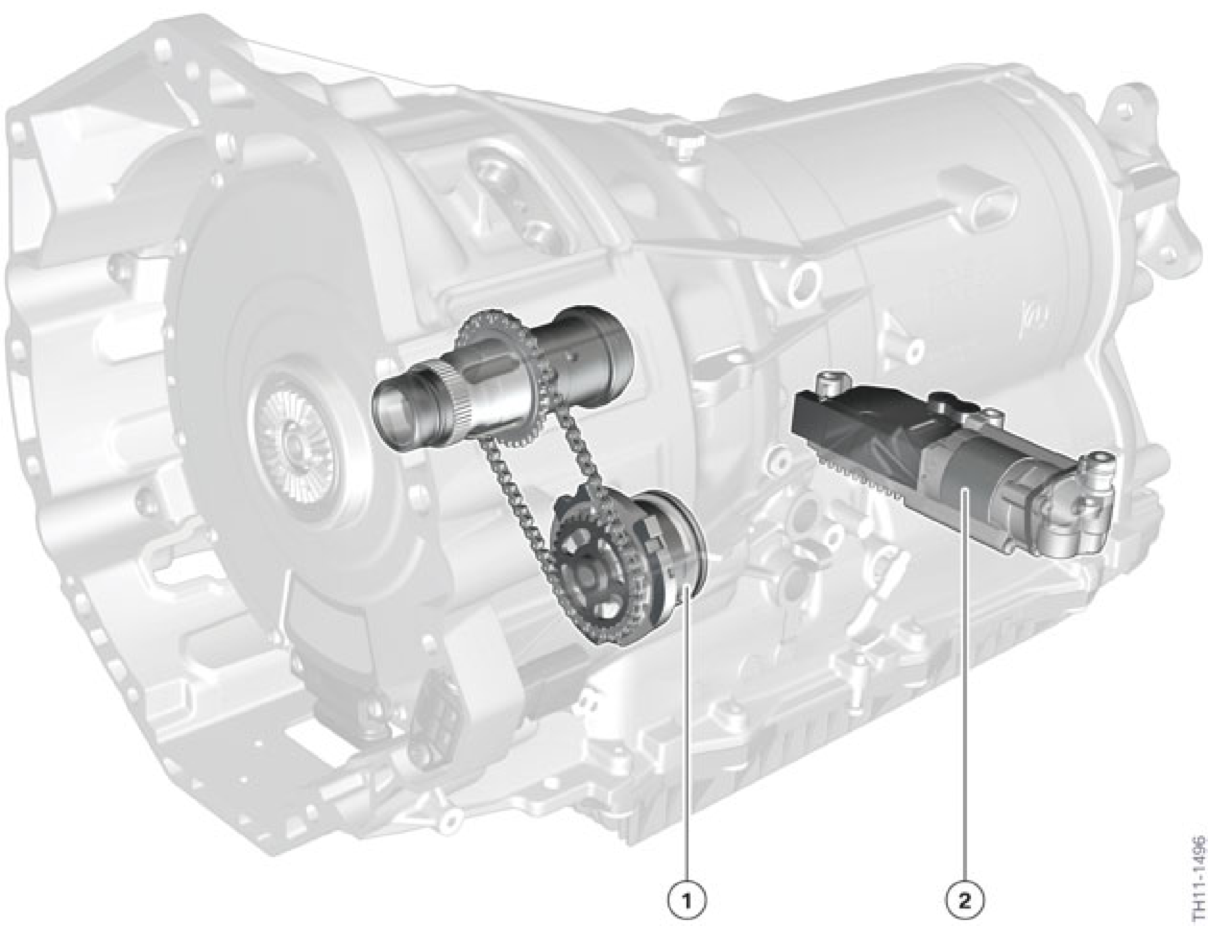
Oil supply

The basic functions of the oil circuit of the GA8P70HZ transmission correspond to those of the GA8HP70Z transmission.

The oil has the following tasks:

- Lubrication
- Control of the shift elements
- Cooling.

It is a conventional pressure circulating system. In addition to the mechanical oil pump known from the GA8HP70Z, an auxiliary electric oil pump has been integrated in the automatic transmission of the F10H.



Oil pumps of the GA8P70HZ transmission

Index	Explanation
1	Mechanical oil pump
2	Auxiliary electric oil pump

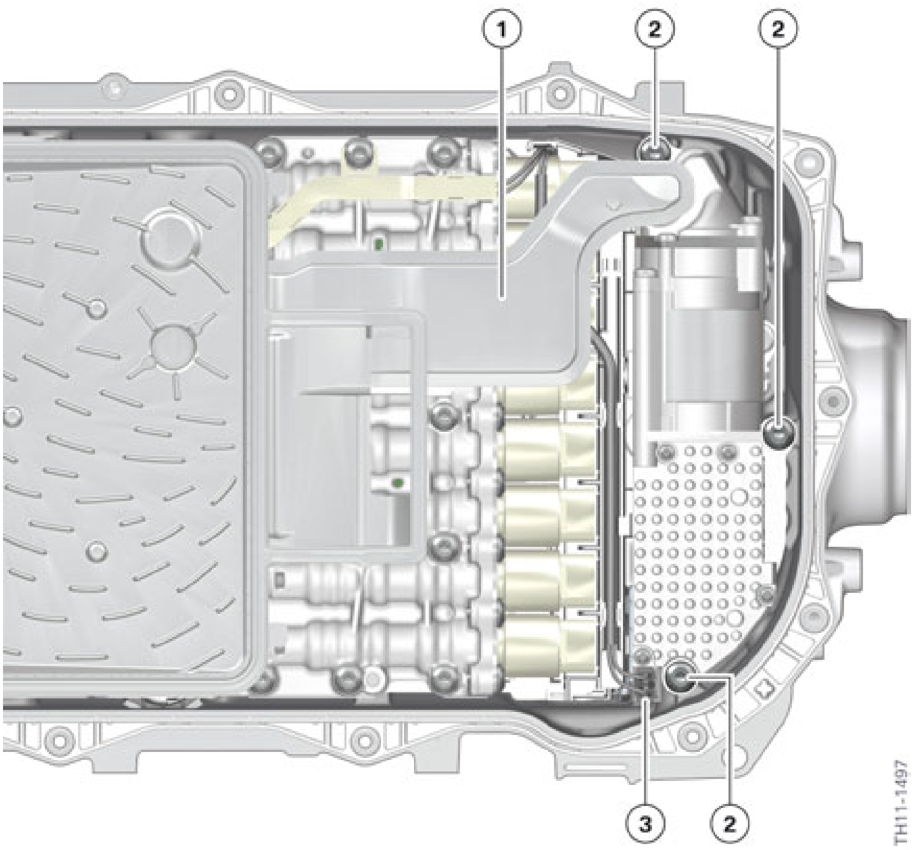
2. Powertrain Components

The mechanical oil pump is driven by a chain via the transmission input shaft. If the decoupler is open, the powertrain is driven via the electrical machine; if the decoupler is closed the powertrain is driven through a combination of a combustion engine and the electrical machine.

In operating phases with too low a speed at the transmission input shaft, the auxiliary electric oil pump can ensure the supply of transmission oil pressure.

The auxiliary electric oil pump is a vane-type compressor, like the mechanical oil pump. It is driven by a brushless direct current motor. The electronic control integrated in the housing of the auxiliary electric oil pump and is activated by the electronic transmission control (EGS). The auxiliary electric oil pump can be operated from a transmission oil temperature of 0° C/32° F.

In the GA8P70HZ transmission it occupies the space of the hydraulic pressure accumulator already known from the F04. Like the hydraulic pressure accumulator, the auxiliary electric oil pump can also be replaced in the event of a fault.



Installation location of the auxiliary electric oil pump

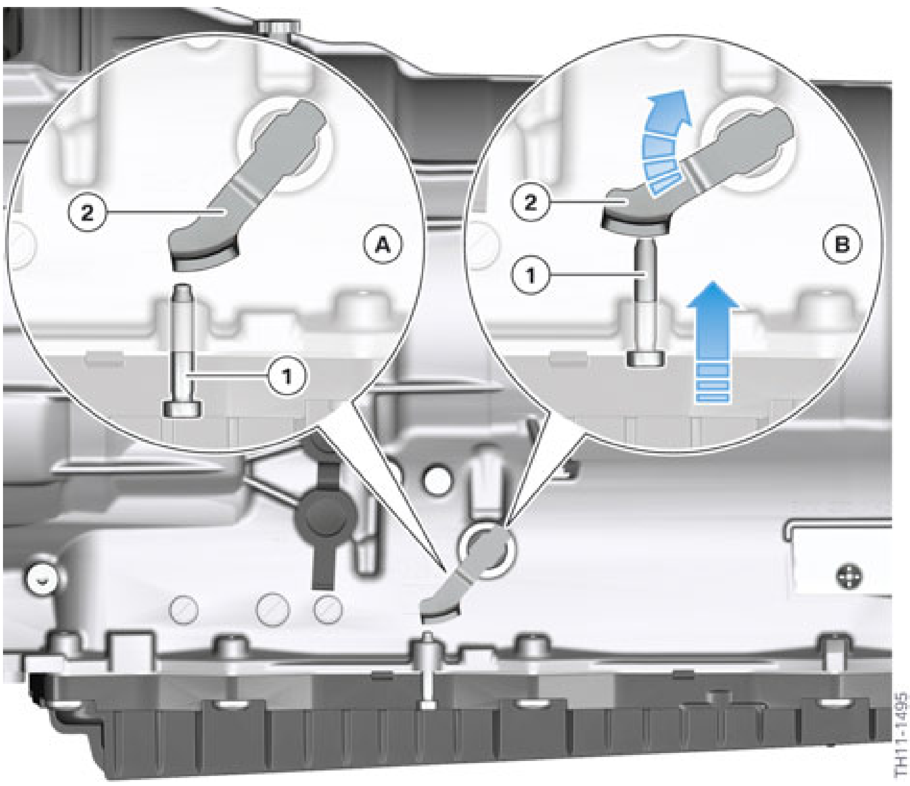
Index	Explanation
1	Intake pipe
2	Bolting points of the auxiliary electric oil pump
3	Electrical connection

Mechanical transmission emergency release

As with previous models, in the event of a fault, the engaged automatic transmission parking lock can be disabled via a mechanical emergency release function by turning an adjusting screw in the vehicle underbody.

⚠

Please refer to the corresponding repair instructions for detailed information on the mechanical transmission emergency release.



Mechanical transmission emergency release

Index	Explanation
A	Transmission parking lock engaged
B	Transmission parking lock released
1	Adjusting screw
2	Parking lock lever

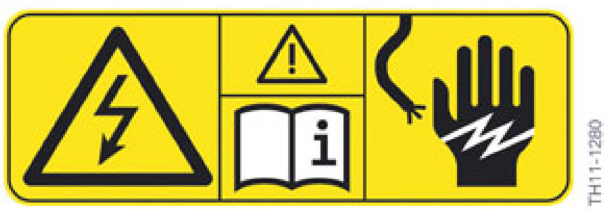
3. Electrical Machine

3.1. Introduction

The electrical machine in the ActiveHybrid 5 is a permanently excited synchronous machine. It can convert electrical energy from the high-voltage battery unit into kinetic energy, by which the vehicle is driven. Both electric driving up to approx. 60 km/h/37 mph, and also the support of the combustion engine, are possible, for example when overtaking (boost function) or for active torque support when changing gears.

In addition, the electrical machine converts kinetic energy into electrical energy during braking and in coasting (overrun) mode to charge the high-voltage battery unit (energy recovery).

The electrical machine is a high-voltage component.



High-voltage component warning sticker

Each high-voltage component has on its housing or casing an identifying label that enables Service technicians and vehicle users to clearly identify the potential hazards associated with the use of high electric voltages.

Only BMW hybrid certified Service technicians are permitted to work on the designated high-voltage components. Certified technicians must: have suitable qualifications, comply with the safety rules and procedures and always follow the repair instructions to the letter.

Work on live high-voltage components is strictly prohibited. Prior to every operation which involves a high-voltage component, it is essential to disconnect the high-voltage system from the voltage supply and to secure it against unauthorized reconnection.

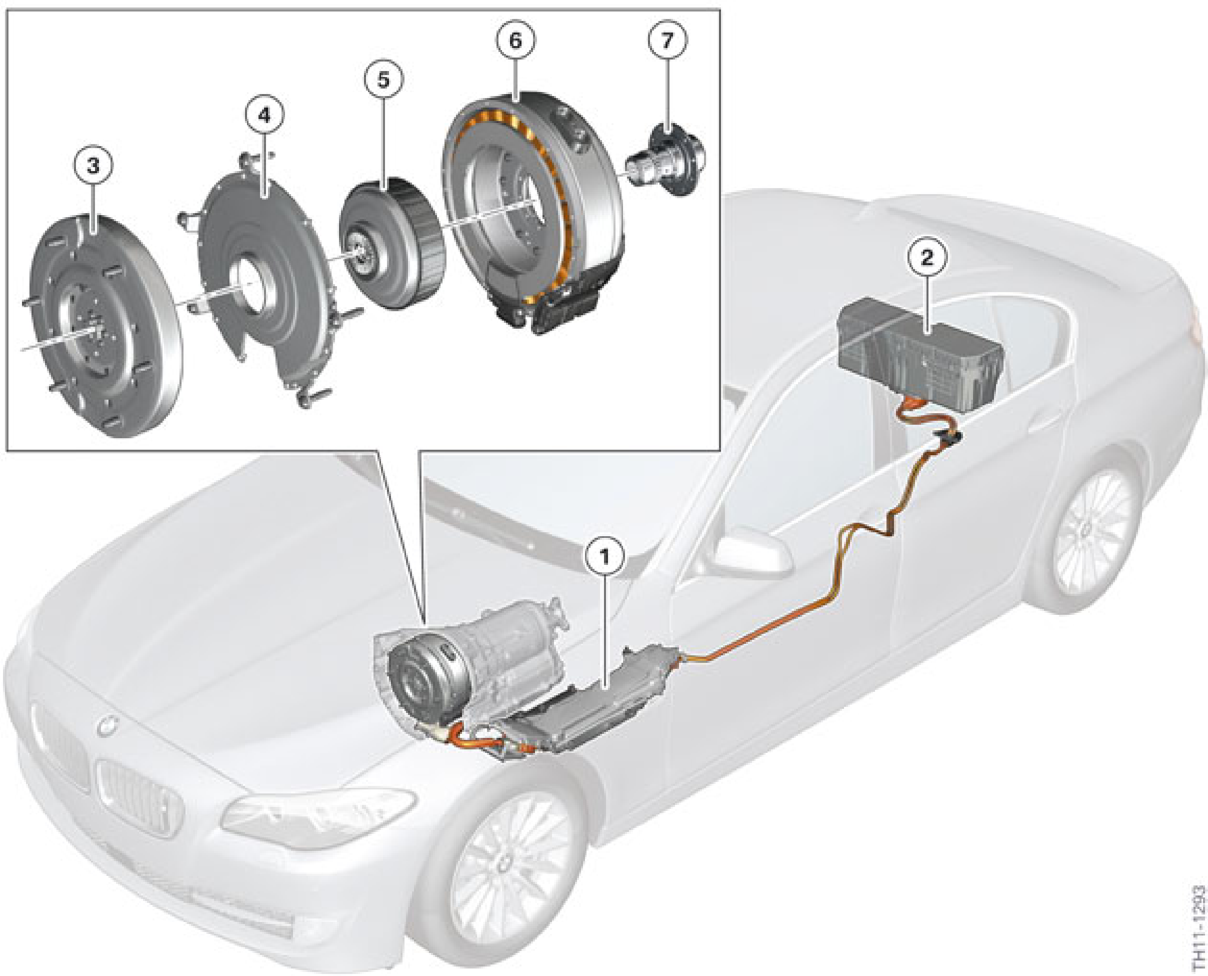
1 Switch off terminal 15.
2 **De-energize** the high-voltage by opening the high-voltage safety switch (Safety Disconnect).
3 **Secure** the high-voltage safety connector from being reconnected.
4 Switch on terminal 15.
5 **Verify** that the "High-voltage system switched-off" Check Control message is displayed in the instrument cluster.
6 Switch off terminal 15 and terminal R.

For high-voltage safety reasons the electrical machine must not be opened or otherwise dismantled.

3. Electrical Machine

In the event of a fault, the complete electrical machine (together with the decoupler/clutch) must always be replaced.

3.2. Installation location

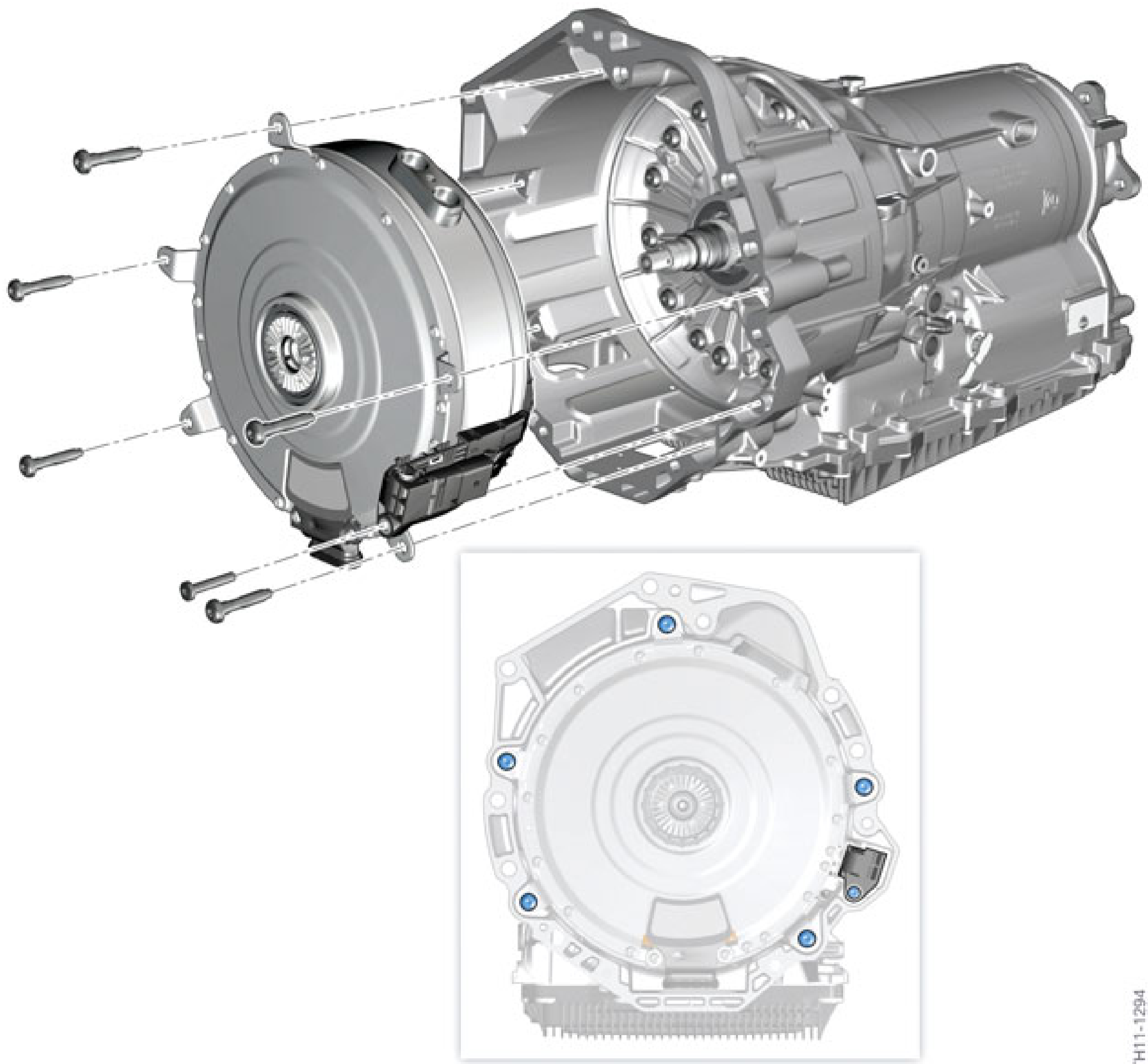


F10H installation location and secondary components of the electrical machine

Index	Explanation
1	Electrical machine electronics
2	High-voltage battery unit
3	Torsional vibration damper
4	Cover for stator support
5	Decoupler (clutch)
6	Electrical machine
7	Hollow shaft

F10H Complete Vehicle
3. Electrical Machine

The electrical machine together with the torsional vibration damper and the clutch form one unit. This unit is integrated into the 8-speed automatic transmission (GA8P70HZ) and weighs only 21 kg/46 lbs. It is installed in the transmission housing in place of the hydraulic torque converter. Thus the automatic transmission in the F10H requires the same space as that in a F10.



F10H mounting of electrical machine

The electrical machine is secured to the transmission housing using five bolts. The connector for the sensor system is also bolted on to the transmission housing.

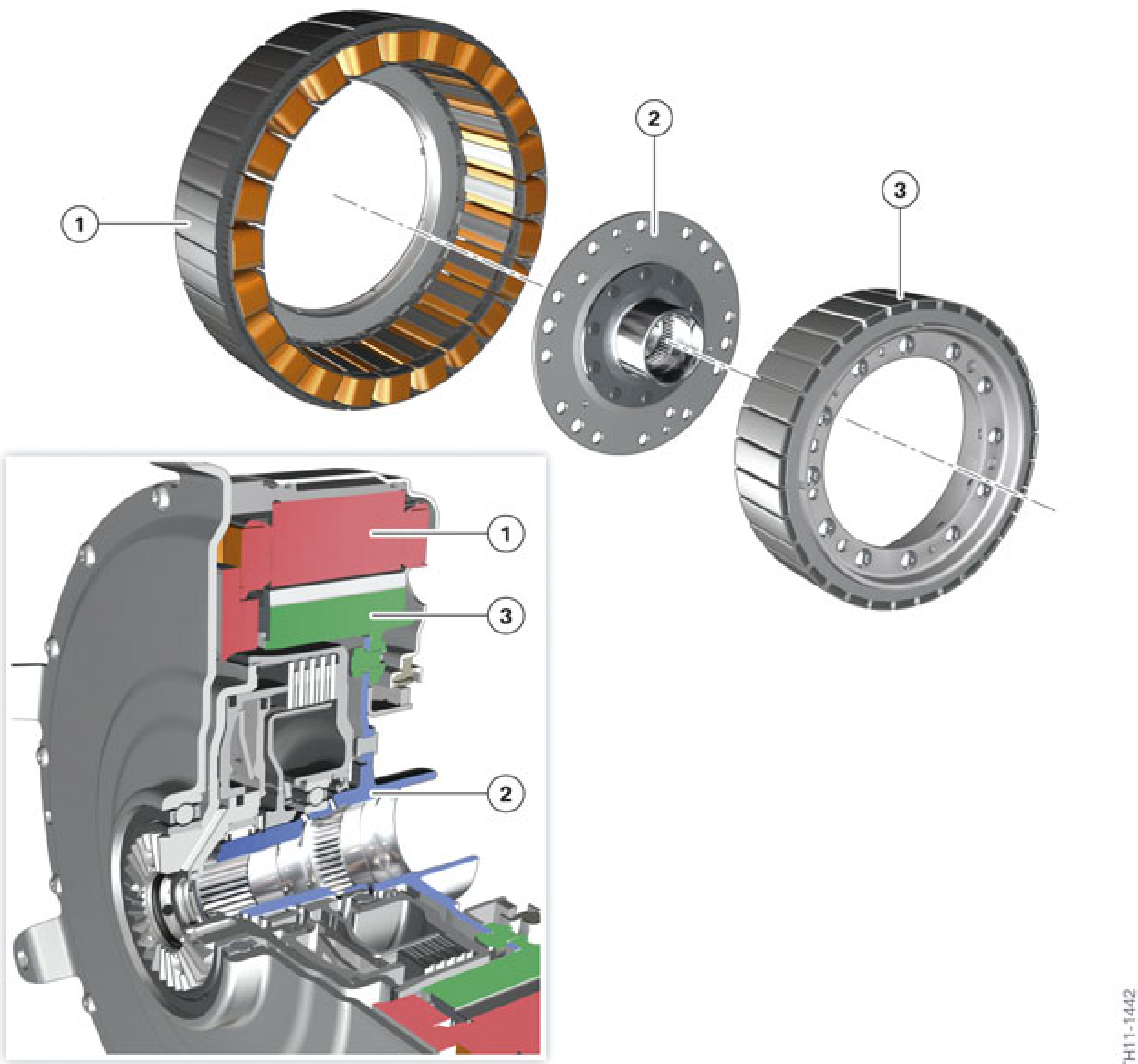
3.3. Design

The main components of the electrical machine are:

- Rotor and stator
- Connections
- Temperature sensor
- Rotor position sensor
- Cooling

The active hybrid 5 is classified as a full hybrid vehicle. In comparison to the "Mild Hybrid" F04, full electric driving is possible in the F10H. For this purpose a clutch is required which can disconnect the combustion engine from the electrical machine and the rest of the drivetrain. This clutch is integrated in the decoupler of the electrical machine.

3.3.1. Rotor and stator



F10H electrical machine, rotor and stator

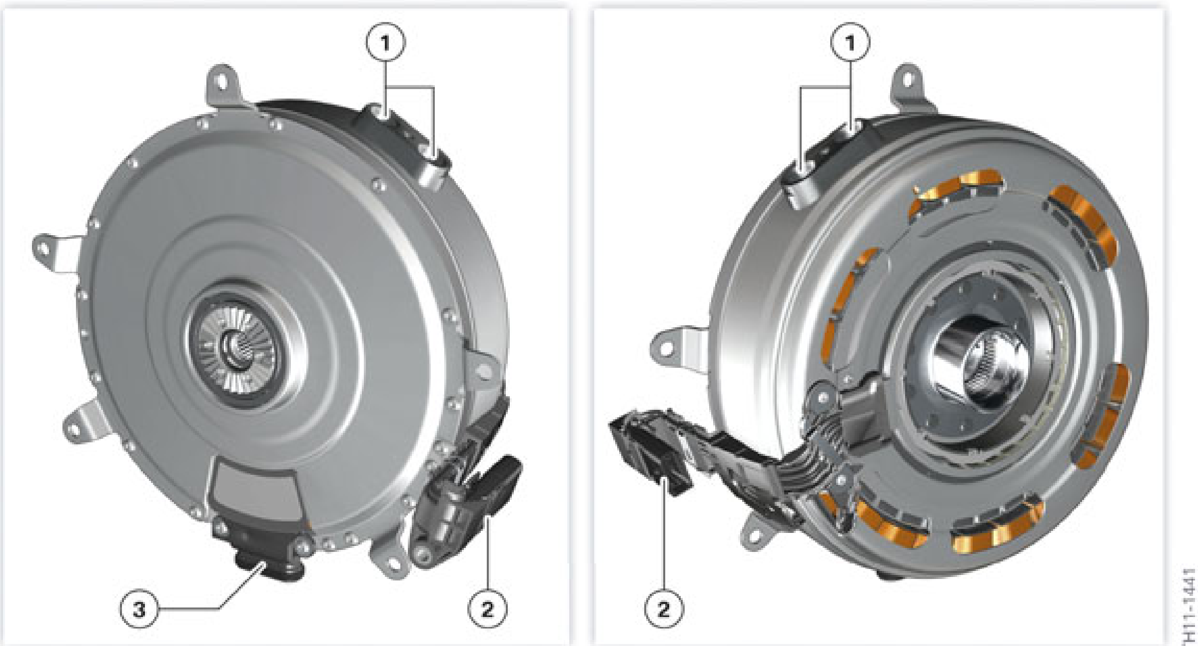
Index	Explanation
1	Stator
2	Hollow shaft with flange
3	Rotor

In contrast to the F04, the electrical machine in the F10H is designed as an internal rotor. "Inner rotor" means that the rotor is arranged with the permanent magnets arranged in a ring shape on the inside. The windings for generating the rotating field are located on the outside and form the stator. This electrical machine has 16 pairs of poles.

3. Electrical Machine

The rotor is mounted on the flange of the rotor hollow shaft, which is connected to the transmission input shaft.

3.3.2. Connections



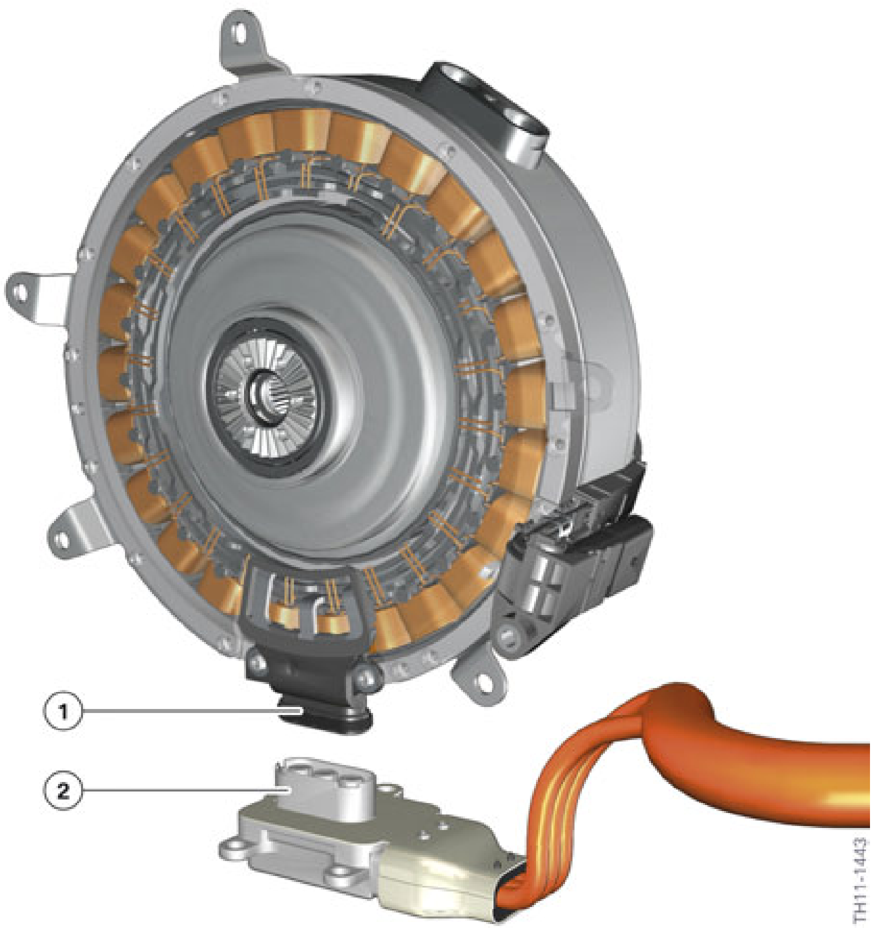
F10H electrical machine connections

Index	Explanation
1	Coolant connection
2	Electrical connection for the sensor system
3	High-voltage connection

There are three connections on the housing of the electrical machine:

- the two coolant lines
- the rotor position sensor and the temperature sensor
- the high-voltage cables.

High-voltage connection



F10H electrical machine, high-voltage connection

Index	Explanation
1	High-voltage connection
2	High-voltage connector

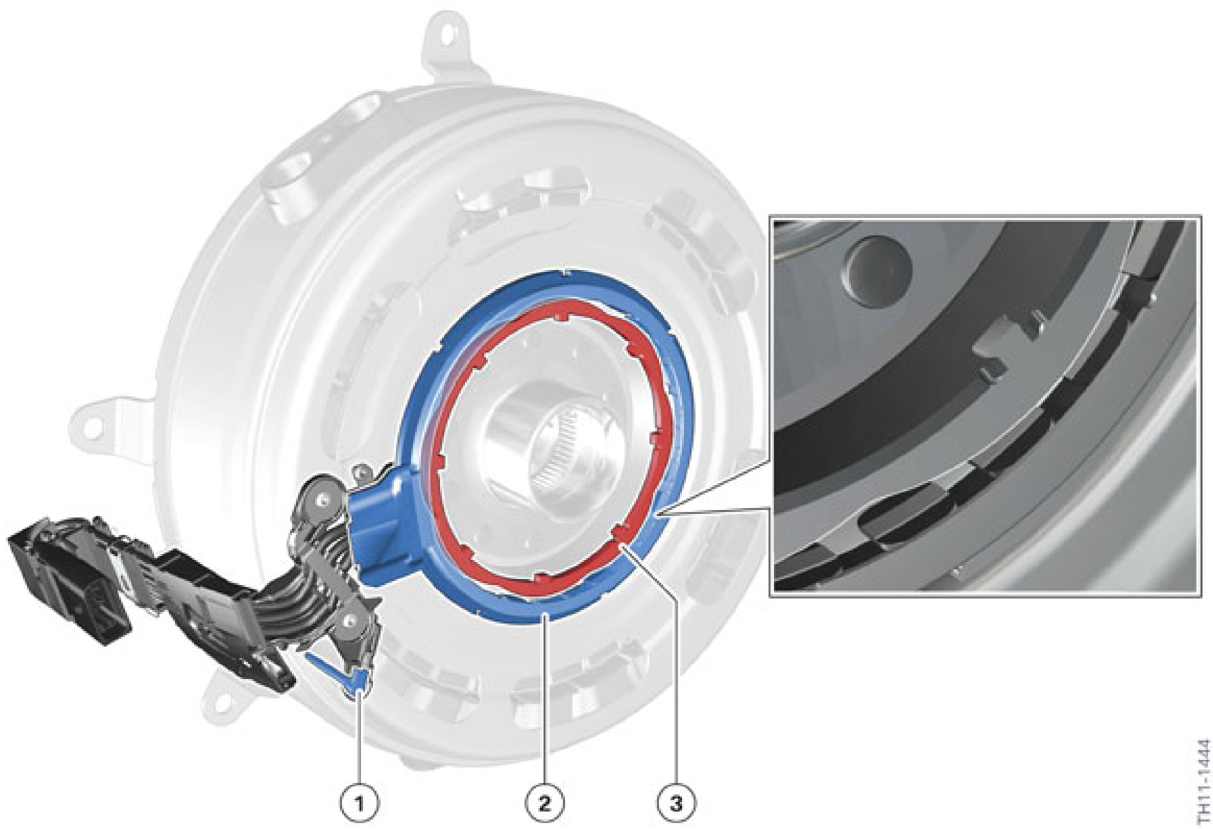
Electrical energy is fed to the windings of the electrical machine via the high-voltage connection. The high-voltage connection connects the electrical machine electronics to the electrical machine via a three-phase, shielded high-voltage cable.

The high-voltage connectors are bolted on to the electrical machine electronics and the electrical machine.

⚠

High-voltage cables must not be repaired! In the event of damage, the line must be replaced.

3.3.3. Sensors



F10H electrical machine sensors

Index	Explanation
1	Temperature sensor
2	Stator of the rotor position sensor
3	Rotor of the rotor position sensor

The rotor position sensor captures the exact position of the electrical machine's rotor. Its structure is similar to that of a synchronous machine and has a specially shaped rotor, which is connected to the rotor of the electrical machine, as well as a stator which is connected to the stator of the electrical machine. The phase voltages induced by rotating the rotor in the windings of the stator are evaluated by the electrical machine electronics to calculate the motor position angle.

⚠

Upon replacement of the electrical machine or the electrical machine electronics, the rotor position sensor must be adjusted using the ISTA BMW diagnosis system.

The rotor position is necessary for a precise, field-oriented control of the electrical machine in order to generate the voltages at the stator windings to match the position of the rotor.

This is required for exact, field-oriented regulation of the electrical machine because the voltages must be generated at the stator windings to match the position of the rotor.

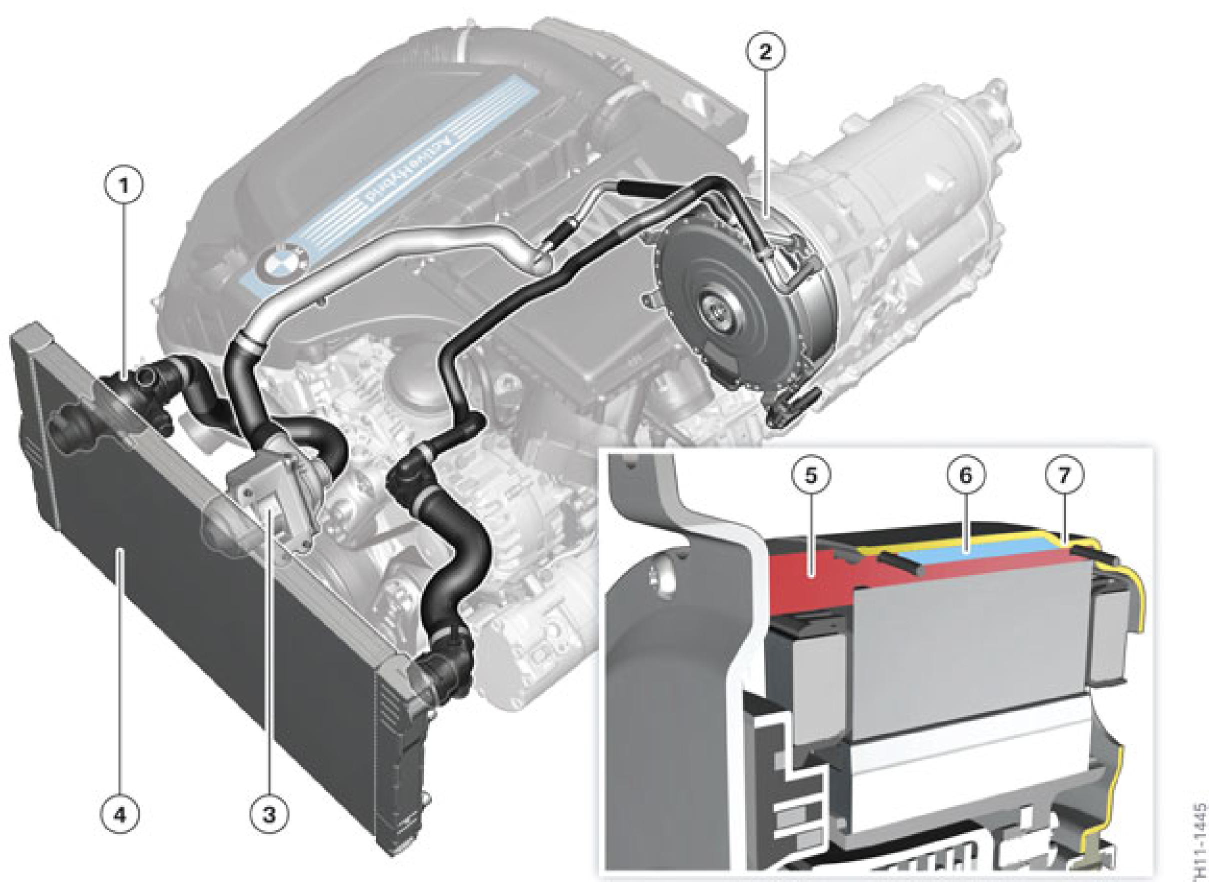
The windings of the electrical machine must not exceed a certain temperature during operation. The temperature is therefore measured in one of the windings with the aid of a temperature sensor. This is a thermistor type sensor with negative temperature coefficient (NTC). Thus the hotter the NTC, the lower its resistance.

The electrical machine electronics evaluates the signals of the temperature sensor and reduces the power of the electrical machine if the temperature of the windings approaches the maximum allowable value.

⚠

A replacement of the rotor position sensor or the temperature sensor is not allowed in the BMW Dealer Service workshop.

3.3.4. Cooling



F10H electrical machine cooling system

Index	Explanation
1	Thermostat
2	Electrical machine
3	Electric coolant pump
4	Radiator
5	Stator support
6	Coolant duct, electrical machine
7	Housing

To cool the electrical machine, there is a coolant duct between the stator support and external housing through which coolant is fed from the engine cooling circuit. The coolant duct is sealed at the front and rear of the housing with two sealing rings.

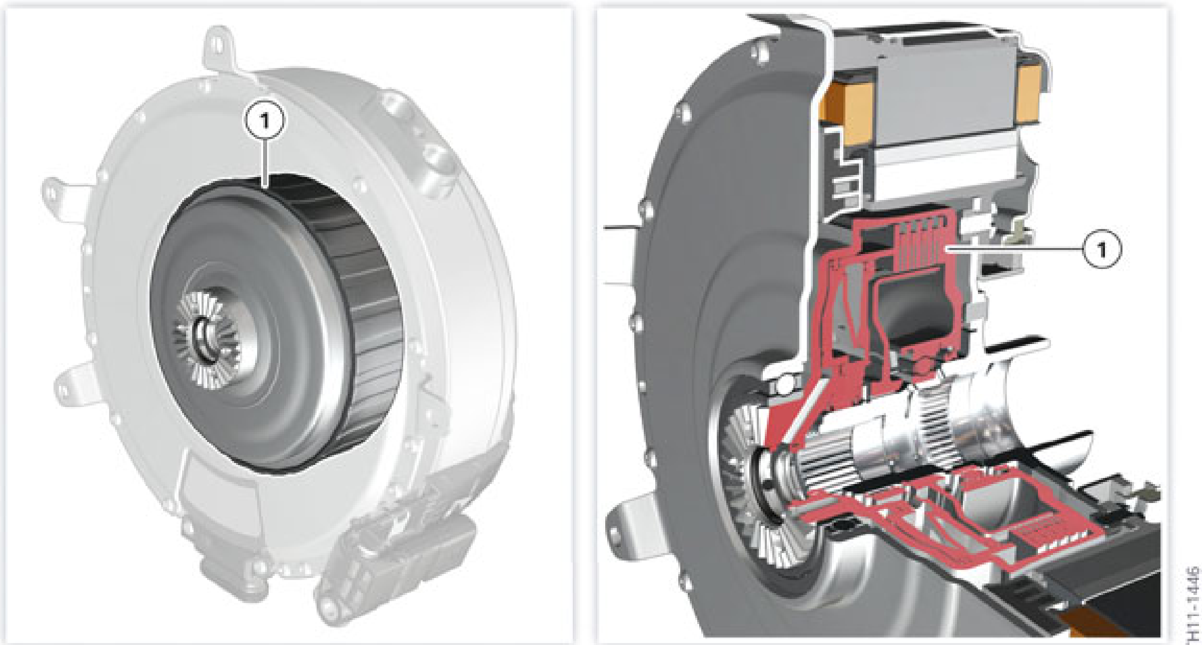
3.3.5. Decoupler

As previously mentioned, decoupler is integrated in the housing of the electrical machine. It is designed as a wet multi-disc clutch and is used to disconnect the combustion engine from the electrical machine and the rear of the drivetrain in certain operating conditions. This is carried out for example in fully electric driving and when "Coasting".

The extreme accuracy of the decoupler makes the connection and disconnection of the combustion engine virtually imperceptible. As soon as the decoupler is closed, the electrical machine, transmission input shaft and the combustion engine rotate at the same speed.

The decoupler is operated by the mechatronics module (as with all clutches and multi-disc brakes of the automatic transmission). It is opened when in a depressurized state. Transmission oil pressure is thus required to close the clutch. This is supplied either by the auxiliary electric oil pump or the mechanical oil pump.

Because the mechanical oil pump is powered by the electrical machine when the decoupler is open, it may not be possible to close the decoupler in the event of a malfunction with the electrical machine and at a transmission oil temperature below 0° C/ 32° F and thus no starting process may take place.



Decoupler of the GA8P70HZ transmission

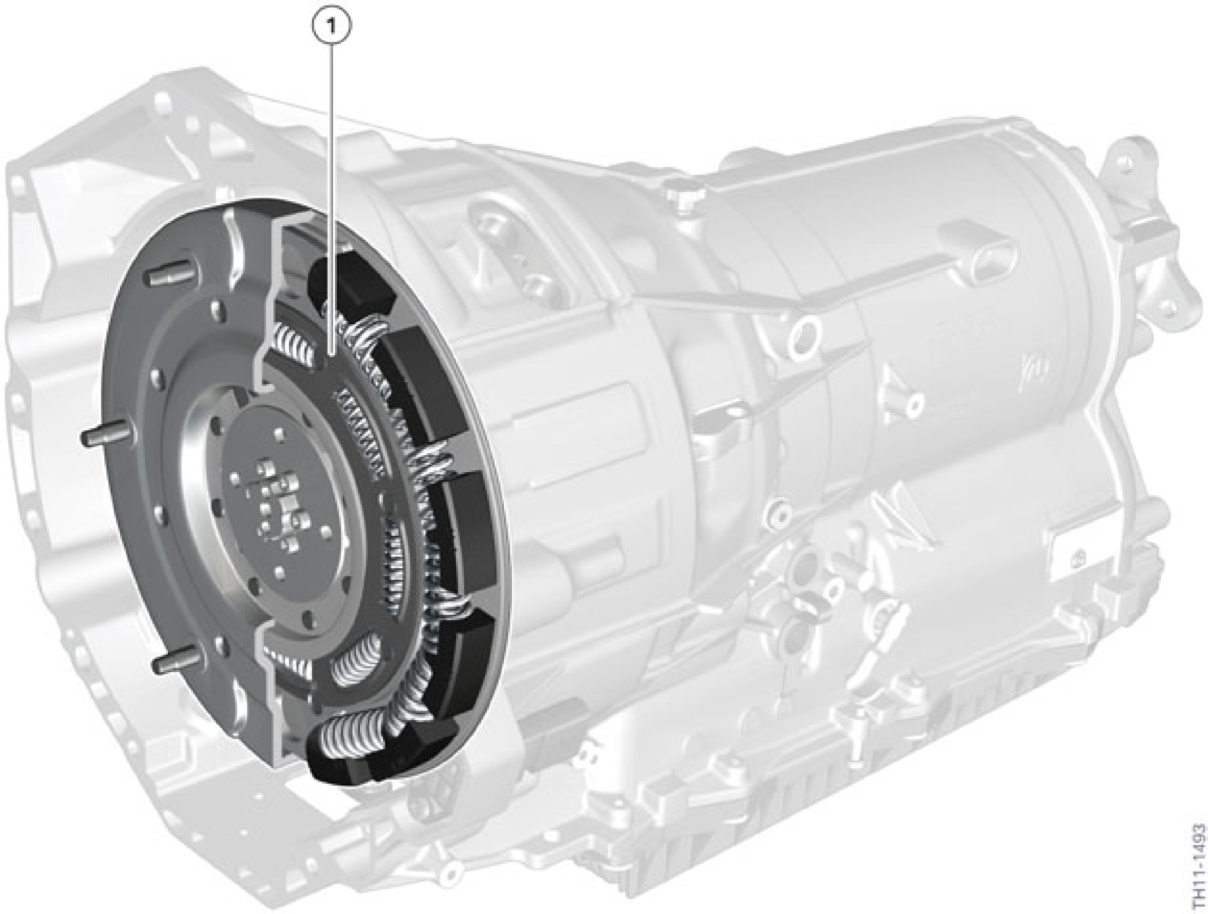
Index	Explanation
1	Decoupler

Similar to the torque converter in the non hybrid (GA8HP70Z) transmission, the decoupler is able to eliminate rotational oscillations of the engine/drivetrain through microslip control. This results in significantly improved acoustic comfort in the vehicle especially at very low engine speeds.

3.3.6. Torsional vibration damper

The unequal running and the resulting torsional vibrations of combustion engines may cause strong droning or rattling noises at certain speeds and operating conditions. To isolate these torsional vibrations a torsional vibration damper is used in the automatic transmission of the F10H. The torsional vibration damper establishes the mechanical connection between the flywheel of the combustion engine and decoupler of the automatic transmission.

The torsional vibration damper can be replaced in the event of a fault.



Torsional vibration damper of the GA8P70HZ transmission

Index	Explanation
1	Torsional vibration damper

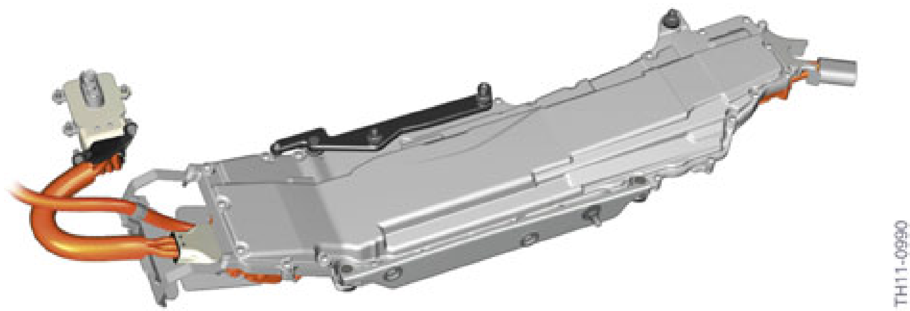
F10H Complete Vehicle
3. Electrical Machine

3.4. Technical data

Supplier	ZF Sachs AG
Maximum torque (< 1 second)	210 Nm (155 lb-ft) at 1300 rpm
(continuous)	80 Nm (59 lb-ft) to 900 rpm
Boost power (< 60 seconds)	38 kW (50 HP) from 1800 rpm
Maximum power (< 1 second)	40 kW (54 HP) at 3000 rpm
Nominal voltage of battery	200 – 385 V
Efficiency	up to 91.8 %
Operating speed range	0 – 7500 rpm
Nominal flow temperature (cooling)	105 °C (221 °F)
Nominal volume flow (cooling)	6 l/min (1.5 gal/min)
Weight (with torsional vibration damper and clutch)	21 kg (46 lbs)
Outer diameter of stator (with support)	285 mm (11.2 in.)
Inner diameter of stator	214 mm (8.4 in.)
Stator length	85 mm (3.3 in.)
Outer diameter of rotor	212 mm (8.3 in.)
Inner diameter of rotor (with support)	172 mm (6.8 in.)

4. Electrical Machine Electronics

4.1. Introduction



F10H Electrical machine electronics

The electrical machine electronics is a high-voltage component whose main task is to convert energy between the high-voltage and low-voltage vehicle electrical system and control the electrical machine. On the one hand direct current voltage from the high-voltage battery is converted into three-phase AC voltage to activate the electrical machine as a motor, and on the other hand when the electrical machine is to work as a generator, the electrical machine electronics convert the three-phase AC voltage of the electrical machine into direct current voltage and can thus charge the high-voltage battery. For these two operating modes a bidirectional DC/AC converter is necessary which can work as both an inverter and a rectifier.

The unidirectional DC/DC converter (also integrated in the electrical machine electronics) ensures the voltage supply to the 12 V vehicle electrical system.

Inside the housing of the electrical machine electronics there is still a control unit which bears the same name, "EME" for short. The EME is a central hybrid control unit which helps reduce consumption and save $C0_2$ by enhancing efficiency and performance through the use of, for example, purely electric driving and brake energy regeneration.

The electrical machine electronics is a high-voltage component!

High-voltage component warning sticker

Each high-voltage component has (on its housing or casing) an identifying label that enables Service technicians and vehicle users to clearly identify the possible hazards associated with working with high electric voltages.

⚠

Only BMW hybrid certified Service technicians are permitted to work on the designated high-voltage components. Certified technicians must: have suitable qualifications, comply with the safety rules and procedures and always follow the repair instructions to the letter.

⚠

Work on live high-voltage components is strictly prohibited. Prior to every operation which involves a high-voltage component, it is essential to disconnect the high-voltage system from the voltage supply and to secure it against unauthorized reconnection.

1. Switch off terminal 15.
2. **De-energize** the high-voltage by opening the high-voltage safety switch (Safety Disconnect).
3. **Secure** the high-voltage safety connector from being reconnected.
4. Switch on terminal 15.
5. **Verify** that the "High-voltage system switched-off" Check Control message is displayed in the instrument cluster.
6. Switch off terminal 15 and terminal R.

⚠

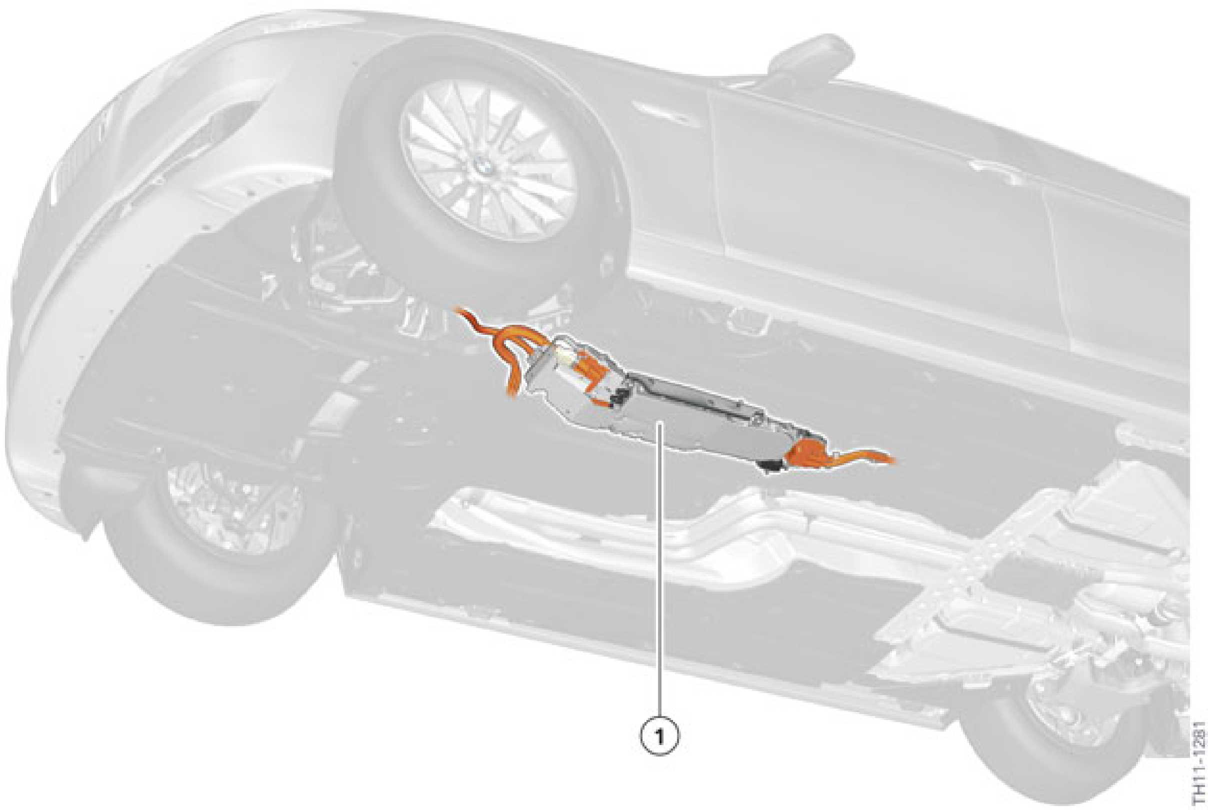
Due to high-voltage safety reasons the electrical machine electronics must not be opened or otherwise dismantled.

In the event of a failure in the electrical machine electronics it is always replaced as a complete unit.

After the electrical machine electronics has been replaced they must be initialized using the BMW diagnosis system. Always follow proper repair instructions and procedures.

F10H Complete Vehicle
4. Electrical Machine Electronics

4.2. Installation location



F10H installation location of the electrical machine electronics

Index	Explanation
1	Electrical machine electronics

The electrical machine electronics is installed under the vehicle to the left of the automatic transmission.

To gain access to the connections of the electrical machine electronics, the corresponding underbody panelling must be removed.

4.3. Functions

The electrical machine preforms the following tasks:

- **Activation and control of the electrical machine:**
  The electrical machine electronics evaluates the rotor position sensor and the temperature sensor in the electrical machine.
- **Control of start-up and shut-down of the high-voltage electrical system:**
  The electrical machine electronics can switch the switch contactors in the high-voltage battery unit and the link discharge on and off. It also evaluates the status of the switch contactor and releases the power for the high-voltage components.

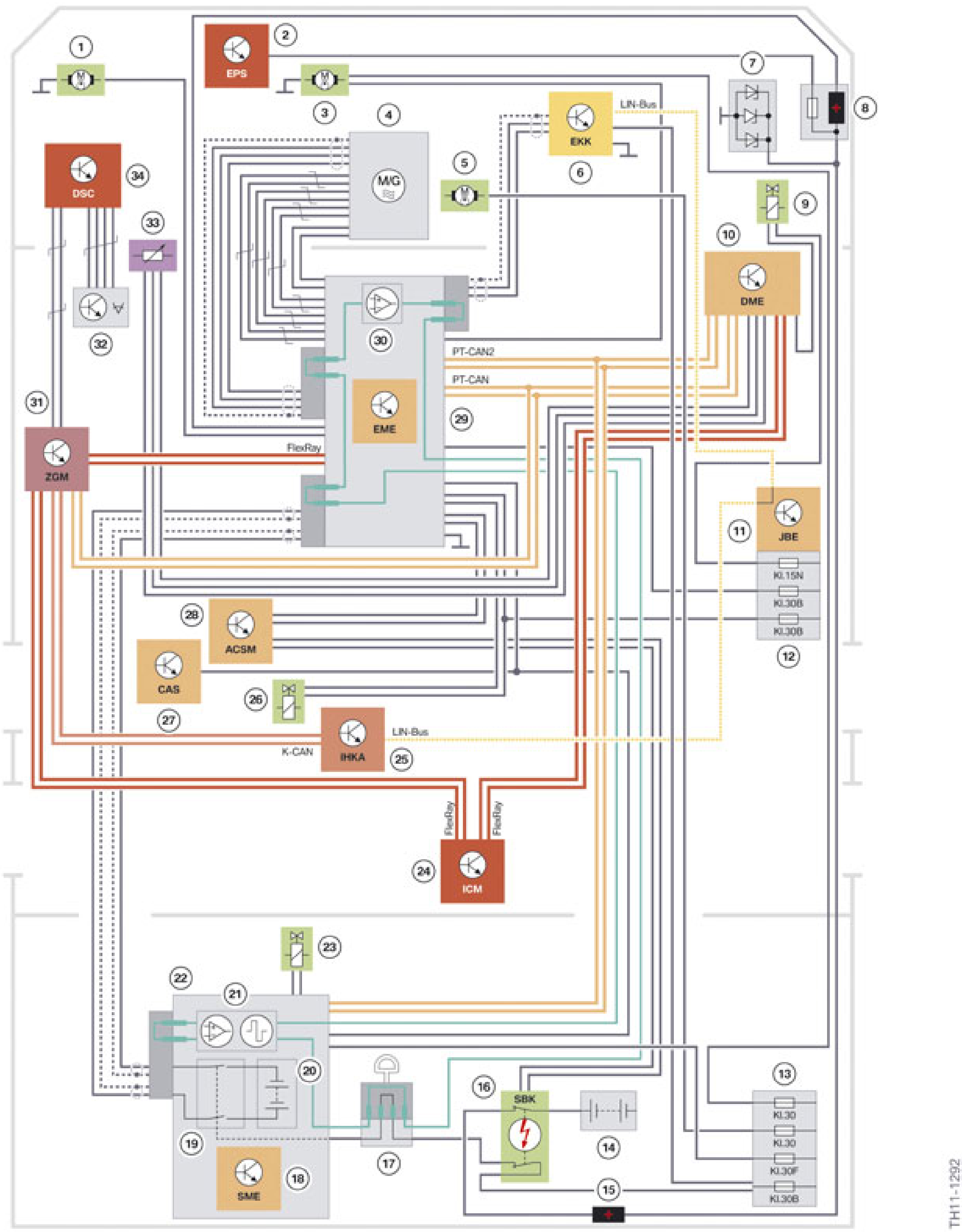
Further information on starting up and shutting down the high-voltage electrical system can be found in the chapter entitled "High-voltage battery unit".

- **Controlling the power of the DC/DC converter:**
  The DC/DC converter in the electrical machine electronics transfers electrical energy from the high-voltage electrical system to the 12 V vehicle electrical system and assumes the function of the alternator in a conventional vehicle. The DC/DC converter in the EME provides a maximum power here of 2.4 kW. The nominal voltage can be specified by the EME in a certain range. Thus the low-voltage vehicle electrical system is supplied with energy. The EME and the electrical machine replace the alternator previously used for this purpose.
- **Control of the power of the electric A/C compressor:**
  Controls the output limit of the electric A/C compressor depending on the current drive function (boost function, energy recovery) and the available power of the high-voltage battery.
- **High-voltage consumer coordination:**
  The power available for the powertrain is specified from the output limits of the high-voltage battery and the current power requirement of the high-voltage components. If relief is requested, the high-voltage consumers can be reduced or switched off depending on their operating condition.
- **High-voltage electrical system diagnosis:**
  This function is used to collect data on the condition of the high-voltage electrical system for the design of future vehicle projects (for example state of charge of high-voltage battery, operating modes, etc.).
- **Activation of the cut-off relay:**
  To avoid overcharging the auxiliary battery, the cut-off relay is closed from a certain engine speed. The auxiliary battery and the battery are thereby connected. This function including the relevant diagnosis is assumed by the electrical machine electronics.
- **Control of switch-off inhibitors/switch-on requests for the automatic engine start-stop (MSA) function:**
  If the load of the 12 V vehicle electrical system is too great, a switch-off inhibitor or a switch-on request is set. A switch-off inhibitor is also set depending on the temperature of the high-voltage battery.
- **High voltage interlock loop:**
  The high-voltage interlock loop is used to protect individuals when working on or around the high-voltage system. It identifies an interrupted plug connection in the high-voltage system, whereby the high-voltage system is switched off immediately. In addition, switching on the high-voltage system is prevented if the high-voltage interlock loop circuit is interrupted.
  The high-voltage interlock loop signal is generated by the battery management electronics and must be guided through all relevant high-voltage components including all high-voltage connections and relevant covers.
  Further information on the high-voltage interlock loop can be found in the chapter entitled "High-voltage battery unit".
- **Activation of the combined shutoff and expansion valve in the refrigerant circuit for cooling the high-voltage battery:**
  The passenger compartment cooling and the cooling of the high-voltage battery can be operated independently of each other.
  For this purpose, two combined expansion and shutoff valves are integrated into the refrigerant circuit. These open only the portion of the circuit that is actually required. This ensures high efficiency and proper control characteristics of the system.

The combined expansion and shutoff valve for the passenger compartment in the heating and air-conditioning unit is activated by the EME; the combined expansion and shutoff valve for the high-voltage battery unit is activated by the battery management electronics (SME).

- **Activation of the electric vacuum pump:**
  In the purely electric driving phase the combustion engine is off and therefore it cannot activate the mechanical vacuum pump. To ensure the supply of a brake vacuum when the engine is off an auxiliary electric vacuum pump is activated by the EME.
  Further information on the electric vacuum pump can be found in the chapter entitled "Hybrid brake system".
- **Operating strategy:**
  Another unique feature of the energy management in the BMW ActiveHybrid 5 is the ability to adapt the operating strategy (not only to the current driving situation) but also to an imminent driving situation. For this purpose, the power electronics evaluates (at an early stage) data which indicates a change in external conditions or the driver's choice and prepares the powertrain components and the vehicle electronics.
  All this operating strategy enhances the vehicle efficiency and performance which has a positive influence on fuel economy and CO_2 emissions.

4.4. System wiring diagram



F10H electrical machine electronics system wiring diagram

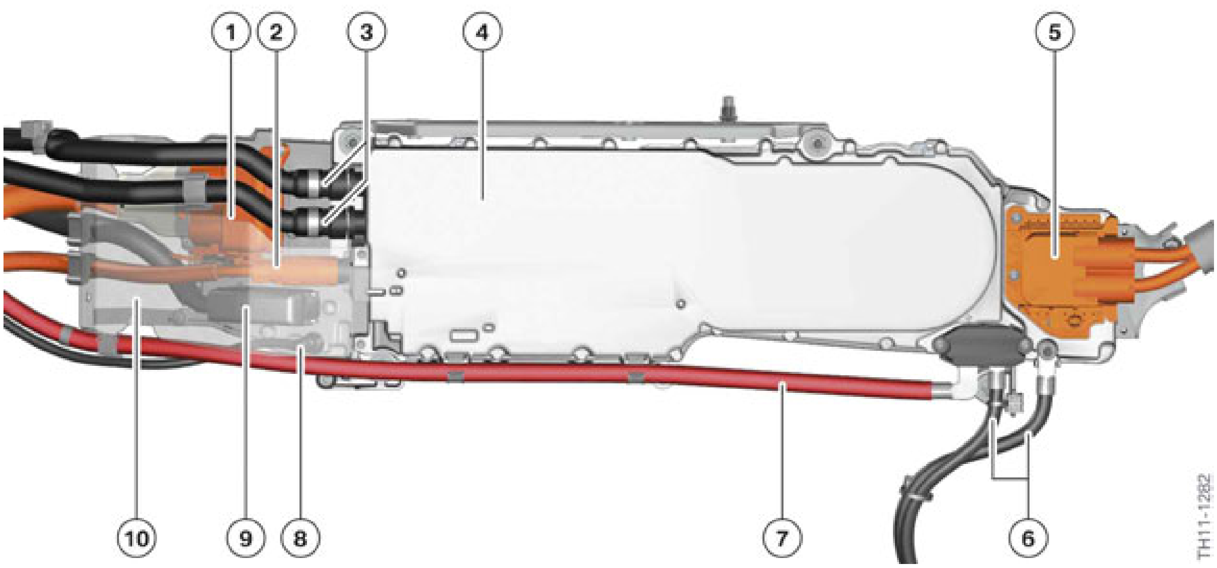
Index	Explanation
1	Electric vacuum pump
2	Electromechanical Power Steering (EPS)
3	Electric coolant pump (cooling circuit, EME)
4	Electrical machine
5	Integrated electric transmission oil pump
6	Electric A/C compressor (EKK)
7	Reverse polarity protection
8	Positive battery connection point
9	Valve, controlled variable-damping motor mounts
10	Digital Engine Electronics (DME)
11	Junction box electronics (JBE)
12	Power distribution box, front
13	Power distribution box, luggage compartment
14	Battery
15	Transition connection point
16	Safety battery terminal (SBK)
17	High-voltage safety connector ("Service Disconnect")
18	Battery management electronics (SME)
19	S-Box with electromechanical switch contactors
20	High-voltage battery
21	Evaluation circuit and signal generator for test signal of the high-voltage interlock loop in the battery management electronics
22	High-voltage battery unit
23	Combined expansion and shutoff valve for high-voltage battery unit
24	Integrated Chassis Management (ICM)
25	Integrated automatic heating/air-conditioning system (IHKA)
26	Combined expansion and shutoff valve for passenger compartment
27	Car Access System (CAS)
28	Crash Safety Module (ACSM)
29	Electrical machine electronics (EME)
30	Evaluation circuit for test signal of the high-voltage interlock loop in the electrical machine electronics
31	Central gateway module (ZGM)
32	Brake pedal travel sensor
33	Low-pressure sensor
34	Dynamic Stability Control (DSC)

4.5. Connections

The connections at the EME can be divided into five categories:

- Low-voltage connections
- High-voltage connections
- Connection for potential compensation lines
- Connections for coolant lines
- Connection for ventilation line.

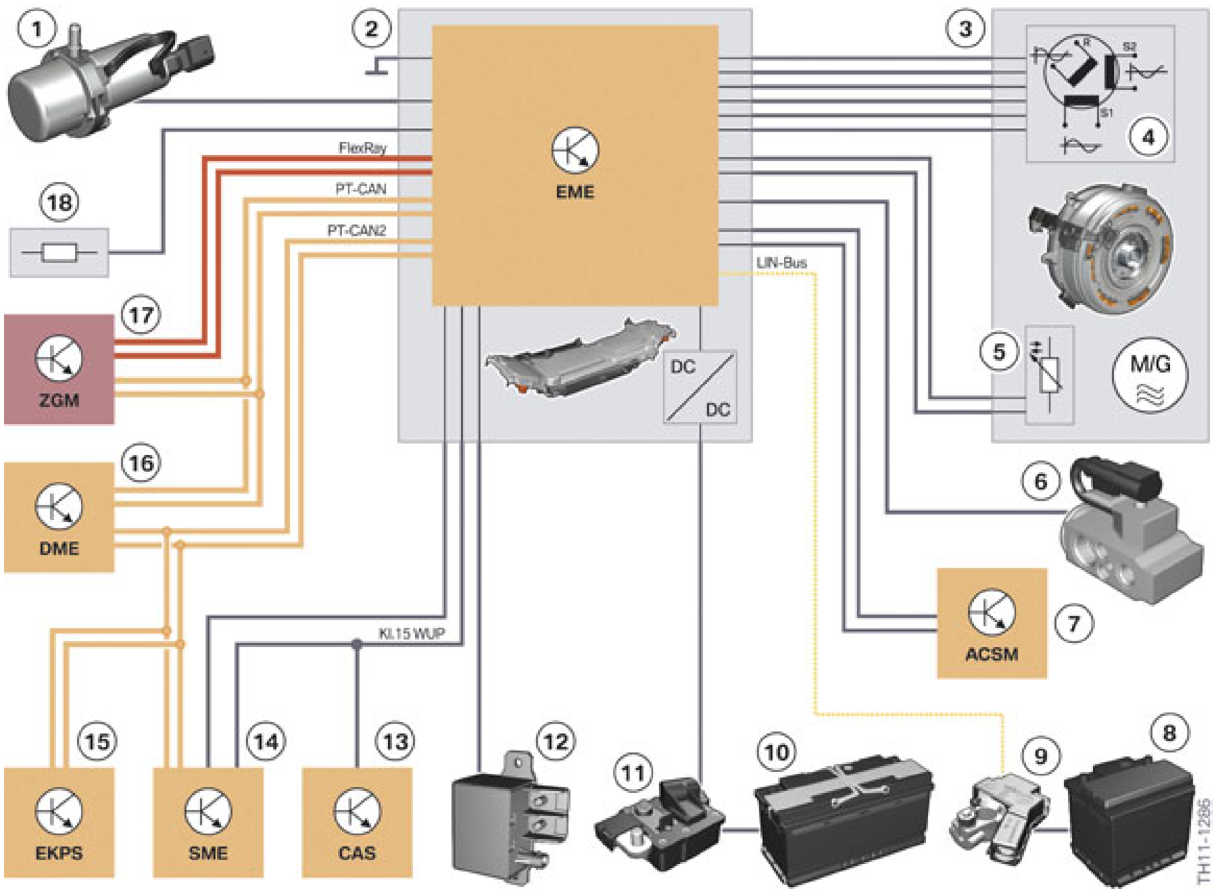
The following graphic shows all connections of the EME along with details of the individual function.



F10H electrical machine electronics connections

Index	Explanation
1	Connector with three-phase high-voltage cable (to the electrical machine)
2	Connector with high-voltage cable (to the electric A/C compressor)
3	Coolant lines
4	Electrical machine electronics (EME)
5	Connector with two high-voltage cables (to the high-voltage battery unit)
6	Grounding cable and potential compensation line
7	12 V connecting line
8	Ventilation line
9	Low-voltage connector
10	Crash protective plate

4.5.1. Low-voltage connections



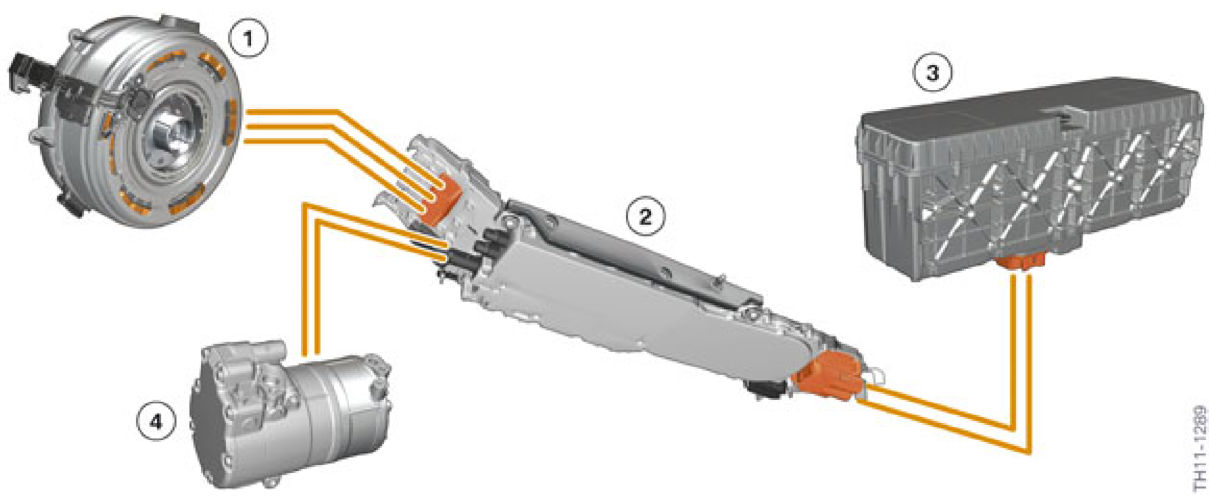
F10H low-voltage connections of the electrical machine electronics

Index	Explanation
1	Electric vacuum pump
2	Electrical machine electronics (EME)
3	Electrical machine
4	Rotor position sensor
5	Temperature sensor
6	Combined expansion and shutoff valve for passenger compartment
7	Crash Safety Module (ACSM)
8	Auxiliary battery (12 V)
9	Intelligent battery sensor (auxiliary battery)
10	Vehicle battery (12 V)
11	Safety battery terminal
12	Cut-off relay
13	Car Access System (CAS)

Index	Explanation
14	Battery management electronics (SME in the high-voltage battery unit)
15	Electronic fuel pump control
16	Digital Engine Electronics (DME)
17	Central gateway module (ZGM)
18	Power distribution box, front

The EME is connected to the 12 V vehicle electrical system (terminals 30 and 31) via a large gauge positive battery cable and grounding cable. The DC/DC converter in the EME supplies the entire 12 V vehicle electrical system with energy via this connection. These two lines are bolted to the EME rather than via a plug connection. All other (low voltage) lines and signals on the control line have a relatively small current level and thus are plugged into the EME.

4.5.2. High-voltage connections



F10H high-voltage connection of the electrical machine electronics

Index	Explanation
1	Electrical machine
2	Electrical machine electronics (EME)
3	High-voltage battery unit
4	Electric A/C compressor (EKK)

There is a total of three high-voltage connections to establish contact between the lines and other high-voltage components at the EME:

- **Connection to the electrical machine:**
  three-phase, AC voltage, 1 shielding for all 3 lines, high-voltage connector bolted on
- **Connection to the high-voltage battery:**

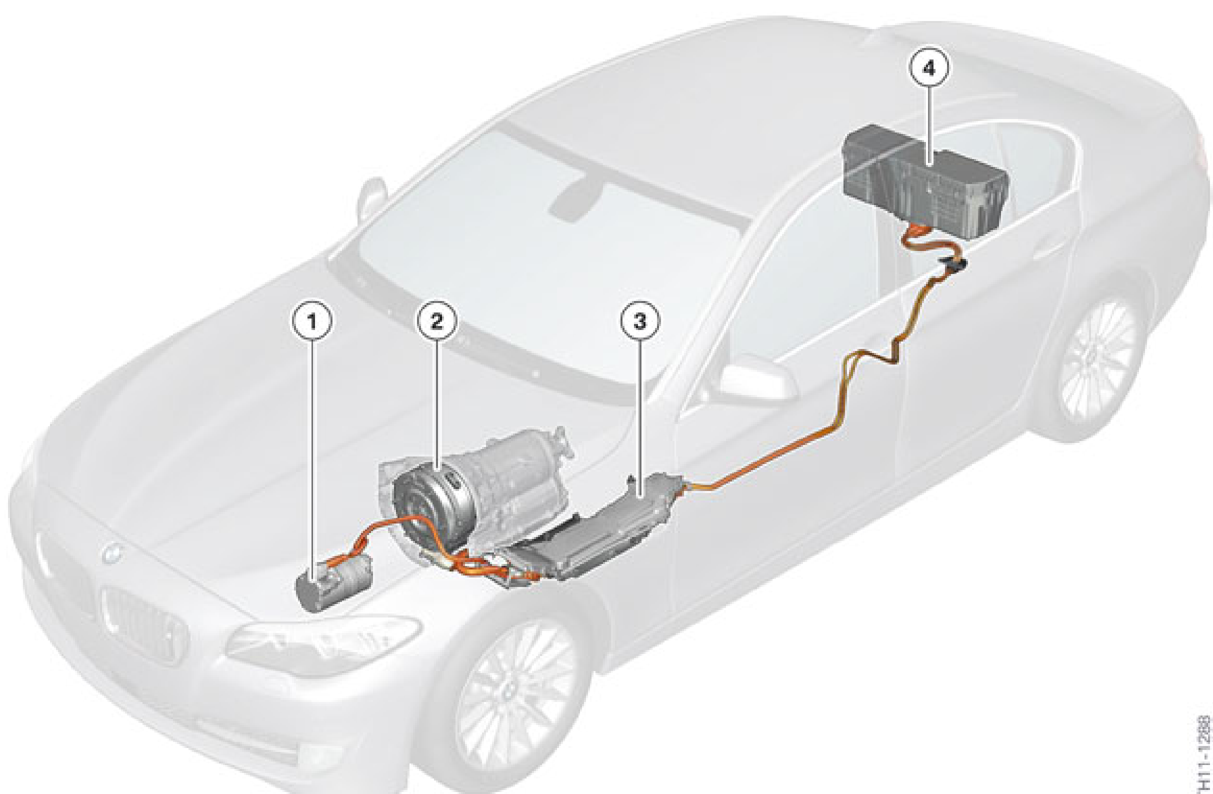
two-pin, direct current voltage, 1 shielding per line, flat high-voltage connector with mechanical locking, contact protection through cover over the contact blades and by bridge of high-voltage interlock loop

- **Connection to the electric A/C compressor:**
  two-pin, direct current voltage, 1 shielding for both lines, round high-voltage connector with mechanical locking, contact protection through cover over the contact blades and by open circuit of the voltage supply for the electric A/C compressor control unit.

A certain sequence must be observed when breaking or establishing the contact connection both for flat and round high-voltage connectors. The individual steps are described in the next section.

4.5.3. High-voltage cables

The high-voltage cables connect the high-voltage components and are clearly identified by their orange color. The manufacturers of hybrid vehicles have agreed on a standard identification of the high-voltage cable with the warning color orange. The high-voltage cables have already been described in the "BMW ActiveHybrid Technology" ST920 training information available on TIS and ICP. Only an overview of the high-voltage cables used in the F10H is provided here.



F10H high voltage cables and components

Index	Explanation
1	Electrical A/C compressor (EKK)
2	Electrical machine
3	Electrical machine electronics (EME)
4	High-voltage battery unit

⚠

High voltage cables must not be repaired!

In the event that a high voltage cable is damage, the entire cable must be replaced.

⚠

Work on live high-voltage components is strictly prohibited. Prior to every operation which involves a high-voltage component, it is essential to disconnect the high-voltage system from the voltage supply and to secure it against unauthorized reconnection.

1 Switch off terminal 15.

2 **De-energize** the high-voltage by opening the high-voltage safety switch (Safety Disconnect).

3 **Secure** the high-voltage safety connector from being reconnected.

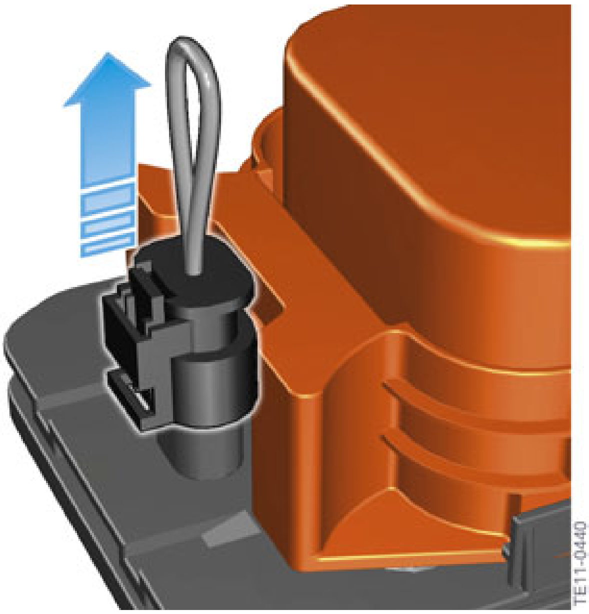
4 Switch on terminal 15.

5 **Verify** that the "High-voltage system switched-off" Check Control message is displayed in the instrument cluster.

6 Switch off terminal 15 and terminal R.

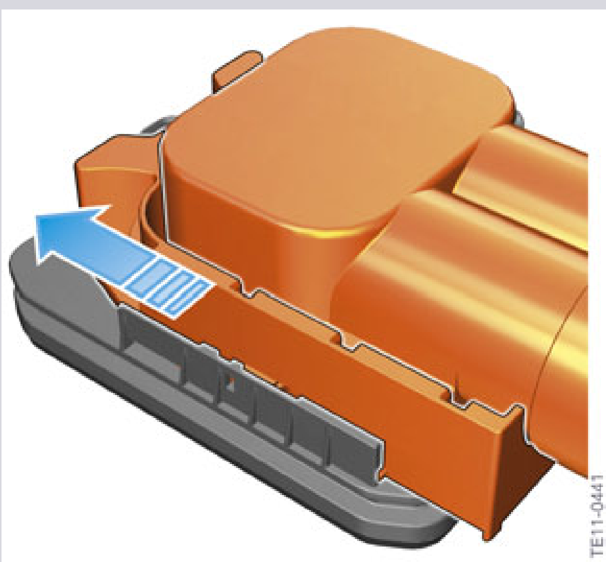
Removing the flat high-voltage connector

The procedure described here applies to the high-voltage connector found on the EME and to the high-voltage connector on the high-voltage battery unit.



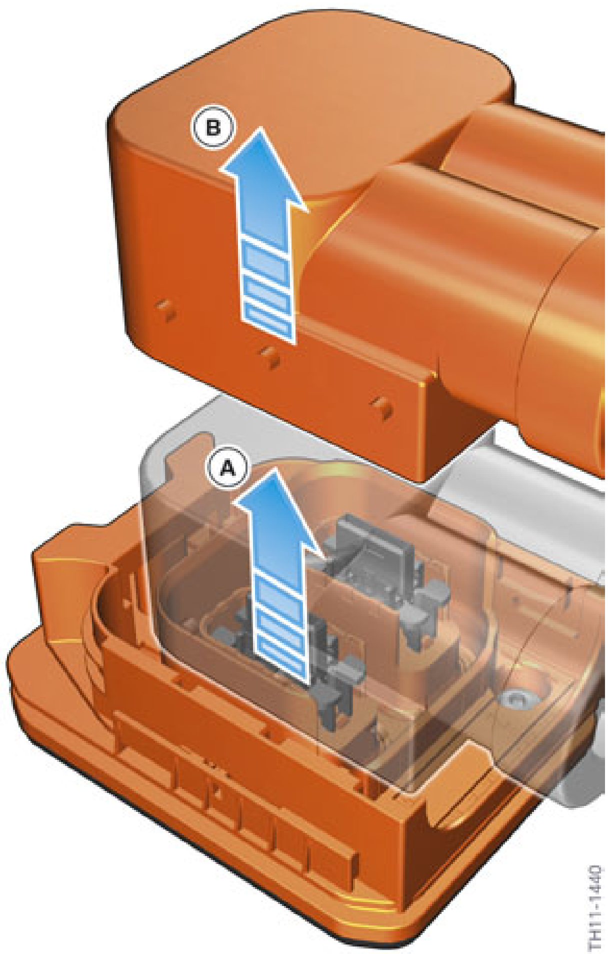
Removing bridge for high-voltage interlock loop

The bridge or jumper closes the circuit of the high-voltage interlock loop in a connected state. The battery management electronics continuously monitors the circuit of the high-voltage interlock loop and only when the circuit is closed is the high-voltage system active. If the circuit of the high-voltage interlock loop is interrupted by removing the jumper, the high-voltage system shuts down automatically. This is an additional safety precaution as the Service technician has already switched off the high-voltage system before beginning work.



Removing the mechanical locking

Only after the high-voltage interlock loop jumper has been removed, can the connector mechanical locking be moved in the direction of the arrow. The mechanical lock is integrated into the high-voltage connector on the high-voltage components. By moving the lock in the direction of the arrow the mechanical guide of the high-voltage connector on the high-voltage cable is released which permits the subsequent disconnection.

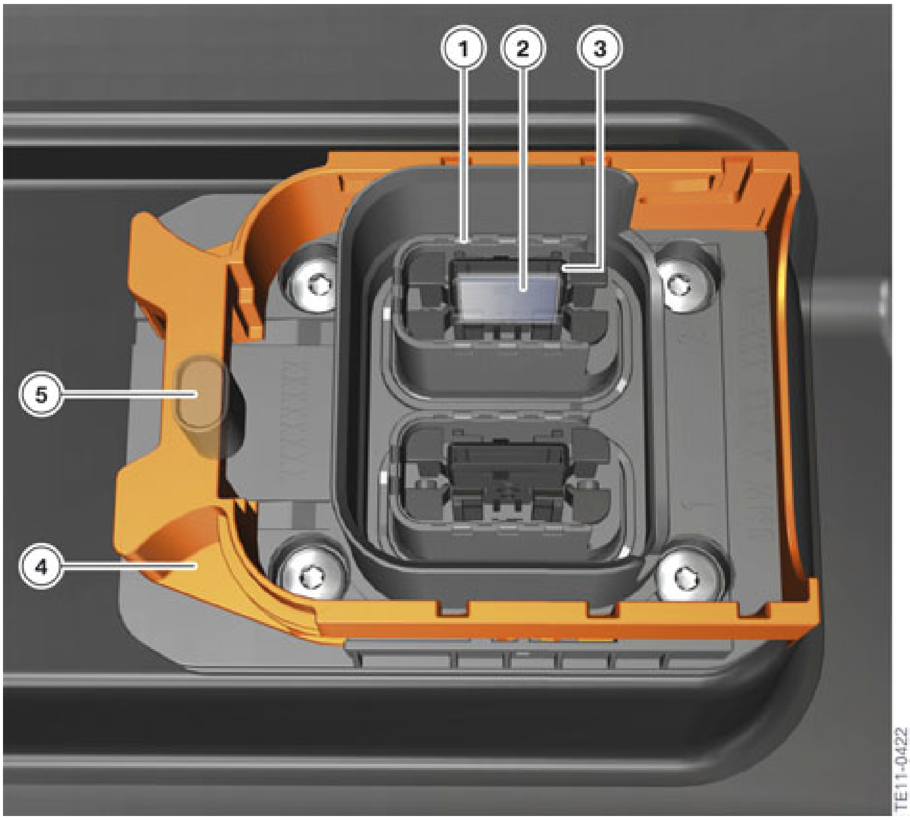


Removing the connector of the high-voltage cable
The connector of the high-voltage cable can now be removed in the direction of the arrow. After the connector has been pulled out a few millimeters (A), one encounters a higher counterforce. The connector must then be pulled out further in the same direction (B). Under no circumstances must the connector be pressed back into the socket on the high-voltage component after reaching position (A). This may damage the connector on the high-voltage components.

⚠

The high-voltage connector of the high-voltage cables must be pulled out at a right angle in two steps and in the same direction. Changing the direction of movement during removal may damage the HV connector.

When re-inserting the high-voltage connector proceed in reverse order. The following graphic shows the complex design of the high-voltage connector on the high-voltage components and explains why one must proceed with care when removing and inserting high-voltage connectors.

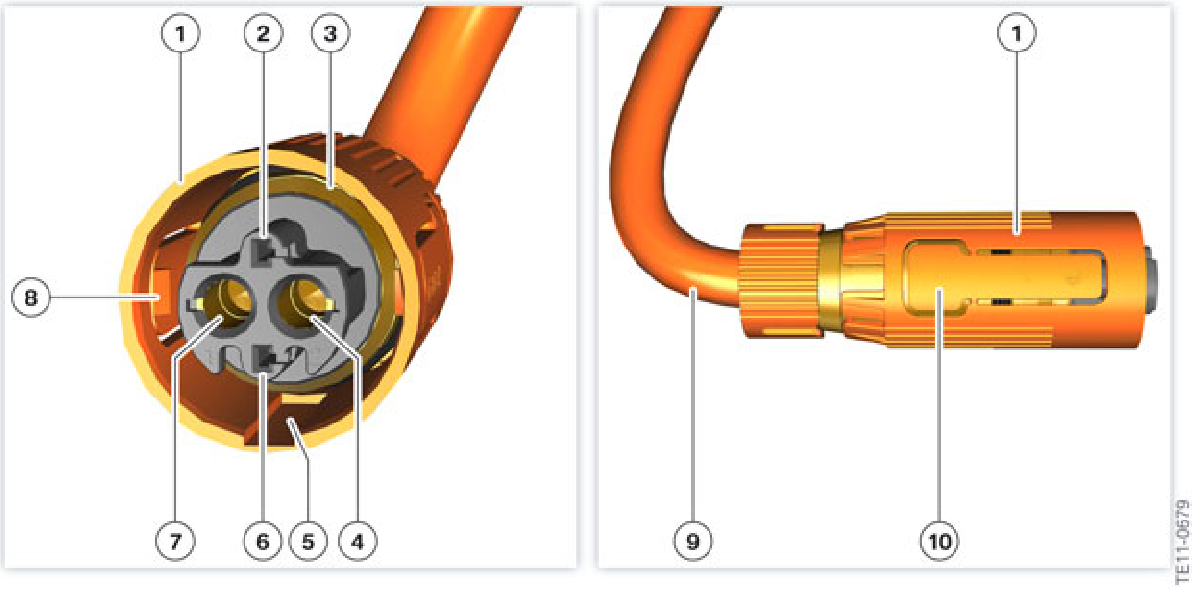


F10H high-voltage connector on electrical machine electronics and high-voltage battery unit

Index	Explanation
1	Electrical contact for shielding
2	Electrical contact for high-voltage cable
3	Contact protection
4	Mechanical locking
5	Socket with connection for bridge in the circuit of the high-voltage interlock loop

Removing the round high-voltage connector

The procedure described here applies to the corresponding high-voltage connector on the EME and to the high-voltage connector on the electric A/C compressor.



F10H high-voltage connector on EME and electric A/C compressor.

Index	Explanation
1	Housing
2	Connection 1 for bridge in the connector
3	Connection for shielding
4	High-voltage connection, pin 2 (DC, minus)
5	Mechanical encoding
6	Connection 2 for bridge in the connector
7	High-voltage connection, pin 1 (DC, plus)
8	Locking element
9	High-voltage cable
10	Actuation points on locking elements

To remove the high-voltage connector the two opposite actuation points (10) of the locking elements (8) must be pressed together. While the locking elements are further pushed together, the connector must be removed lengthways.

When reconnecting the high-voltage cable it is recommended to also first push the locking elements together. This ensures that the locking elements glide past the socket on the outside. If the locking elements are not first pushed together, there is a risk that they may slip inwards when positioning and suffer damage. At the end of the positioning please ensure that the locking elements engage (listen for clicking noise). In addition, the engaging of the locking elements should be checked by subsequent light pulling on the connector.

The bridge in the high-voltage connector serves for electrical safety. The signal of the high-voltage interlock loop runs through this bridge when the high-voltage cable is connected to the EME. If the circuit is interrupted, this also results in an automatic interruption to the current flow in the respective high-voltage cable. As the two contacts of the bridge (opposite the high-voltage contacts) disconnect first, this constitutes protection against the formation of an electric arc when removing the high-voltage connector.

4.5.4. Connection for potential compensation lines

The isolation monitoring determines whether the insulation resistance between active high-voltage components (e.g. high-voltage cables) and ground is above or below a required minimum value. If the insulation resistance falls below the minimum value, the danger exists that the vehicle parts will be energized with dangerously high-voltage. If a person were to touch a second active high-voltage component, he or she would be at risk of electric shock.

There is therefore fully automatic isolation monitoring for the F10H high-voltage system. It is performed by the battery management electronics at regular intervals while the high-voltage system is active. Ground serves as the reference potential. Without additional measures only local isolation faults in the high-voltage battery unit could be determined in this way. However, it is equally important to identify isolation faults between the high-voltage cables and the vehicle to ground. For this reason all the electrically conductive housings of high-voltage components are (galvanically) connected to ground. In this way isolation faults in the entire high-voltage electrical system can be identified from a central point by the isolation monitoring.

⚠

The high-voltage system must not be operated if the potential compensation cables are not properly connected to the high-voltage components.

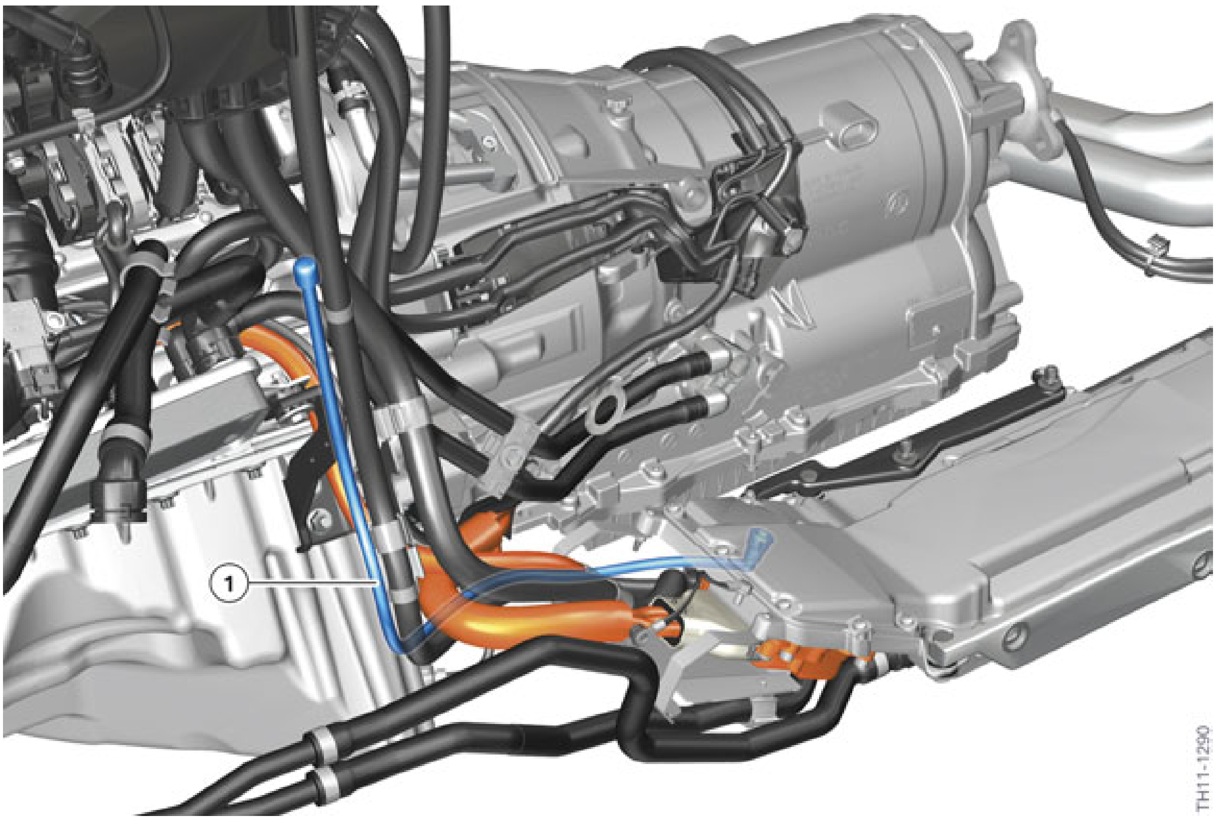
⚠

If in the event of a repair the high-voltage components or the body components are replaced, the following must be observed during assembly: The connection between the housing of the component and the body must be properly re-established. The repair instructions must be followed especially with regard to the tightening torque, self-tapping screws, etc.

4.5.5. Connections for coolant lines

The electrical machine electronics is cooled by its own cooling circuit. This cooling circuit is described in the chapter entitled "Electrical machine electronics > Cooling".

4.5.6. Connection for ventilation line



F10H electrical machine electronics ventilation line

Index	Explanation
1	Tank ventilation line

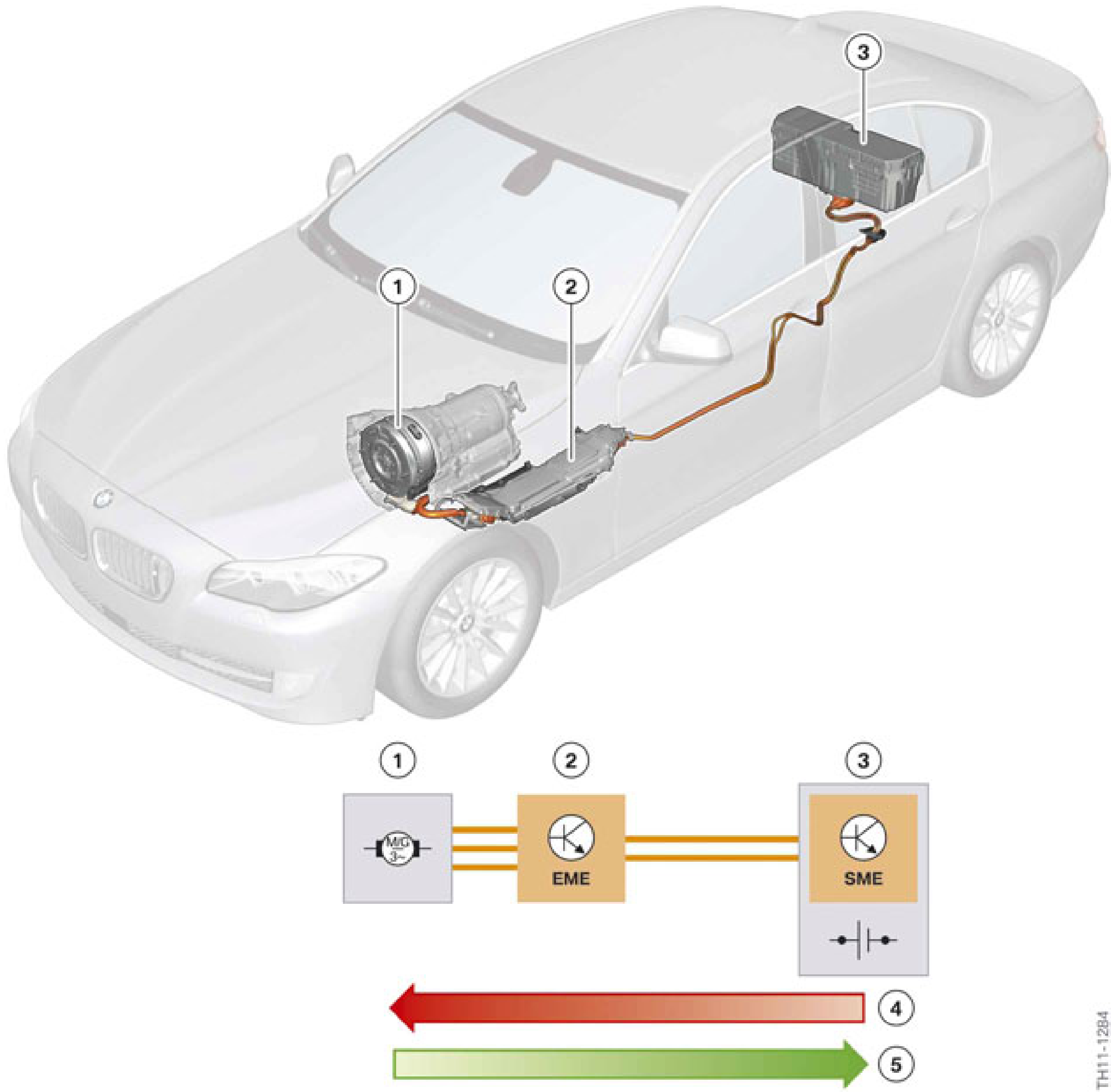
A ventilation line is required to prevent water (resulting from temperature changes and thus possible condensation of air moisture) collecting inside the EME. The end of the ventilation line is located higher, above the electrical machine electronics.

4.6. Energy flow

4.6.1. High-voltage battery – Electrical machine

The following graphic shows the energy flow between high-voltage battery and electrical machine.

4. Electrical Machine Electronics



F10H energy flow between high-voltage battery and electrical machine

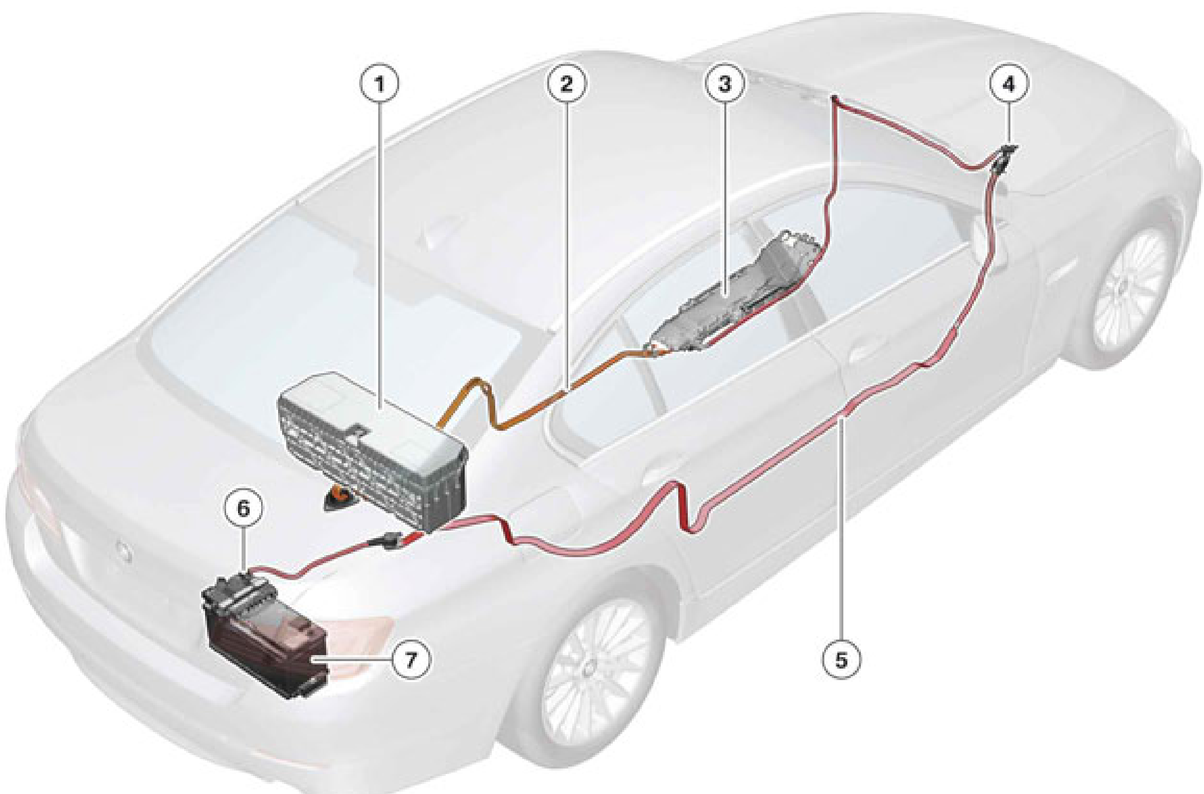
Index	Explanation
1	Electrical machine
2	Electrical machine electronics
3	High-voltage battery unit
4	Power electronics of the EME work as an inverter and convert high-voltage DC into three-phase high-voltage AC
5	Power electronics of the EME work as a rectifier and convert three-phase high-voltage AC into high-voltage DC

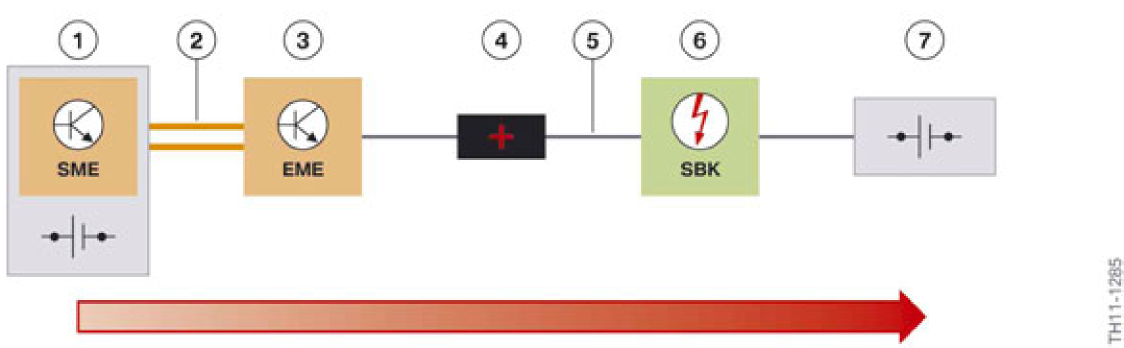
A bidirectional inverter in the EME converts the direct current (DC) voltage from the high-voltage battery into three-phase alternating current for activating and controlling the electrical machine.

In generator mode the high-voltage battery is charged via the inverter (energy recovery).

4.6.2. High and Low-voltage systems

The following graphic shows the energy flow between the low-voltage vehicle electrical system and the high-voltage electrical system in the F10H.





F10H energy flow between high-voltage battery and 12 V vehicle electrical system

4. Electrical Machine Electronics

Index	Explanation
1	High-voltage battery unit
2	High-voltage cable
3	Electrical machine electronics (EME)
4	Positive battery connection point
5	Main current line
6	Safety battery terminal (SBK)
7	Battery

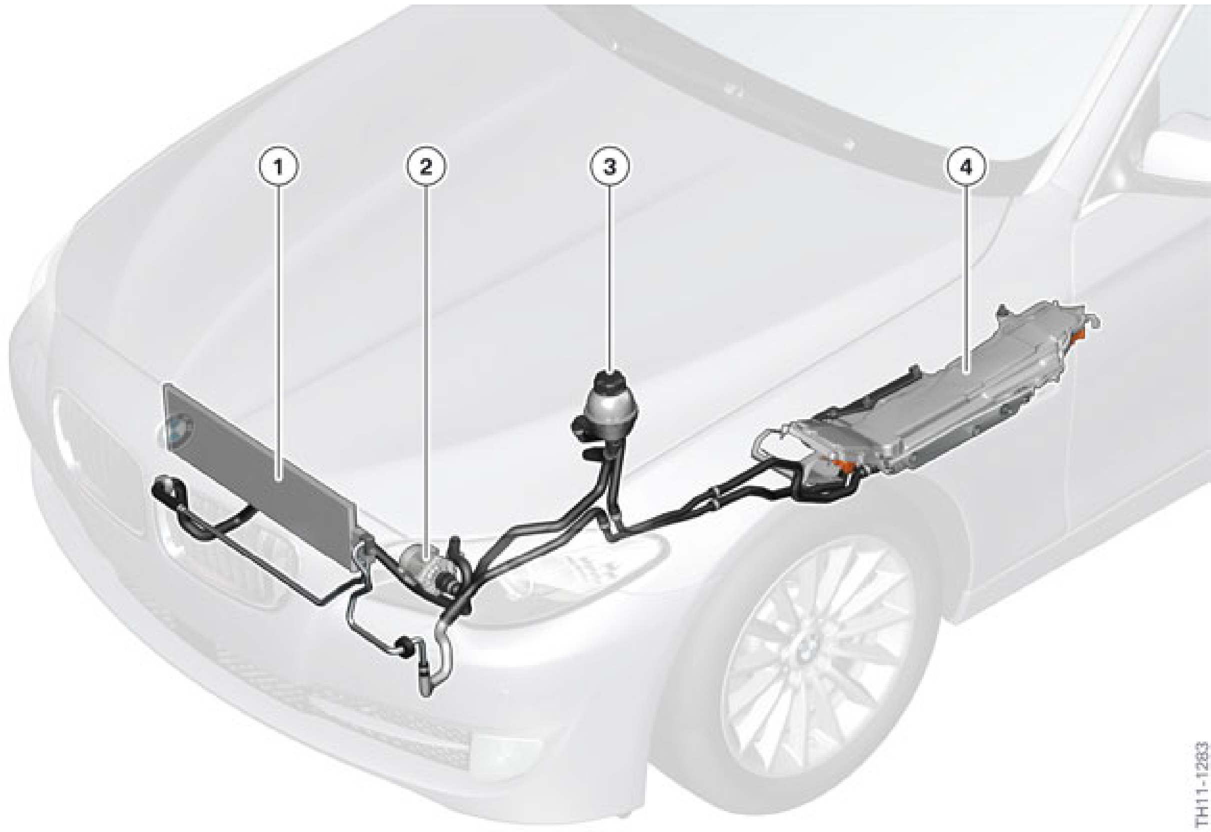
In comparison to conventional vehicles, the 12 V vehicle electrical system is not supplied with energy via a generator but via a DC/DC converter from the high-voltage electrical system.

In contrast to the F04, the EME in the F10H has no bidirectional DC/DC converter. This means that only energy from the high-voltage electrical system can be transferred to the 12 V vehicle electrical system. Charging of the high-voltage battery from the 12 V vehicle electrical system is not possible. As the combustion engine in the F10H can be started at any time from the 12 V vehicle electrical system via the conventional starter motor, this is also not necessary.

The high-voltage battery is charged when the engine is running the electrical machine.

F10H Complete Vehicle
4. Electrical Machine Electronics

4.7. Cooling



F10H cooling system of the electrical machine electronics

Index	Explanation
1	Coolant/air heat exchanger
2	Electric coolant pump
3	Expansion tank
4	Electrical machine electronics (EME)

The EME is cooled by its own cooling circuit.

The cooling circuit comprises:

- High-performance low-temperature cooler
- 50 W coolant pump
- Expansion tank
- Coolant lines.

The coolant/air heat exchanger is integrated in the cooling module. The electric coolant pump and the electrical fan are activated according to cooling requirements of the EME thus contributing to a reduction in energy consumption.

4. Electrical Machine Electronics

Due to the demand-driven activation of the electrical fan and the electric coolant pump, strong temperature deviations which would have a negative effect on the service life of the electronics are avoided and energy-optimized cooling achieved.

4.8. Technical data

Electrical machine electronics	
Supplier	Robert Bosch GmbH
Weight	10 kg (22 lbs)
Length	875 mm (34.4 in.)
Height	151 mm (5.9 in.)
Width	249 mm (9.8 in.)
Operating temperature range	–40 °C (–40 °F) to +80 °C (176 °F)

Power electronics	
Operating voltage range	Max. 385 V DC
Maximum current	150 A (continuous); 400 A (0.3 second)

DC/DC converter	
Nominal output voltage	14 V DC
Maximum output current	170 A (continuous)
Maximum output power	2.4 kW (continuous); 3 kW (peak value 100 ms)

5. High-voltage Battery Unit

5.1. Overview

The high-voltage battery unit is a complete system that contains not only the high-voltage battery itself, but also the following additional components:

- Battery management electronics (SME) control unit
- Electromechanical switch contactors
- Connections for high-voltage cables
- Connections for signal lines
- Connection for potential compensation line
- Connections for refrigerant line and condensation drain line
- Cell supervision circuits
- Degassing pipe.

The primary task of the high-voltage battery unit is to draw electrical energy from the high-voltage electrical system, store it and make it available again later as necessary. In addition, it assumes important tasks that contribute to the safety of the high-voltage system, such as the high-voltage interlock loop. The high-voltage safety connector (also called the "Service Disconnect") is not an integral part of the high-voltage battery unit in the F10H. Instead is located in the luggage compartment to the right, near the auxiliary 12V battery.

The high-voltage battery unit of the F10H is manufactured by BMW AG. The cells of the high-voltage battery unit are manufactured by A123 Systems. The high-voltage battery unit was also developed by BMW AG.

The high-voltage battery unit of the F10H belongs to hazard class 9 (UN 3090) and cannot be transported by air freight.

⚠

The high-voltage battery unit is a complex, high-voltage component thus it is imperative to follow the handling and safety regulations. In particular, lithium-ion batteries must not be overcharged or subjected to excessively high temperatures which may increase the risk of fire. The high-voltage battery unit must never be opened, otherwise there is a risk of fatal injury due to the high currents and voltages involved.

⚠

Work on live high-voltage components is strictly prohibited. Prior to every operation which involves a high-voltage component, it is essential to disconnect the high-voltage system from the voltage supply and to secure it against unauthorized reconnection.

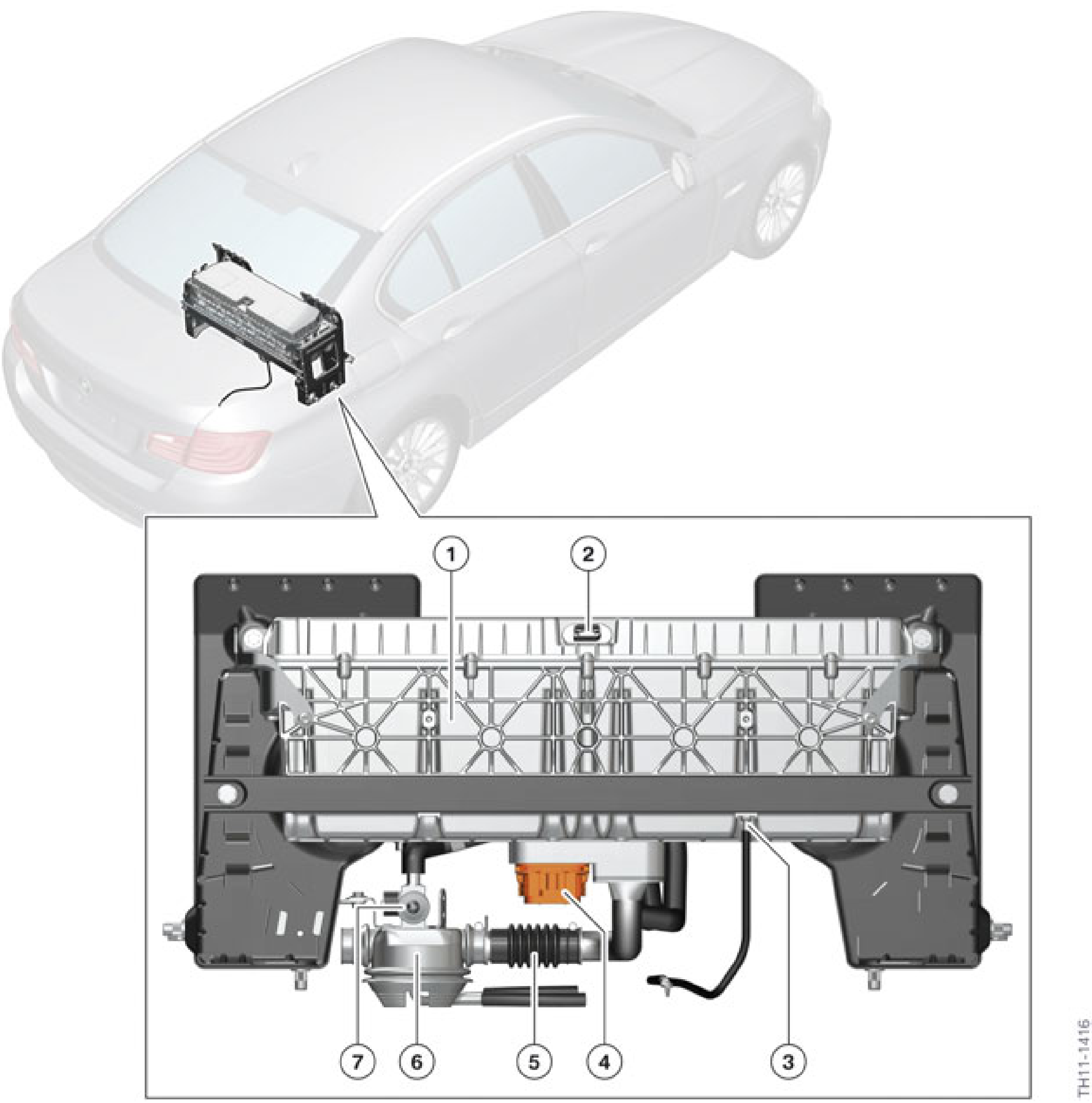
1 Switch off terminal 15.
2 **De-energize** the high-voltage by opening the high-voltage safety switch (Safety Disconnect).
3 **Secure** the high-voltage safety connector from being reconnected.
4 Switch on terminal 15.

5 **Verify** that the "High-voltage system switched-off" Check Control message is displayed in the instrument cluster.

6 Switch off terminal 15 and terminal R.

5.1.1. Installation location and external characteristics

The high-voltage battery unit is installed in the luggage compartment behind the rear seat. As a result, the F10H cannot be ordered with the "Through-loading system" or "Ski bag" optional equipment. The high-voltage battery unit is covered by a panel and is therefore not directly visible. When this panel is removed the features and connections shown in the following graphic can be accessed.



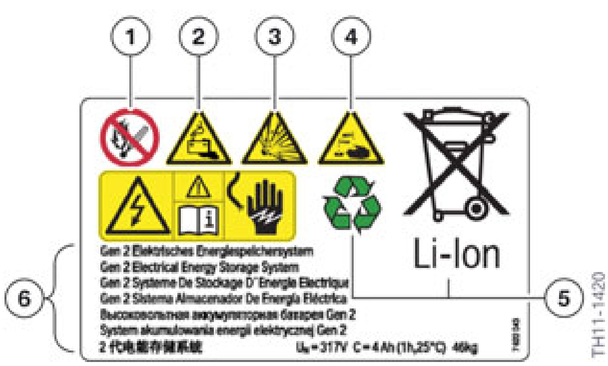
Installation location of the high-voltage battery unit in the F10H

Index	Explanation
1	High-voltage battery unit
2	Signal connection for battery management electronics control unit
3	Connection for potential compensation line
4	High-voltage connection
5	Degassing pipe
6	Grommet
7	Combined expansion and shutoff valve in the refrigerant circuit

The high-voltage battery unit is connected to the vehicle body on the left and right using two holders. There is a special tool available to the Service workshops to facilitate removal and installation of the high-voltage battery unit.

The required electrical connection between the high-voltage battery unit housing and ground is provided (as shown in the graphic) by a separate electrical cable.

There is a safety data warning sticker on the top of the high-voltage battery unit which warns against the dangers associated with working on high-voltage components.



Warning sticker on the high-voltage battery unit

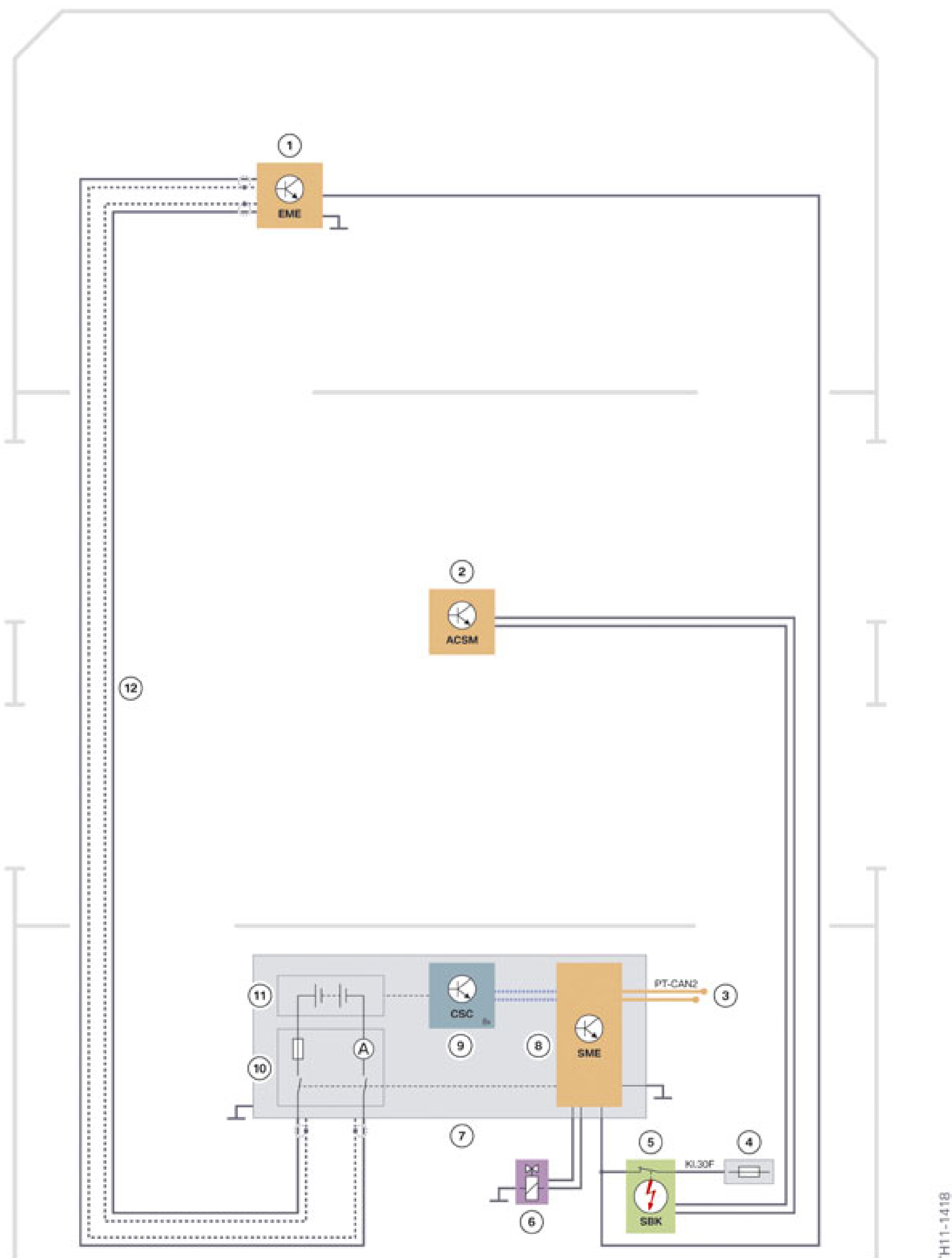
Index	Explanation
1	Prohibited symbol: No open flames, sparks or smoking
2	Warning symbol: Warning against dangers of batteries
3	Warning symbol: Warning against explosive substances / materials
4	Warning symbol: Warning against caustic substances / materials
5	Reference to disposal of high-voltage battery unit: Recycling by specialist personnel possible, not to be disposed of in domestic waste
6	Multilingual text: "Gen 2 Electrical Energy Storage System U_N = 317 V, C = 4 Ah (1 h, 25 °C), 46 kg"

5. High-voltage Battery Unit

⚠

Always observe the safety data sheet for the high-voltage battery when performing repair work on these components. Use the personal safety equipment prescribed there. If there is any uncertainty regarding a possible hazard, contact the technical support (PUMA) of the BMW Group.

5.1.2. System wiring diagram



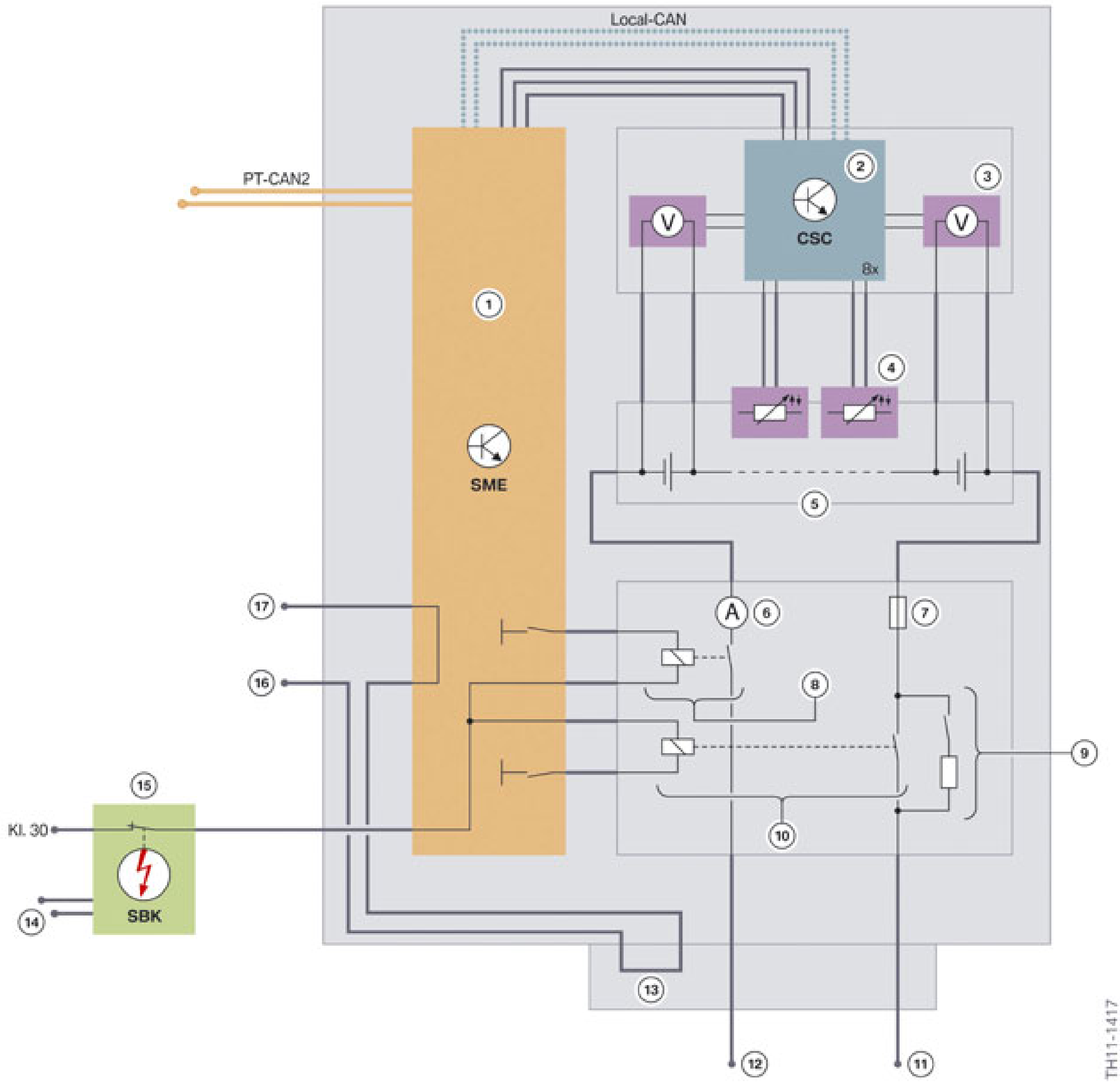
System wiring diagram of high-voltage battery unit

Index	Explanation
1	Electric motor electronics (EME)
2	Crash Safety Module (ACSM)
3	PT-CAN2 cables
4	Fuse for battery management electronics in luggage compartment fuse holder
5	Safety battery terminal
6	Combined expansion and shutoff valve
7	High-voltage battery unit
8	Battery management electronics (SME)
9	Cell supervision circuits (CSC)
10	External interface with electromechanical switch contactors (S-Box)
11	Cell modules of the high-voltage battery
12	High-voltage cables with shielding

The system wiring diagram of the components of the high-voltage interlock loop can be found in the chapter entitled "Monitoring functions" in this training material.

5.1.3. General structure

The following wiring diagram shows the internal electrical structure of the high-voltage battery unit.



Wiring diagram of the internal structure of the high-voltage battery unit

Index	Explanation
1	Battery management electronics
2	Cell supervision circuits (CSC)
3	Sensors for measuring the cell voltage
4	Sensors for measuring the cell temperature
5	Separate high-voltage battery, comprising several cell modules

Index	Explanation
6	Sensor for measuring the current level in the minus line of the high-voltage battery unit
7	Overcurrent fuse in the plus line of the high-voltage battery unit
8	Electromechanical switch contactor in the negative (ground) line of the high-voltage battery unit
9	Pre-charge switch
10	Electromechanical switch contactor in the positive line of the high-voltage battery unit
11	Positive line of the high-voltage battery unit
12	Negative line of the high-voltage battery unit
13	Bridge in the circuit of the high-voltage interlock loop (element of the high-voltage connection)
14	Control lines of the multiple restraint system for blasting the safety battery terminal
15	Safety battery terminal
16	Output of the circuit of the high-voltage interlock loop
17	Input of the circuit of the high-voltage interlock loop

The wiring diagram shows that the high-voltage battery unit (beside the actual battery cells which are compiled in cell modules) contains the following electrical/electronic components:

- Control unit Battery management electronics SME
- Eight cell supervision circuits (CSC)
- Interface box with switch contactors, sensors and overcurrent fuse.

These internal components, as well as the high-voltage connection and the high-voltage safety connector, are described in the following chapters.

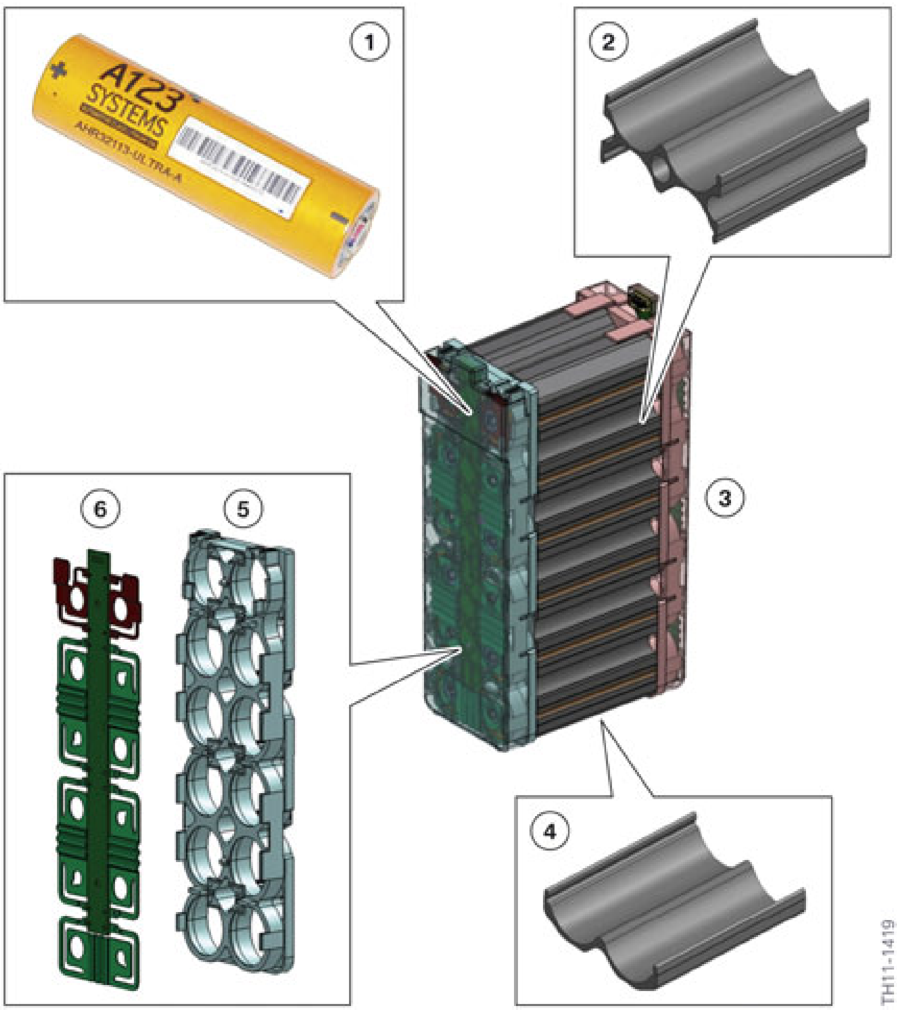
5.1.4. High-voltage battery

The term "high-voltage battery" refers to the actual energy accumulator for the high-voltage system. The nominal voltage of 316.8 V is achieved by connecting a total of 96 battery cells in series (each with a nominal voltage 3.3 V).

The high-voltage battery unit is made up of 8 cell modules.

Each cell module comprises the following:

- 12 cells
- Cooling extruded profiles (5 center pieces and 2 side pieces)
- Cell contact system (ZKS)



Structure of cell module

Index	Explanation
1	Cell
2	Cooling extruded profile, center piece
3	Cell module
4	Cooling extruded profile, side piece
5	Bracket for cell contacting system
6	Cell connector, cell contacting system (ZKS)

The individual lithium-iron-phosphate cells from A123 Systems are cylindrical in shape. These cells are lithium-ion batteries whose cathodes are made from lithium-iron-phosphate. They have impressive high load capacity and toughness. Another advantage of the lithium-iron-phosphate cells as opposed to other lithium-ion batteries is that there is no oxidation material in the cell therefore any exothermic (chemical/thermal) reactions are less severe.

5. High-voltage Battery Unit

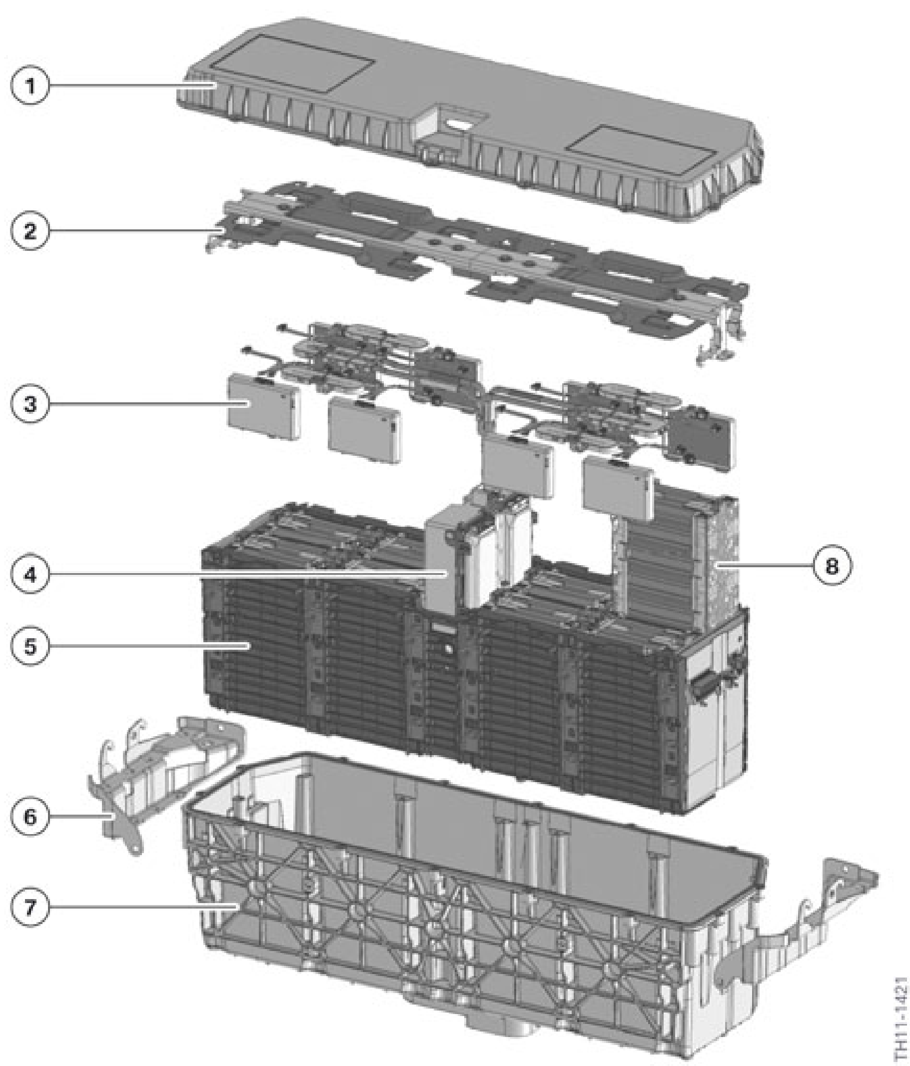
The cells are bonded in the cooling extruded profiles. In general, the adhesive tape should have high thermal conductivity to ensure good heat dissipation from the cell to the extruded profile. In the event of damage to the plastic sleeve of the cell the adhesive must also have an insulating effect.

The two parts of the cell contacting system (ZKS) are welded to the cell terminal by means of a laser welding process. This ensures the series connection of the individual cells.

There are two power taps in each module to discharge the current from the module. The power taps are attached at the front cell contacting system. The power taps of neighboring modules are connected to the module connectors.

Lithium-ion technology cells are sensitive to overloading, overvoltage, overcurrent and excess temperature. For this reason printed circuit boards for voltage taps are fitted at the cells with integrated temperature sensors on both sides of the cell module. The monitoring of the cells is done by the cell supervision circuit (CSC).

The cell modules are installed in a vaporizing unit to cool the cells. The refrigerant is routed along the cells through channels in the heat sink.



Structure of the high-voltage battery unit

Index	Explanation
1	Case Covers
2	Shield including current collector
3	Cell supervision circuit (CSC)
4	Battery management electronics (SME)
5	Heat sink and crash plates
6	Retaining bracket
7	Housing
8	Cell module

The crash plates are located between the cell modules. In the event of a crash the forces pass through the heat sink and crash plates thus protecting the cell modules.

If extremely high temperature or high pressure arises in a cell, the excess pressure can be decreased via pressure relief valves which are integrated in both covers of each cell. The gases released in the process are discharged from the vehicle through the degassing pipe.

The high-voltage battery can be connected to or isolated from the high-voltage electrical system using the electromechanical switch contactors inside the high-voltage battery unit. These contacts are located on the positive terminal and the negative terminal, before the terminals of the high-voltage battery are routed outwards. The electromechanical switch contactors are activated by the battery management electronics (SME).

5.1.5. Battery management electronics

High demands are placed on the service life of the high-voltage battery (service life of the vehicle). In the interest of satisfying these demands the high-voltage battery is operated in a precisely defined range so as to maximize its service life. This includes the following marginal conditions:

- Protecting the cells against overheating (by cooling and/or limiting the current)
- Adjusting the state of charge of the individual cells (where necessary) with regard to one another
- Not fully depleting the battery's amount of storable energy.

The high-voltage battery of the F10H incorporates its own control unit, which monitors these marginal conditions and intervenes where necessary. It is referred to as the "battery management electronics (SME)" and is located inside the high-voltage battery unit, which is why it is not accessible from the outside. Therefore the battery management electronics control unit **cannot** be replaced in the BMW Dealer Service workshop. The SME control unit in particular must perform the following tasks:

- Controlling cooling
- Determining the state of charge (SoC) and the state of health (SoH) of the high-voltage battery
- Determining the available power of the high-voltage battery and where necessary requesting a limitation from the electric motor electronics
- Controlling the starting and shutting down of the high-voltage system at the request of the EME

- Safety functions (e.g. high-voltage interlock loop, isolation monitoring)
- Monitoring the voltage and temperature of the battery cells as well as the current
- Communicating fault states to the EME.

The SME control unit has its own fault memory which can be read out using the ISTA BMW diagnosis system.

The electrical interfaces of the SME control unit are:

- 12 V supply voltage (terminal 30F for supplying voltage to SME, ground connection, terminal 31)
- PT-CAN2
- Wake-up line
- Forward and return lines for high-voltage interlock loop
- Line for activating the combined shut-off and expansion valve.
- Local Controller Area Network connection for CSCs

5.1.6. Cell monitoring

Certain conditions must be observed for fault-free operation of the lithium-ion cells in the F10H: The cell voltage and the cell temperature cannot exceed or drop below certain values as otherwise the battery cells may suffer long-term damage. For this reason each high-voltage battery unit has several cell supervision circuits (CSC).

Cell monitoring electronics

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The cell supervision circuits are located inside the high-voltage battery unit. For this reason they are not accessible to Service Technicians. Working on the cell supervision circuits in dealer workshop is not permitted.

The cell supervision circuits perform the following functions:

- Measurement and monitoring of the voltage of each individual battery cell
- Measurement and monitoring of the temperature at a point on each cell module
- Communication of the measured variables to the battery management electronics control unit (SME)
- Implementation of the process for adjusting the cell voltage of the battery cells.

The measurement of the cell voltage is done at a very high sampling rate. The end of the charging procedure, as well as the discharging procedure, can be identified using the voltage measurement. Using the cell temperature an overload or electrical fault can be identified. In such a case the current level must be reduced immediately or the high-voltage system shut down completely in order to avoid progressive damage to the battery cells. In addition, the measured temperature is used to control the cooling system in order to constantly operate the battery cells in the temperature range which is optimal for their performance and service life.

Local CAN

Communication between the 8 CSCs is via a Local Controller Area Network. The Local CAN combines all CSCs together and is used for the communication of the CSCs with the battery management electronics. The battery management electronics control unit assumes the master function here. The Local CAN is a low-voltage wiring harness with a maximum of 12 V.

Adjusting the cell voltages

If one or several of the battery cells has a significantly lower cell voltage than all other battery cells, the usable energy content of the high-voltage battery is restricted. The withdrawal of energy is in fact determined by the "weakest" battery cell: If the voltage of the weakest cell has fallen to the discharge limit, the discharging procedure must be terminated, even if the other battery cells still have enough energy stored. However, if the discharging procedure is continued the weakest battery cell may be permanently damaged. For this reason there is a function to adjust the cell voltage to approximately the same level. This process is also called "cell balancing" or "cell equilibrium".

For this purpose, the battery management electronics control unit regularly wakes up during rest phases and compares the cell voltages of all cells. As the adjustment of the cell voltages can only be done by the targeted discharge of individual battery cells, those with a considerably higher cell voltage are selected over the weakest battery cells. The discharging procedure starts upon a request via the Local Controller Area Network to the cell supervision circuits belonging to these battery cells and is carried out until the voltage level is adjusted. The discharge current flows via an ohmic resistor which is integrated in the cell supervision circuits.

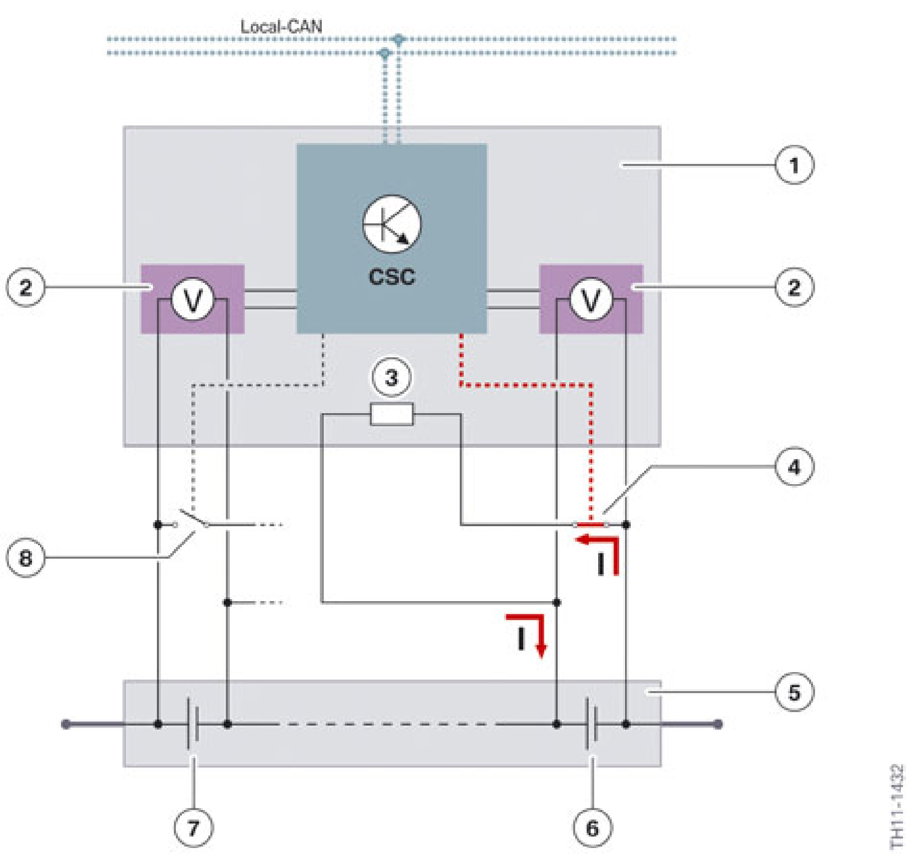


Illustration of the process for adjusting the cell voltages

Index	Explanation
1	Cell supervision circuit (CSC)
2	Sensors for measuring the cell voltage
3	Discharge resistor
4	Closed (active) contact for discharging a battery cell
5	High-voltage battery
6	Battery cell whose cell voltage is reduced by discharging
7	Battery cell not being discharged
8	Open (inactive) contact for discharging a battery cell

The adjustment of the cell voltage thus results in energy loss; however it is reasonable and necessary to maximize the use and service life of the battery. This procedure can be implemented only when the vehicle is at a standstill.

The following conditions are necessary for adjusting the cell voltages:

- Terminal 15 switched off and vehicle or vehicle electrical system asleep
- High-voltage system shut down
- Deviation of the cell voltages or the individual states of charge of the cells are greater than a threshold value
- Total SoC of the high-voltage battery is greater than a threshold value.

The adjustment of the cell voltages is done automatically when the specified conditions are fulfilled. The customer therefore does not receive a Check Control message nor does he have to implement a special measure.

If the cell voltages however have too great a deviation or the adjustment of the cell voltage is not successful, a fault code entry must be created in the battery management electronics control unit. The customer is made aware of this fault status by a Check Control message. With help of the BMW diagnosis system the fault memory must then be evaluated and remedies deduced.

5.1.7. External interface (S-box)

In the high-voltage battery unit there is an interface unit with its own housing, which is also called a "shift box" or "S-box" for short. As it is located inside the high-voltage battery unit, it **cannot** be replaced in the BMW Dealer Service workshop.

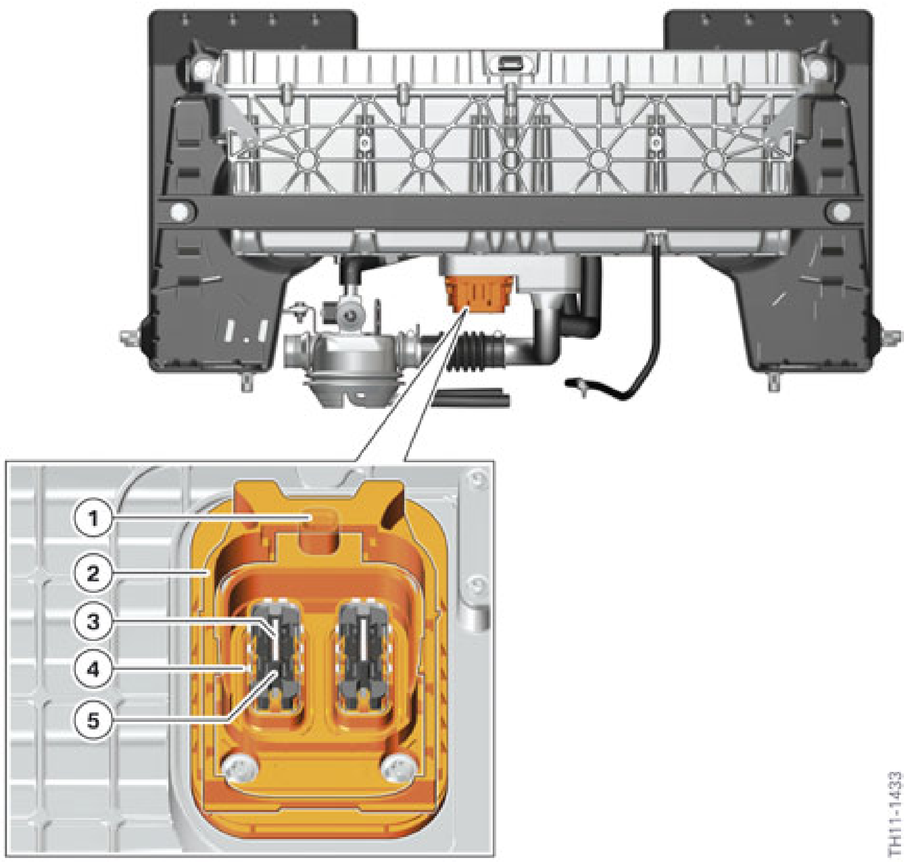
The following components are integrated in the interface unit:

- Current sensor in the current path of the negative battery terminal
- Safety fuse in the current path of the positive battery terminal
- Two electromechanical switch contactors (one switch contactor per current path)
- Pre-charge switch for slow start-up of the high-voltage system
- Voltage sensors for monitoring the switch contactors and for measuring the total battery voltage.

These components and their electrical connection are shown in the internal wiring diagram of the high-voltage battery unit at the start of this chapter.

5.1.8. High-voltage connection

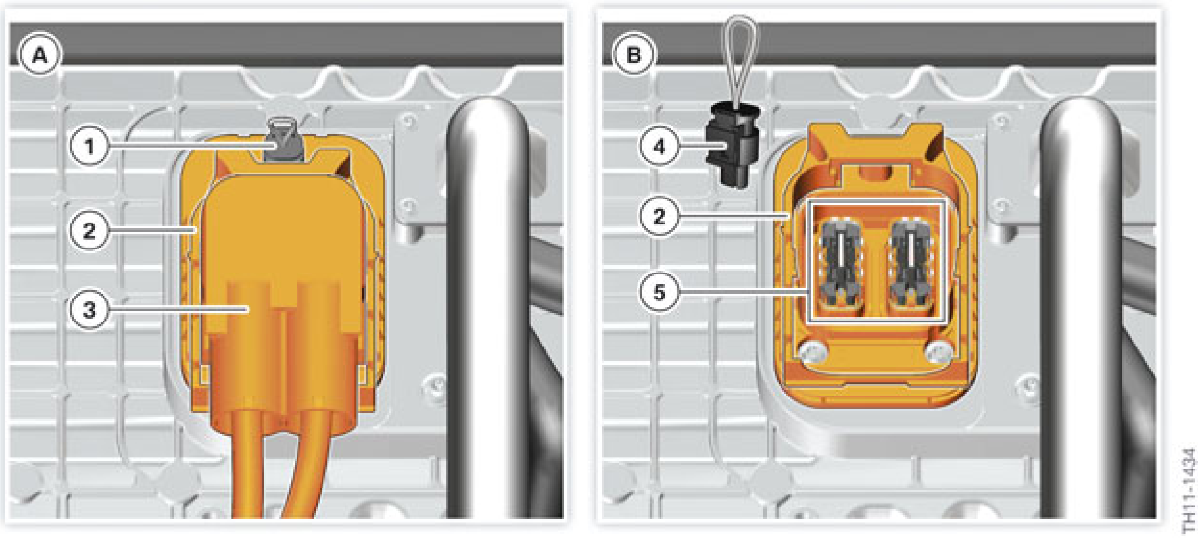
There is a 2-pin high-voltage connection at the high-voltage battery unit.



High-voltage connector on the high-voltage battery unit

Index	Explanation
1	Socket connection for bridge in the circuit of the high-voltage interlock loop
2	Mechanical locking
3	Electrical contact for high-voltage cable
4	Electrical contact for shielding
5	Contact protection

The high-voltage connection has the primary task of connecting the high-voltage battery unit to the high-voltage cables. In addition, the high-voltage connection provides protection against contact with live parts: The actual contacts are covered in plastic so that nobody can touch them directly. Only when the cable is connected is the covering pushed away and the contact established. The plastic slide serves as the mechanical latch mechanism of the connector. It is also performs a safety function: If the high-voltage cable is not connected, the slide covers the connection for the bridge of the high-voltage interlock loop. Only when the high-voltage cable is properly connected and the connector is locked, is this connection accessible so that the bridge can be inserted. This guarantees that only when a high-voltage cable is connected is the circuit of the high-voltage interlock loop also closed. This principle applies to the high-voltage connection on the high-voltage battery unit and on the electrical machine electronics. The high-voltage system can thus only be active if the high-voltage cable is connected and the circuit of the high-voltage interlock loop is closed.

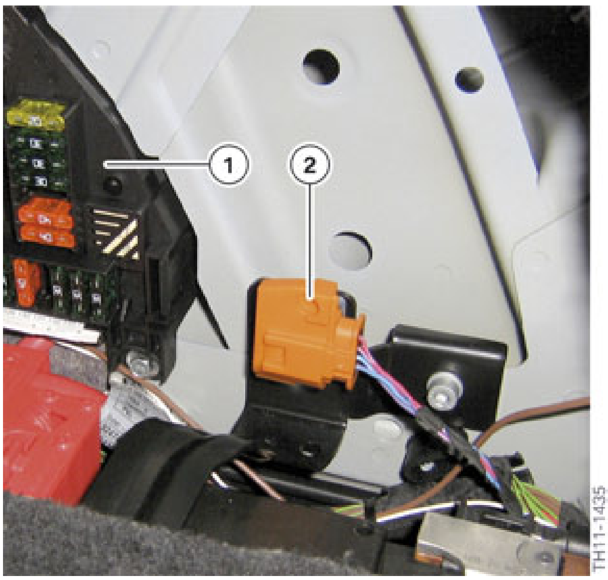


High-voltage connection

Index	Explanation
A	High-voltage connection with connected high-voltage cable
B	High-voltage connection with disconnected high-voltage cable
1	High-voltage interlock loop bridge (connected)
2	Mechanical slide
3	High-voltage connector of the high-voltage cable
4	High-voltage interlock loop bridge (disconnected)
5	High-voltage connection on the high-voltage battery unit

5.1.9. High-voltage safety connector (Service Disconnect)

The high-voltage safety connector ("Service Disconnect") of the F10H is not a direct component of the high-voltage battery unit. Instead it is installed separately near the auxiliary 12V battery. It is only accessible by removing a cover located on the right side of the luggage compartment.



Installation location of the high-voltage safety connector

Index	Explanation
1	Fuse carrier in the luggage compartment
2	High-voltage safety connector (Service Disconnect)

As in the previous ActiveHybrid vehicles, the F10H high-voltage safety connector also fulfills two tasks:

- Disconnecting the high-voltage system from the supply
- Providing a safeguard to prevent unintentional restarting.

The high-voltage safety connector or the connected bridge is part of the high-voltage interlock loop circuit. If one disconnects the connector and socket of the high-voltage safety connector, the circuit of the high-voltage interlock loop is interrupted. The high-voltage system automatically shuts down and is thus switched to a de-energized state.

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The connector and socket of the high-voltage safety connector cannot be disconnected from each other completely. Both parts are mechanically secured against complete separation. To interrupt the circuit of the high-voltage interlock loop, it is sufficient to separate the two parts far enough so that the padlock can be used to secure against reconnection.

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Before working on the high-voltage connection of the high-voltage battery unit or another high-voltage component, it is essential to **disconnect** the high-voltage system from the supply, **secure** it against unintentional reconnection and **verify** that it is safely isolated from the supply.

The exact procedure to put the system to a de-energized state is described in the chapter entitled "Safe working on the high-voltage system".

5.1.10. Ventilation

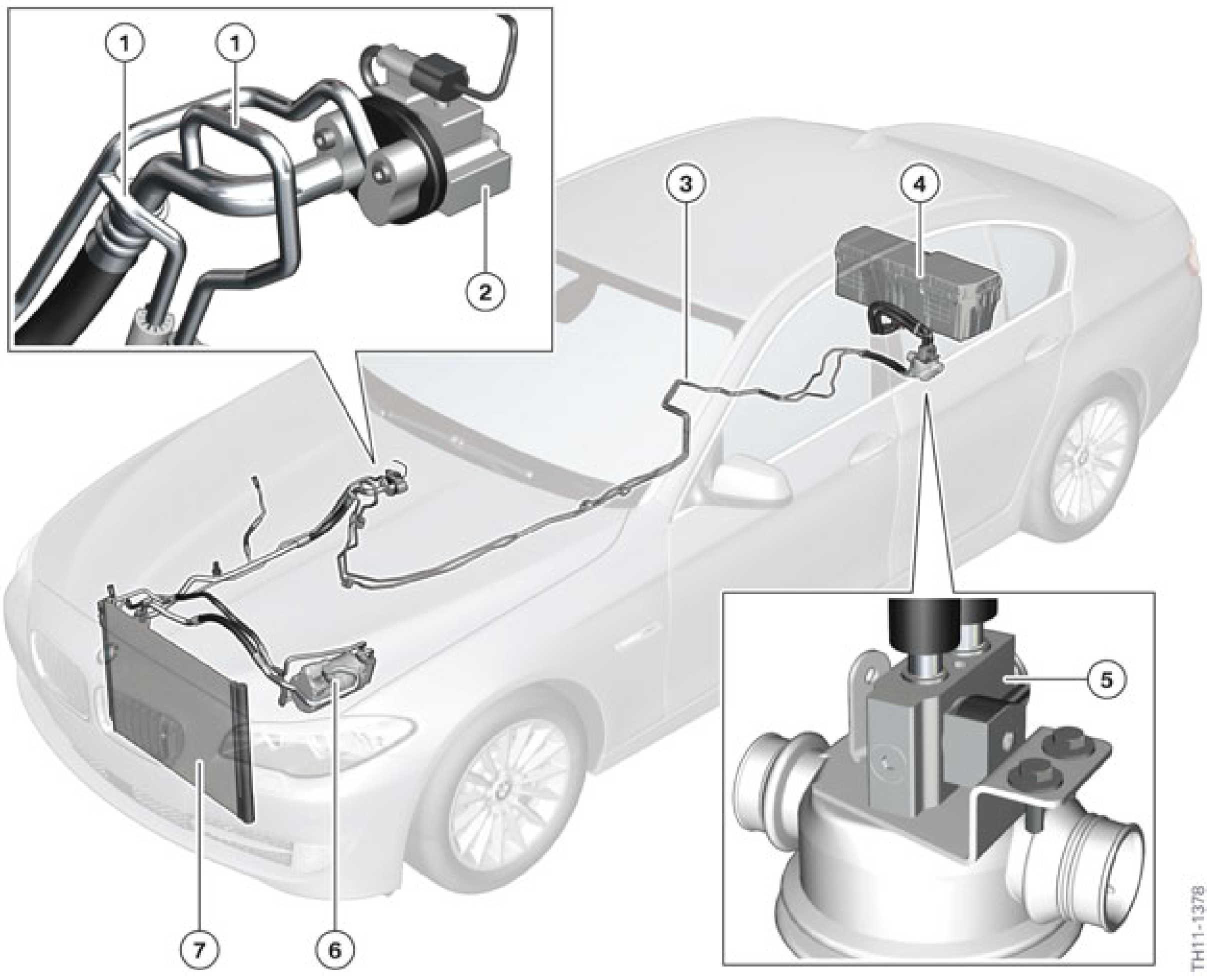
The vent orifice is used to balance large pressure differences between the inside and outside of the high-voltage battery unit and thus prevent dirt contamination of the interior of the high-voltage battery unit. Such pressure differences can only arise in the event of a damaged battery cell. The vent is also used for draining condensation from the high-voltage battery unit which develops due to temperature deviations.

The gases and condensation can escape the housing of the high-voltage battery unit into the atmosphere via a vent pipe connected to a grommet.

5.1.11. Cooling system

To maximize the service life of the high-voltage battery and obtain the greatest possible power, it is operated in a defined temperature range. The high-voltage battery unit is essentially operational in the range of -25 °C (-13 °F) to +55 °C (131 °F). These temperature limits, however, relate to the actual temperature of the cells, not the outside temperature. In terms of temperature behavior the high-voltage battery unit is a slow-action system, i.e. it takes several hours until the cells assume ambient temperature. Therefore having the battery spend a short period of time in an extremely hot or cold environment does not mean that the cells will have already assumed this temperature.

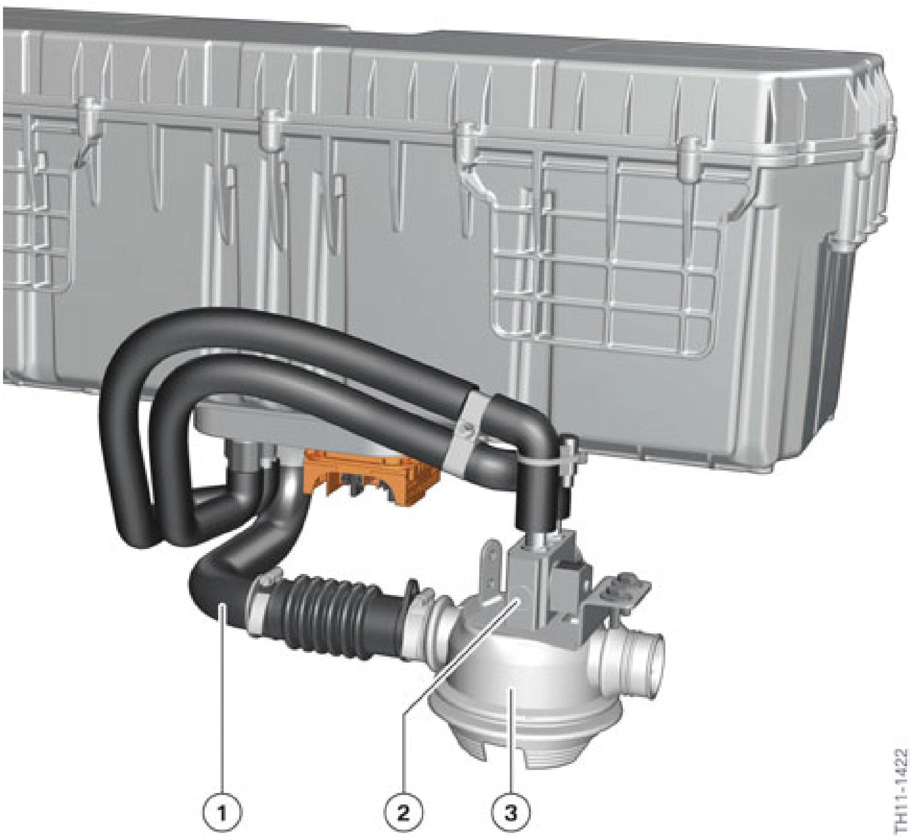
The upper temperature limit does not pose any restriction with regard to use of the hybrid functions since temperatures higher than +55 °C (131 °F) at the installation location of the high-voltage battery unit cannot be reached. Cooling the high-voltage battery unit keeps the cell temperature at a clearly lower level. The hybrid functions are thus available even when the electric drive is subject to intensive use and at high outside temperatures. Furthermore, cooling further increases the service life of the high-voltage battery unit.



F10H refrigerant circuit

Index	Explanation
1	Branching of refrigerant lines to high-voltage battery unit
2	Combined expansion and shutoff valve for passenger compartment
3	Refrigerant lines to high-voltage battery unit
4	High-voltage battery unit
5	Combined expansion and shutoff valve
6	Electrical A/C compressor
7	Condenser

The high-voltage battery in the F10H is cooled by refrigerant. The refrigerant circuit of the air conditioning has therefore been extended to cool the high-voltage battery unit. The expansion valve for climate control of the passenger compartment and that for the high-voltage battery are connected in parallel. The battery management electronics can activate and open the combined shut-off and expansion valve for the high-voltage battery unit via a pulse-width modulated signal. In this way refrigerant can flow to the high-voltage battery, expanding, evaporating and cooling in the process. The cooling of the high-voltage battery unit can be carried out completely independently of the cooling in the passenger compartment as these are controlled by a separate expansion and shut-off valve.



Components for cooling the high-voltage battery unit

Index	Explanation
1	Degassing pipe
2	Combined expansion and shutoff valve
3	Grommet

Due to the cooling of the high-voltage battery unit with refrigerant and the resulting large temperature differences, water vapor can condense inside the high-voltage battery unit. This condensation is vented from the passenger compartment via a degassing pipe and a grommet. The degassing pipe serves to compensate large pressure differences between the interior and the exterior of the high-voltage battery. Pressure differences of this nature can only arise if a cell is damaged – for safety reasons the housing of the damaged cell then opens in order to reduce the pressure. The escaping gases are then directed out of the vehicle through the degassing pipe.

In the temperature range in which the high-voltage battery unit does not need to be cooled the combined expansion and shutoff valve for the high-voltage battery unit remains closed.

Depending on the cooling requirement and availability of cooling power, the cooling of the high-voltage battery unit is controlled by the battery management electronics control unit. The availability of cooling power is requested by the IHKA control unit. The combined expansion and shutoff valve is activated by the SME control unit.

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Because the high-voltage battery unit is connected to the refrigerant circuit, preliminary work similar to that conducted on the air- conditioning of conventional vehicles (e.g. evacuation) will be required prior to removal and installation work. Please refer to the repair instructions.

5.2. Functions

5.2.1. Starting the high-voltage system

Starting the high-voltage system occurs in the interaction between the electric motor electronics (EME) and battery management electronics (SME) control units. The EME control unit performs the role of the primary, the SME control unit that of the secondary. The associated commands are transferred as bus signals via the PT-CAN2.

The EME control unit requests starting of the high-voltage system when either terminal 15 is switched on or a request for independent air conditioning is present. The start takes place in multiple steps, each of which takes place only if the previous step has been completed successfully.

1 Testing the high-voltage electrical system
2 Raising the voltage
3 Closing the contacts of the switch contactors.

The first step, testing the high-voltage system, involves checking whether the high-voltage battery unit and the entire high-voltage electrical system are operational. This includes the requirement that the circuit of the high-voltage interlock loop be closed in order to start the high-voltage system.

Even after the tests have been completed successfully, the contacts of the switch contactors still cannot be closed. Because of the capacities in the high-voltage circuit (link capacitors) a very high switch-on current would flow which would permanently damage both the link capacitors and the switch contactors. Therefore, the voltage is raised slowly beforehand. To do so, the contact of the switch contactor for the negative lead is initially closed. A low constant current is set in the positive wire via an electronic circuit to charge the link capacitors. In this way the voltage in the high-voltage electrical system increases within a few hundred milliseconds to a value which is only slightly below the voltage of the high-voltage battery unit. Then the contact of the switch contactor in the positive lead is also closed.

The SME control unit communicates successful starting via the PT-CAN2 to the EME control unit. In the same way, fault states are signalled if the start was not successful.

Although the starting of the high-voltage system contains the above-mentioned steps, there is no noticeable delay when starting the car as far as the customer is concerned. An F10H starts practically as quickly as an F10.

5.2.2. Regular shut-down of the high-voltage system

When it comes to shutting off the high-voltage system a distinction is made between regular shut-down and fast shut-down. In the case of regular shut-down the emphasis is on protecting the electrical components and checking the high-voltage system. For example, the contacts of the electromechanical switch contactors are to be opened only when the current has dropped to a value close to 0 A, as they are otherwise subjected to heavy load.

The conditions which trigger a regular shut-down are:

- Deactivation of terminal 15
- End of stationary cooling
- End of programming procedure of a high-voltage control unit.

The basic sequence of regular shut-down is then (irrespective of the triggering condition) the same:

1 Regulating currents in the high-voltage electrical system down to zero (EME)
2 Opening the switch contactors in the high-voltage battery unit (SME)
3 Discharging the high-voltage circuit, i.e. active discharging of the link capacitors and short-circuiting of the windings in the electric motor (EME)
4 Checking the high-voltage system, e.g. as to whether the contacts of the electromechanical switch contactors were correctly opened.

Both the after-running period (after switching off terminal 15) and the shut-down itself can last a few minutes. The automatic monitoring functions are a reason for this, for example. The regular shut-down is interrupted if in the meantime either a request for a renewed start-up is made or a condition has arisen to request a quick shut-down.

5.2.3. Fast shut-down of the high-voltage system

The main purpose here is to shut down the high-voltage system as quickly as possible. This quick shut-down is then always carried out if for safety reasons the voltage in the high-voltage system has to be reduced to a safe value as quickly as possible.

The following list describes the triggering conditions and the functional chain leading to the quick shut-down.

- **Accident**
  Crash Safety Module ACSM identifies an accident. Depending on the severity of the accident, the shut-down is requested via data bus signals or forced by disconnecting the safety battery terminal from the positive terminal of the 12 V battery. In the second scenario the voltage supply of the electromechanical switch contactors is automatically interrupted and their contacts open automatically.
- **Overload current monitoring:**

The current level in the high-voltage electrical system is monitored via a current sensor in the high-voltage battery unit. If too high a current level is identified, the battery management electronics control unit causes a hard opening of the electromechanical switch contactor. Considerable wear occurs to the contacts of the switch contactors as a result of this opening under a high current, which must be accepted to protect the other components from damage.

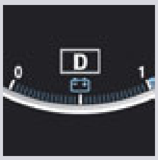
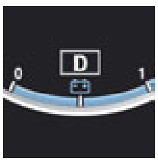
- **Protection in the event of a short circuit:**
  In the high-voltage battery unit there is an overload current fuse which interrupts the high-voltage circuit in the event of a short circuit.
- **Critical cell state:**
  If a cell supervision circuit (CSC) identifies extreme undervoltage, overvoltage or excess temperature at a battery cell, this also leads to a hard opening of the electromechanical switch contactors - controlled by the battery management electronics control unit (SME). Although this may lead again to increased wear at the contacts, this quick shut-down is necessary to prevent destroying the respective battery cells.
- **Malfunction of the 12 V voltage supply of the high-voltage battery unit:**
  In this case the battery management electronics control unit (SME) no longer works and it is no longer possible to monitor the battery cells. For this reason the contacts of the electromechanical switch contactors also open here automatically.
- **High voltage interlock loop:**
  The SME evaluates the signal of the high-voltage interlock loop and checks whether there is an open circuit with this circuit. In the event of an open circuit, the SME can cause the high-voltage system to perform a quick shut-down. If the high-voltage interlock loop is disconnected at the high-voltage safety connector ("Service Disconnect"), the shut-down is no longer effected via the SME , but the switch contactors are opened directly.

In all the cases mentioned the contacts of the electromechanical switch contactors are opened and the high-voltage circuit discharged immediately. This ensures the fastest possible shut-off, as is appropriate for these safety-relevant cases.

5.2.4. Charging strategy and operational strategy

The charging strategy for the high-voltage battery unit has two primary objectives: to maximize the service life of the high-voltage battery unit and to maintain reserves with regard to both energy consumption and energy draw. The charging strategy is geared towards achieving these two primary objectives and the high-voltage battery unit thus helps in many driving situations to increase the efficiency and dynamics of the vehicle.

In order to achieve the first objective, the strategy involves the already mentioned regulation of the cell temperature and the regulation of the useful energy content. Instead of using the full state of charge range from zero to 100%, only a range between approximately 25 and 70% is actually used. Depending on the cell temperature and the calculated state of health of the high-voltage battery, this useful range can be limited further. However, a sufficiently large useful energy content remains available over the full service life in order to implement the hybrid-specific functions and advantages. The state of charge of the high-voltage battery unit is indicated to the driver in the instrument cluster (KOMBI) in the form of bar graph gauge and in the Central Information Display in the form of bar graph. The value displayed there corresponds to the value used by the charging strategy of roughly 25 to 70%. There are therefore still reserves below the minimum displayed value and above the maximum displayed value.

Display in instrument cluster	Displayed state of charge	Actual state of charge
	0%	About 25%
	100%	About 70%

During brake energy regeneration or load point increase of the combustion engine excess energy is converted into electrical energy by the electric motor. The high-voltage battery unit stores this electrical energy and makes it available again as required for the following purposes:

- To supply the 12 V vehicle electrical system
- To relieve and support the combustion engine (efficiency increase)
- For the boost function (increase in dynamics)
- For electric driving
- For independent air conditioning

5.2.5. Monitoring functions

There is a large number of monitoring functions in which the high-voltage battery unit and the battery management electronics play a substantial role.

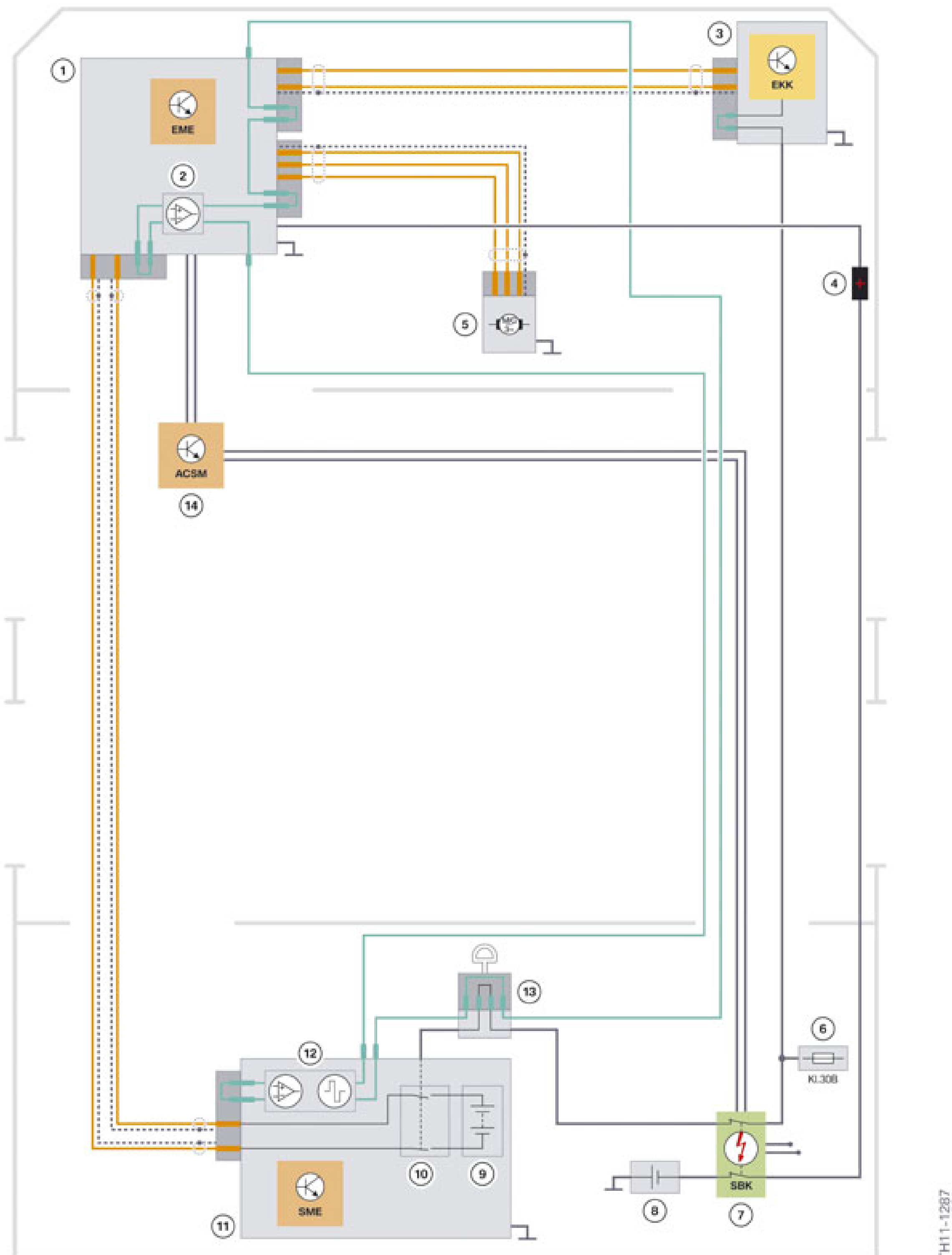
This includes the following:

- Monitoring functions to ensure the safety of the high-voltage system
- Monitoring functions to ensure optimal operating conditions of the high-voltage battery

For the safety-related monitoring functions, we will specifically discuss the role of the high-voltage battery unit in the high-voltage interlock loop and the isolation monitoring.

The **high-voltage interlock loop** is a fail-safe circuit that ensures safety when working on high-voltage component if the high-voltage electrical system has not been switched off properly beforehand. If this circuit is interrupted, the voltage supply of the high-voltage system is switched off and switching on the supply of the high-voltage system is prevented.

The principle of the high-voltage interlock loop is explained in the "Principles of Hybrid Technology" training information available on TIS and ICP. In the F10H the high-voltage interlock loop is made up of the high-voltage components illustrated below.



F10H System wiring diagram of high-voltage interlock loop

5. High-voltage Battery Unit

Index	Explanation
1	Electrical machine electronics (EME)
2	Evaluation circuit for test signal of the high-voltage interlock loop in the EME
3	Electrical A/C compressor
4	Jump start terminal point
5	Electrical machine
6	Fuse carrier in the luggage compartment
7	Safety battery terminal
8	12 V battery
9	High-voltage battery
10	S-Box with electromechanical switch contactors
11	High-voltage battery unit
12	Evaluation circuit and signal generator for test signal of the high-voltage interlock loop in the battery management electronics
13	High-voltage safety connector ("Service Disconnect")
14	Crash Safety Module (ACSM)

The electronics for controlling and generating the test signal for the high-voltage interlock loop is integrated in the F10H in the battery management electronics. Generating the test signal starts when the high-voltage system is to be started and ends when the high-voltage system has been shut down. A square wave alternating current signal is generated as the test signal by the battery management electronics and supplied to the test lead. The test lead has a ring topology (similar to that of the MOST bus). The signal of the test lead is evaluated at two points in the ring: in the electrical machine electronics and finally right at the end of the ring, in the battery management electronics. If the signal is outside a permanently defined range, an interruption of the circuit or a short circuit in the test lead is detected and the high-voltage system is shut off immediately. If the high-voltage interlock loop at the high-voltage safety connector (“Service Disconnect”) is disconnected, then the switch contactors are opened directly. In addition, all high-voltage components are switched off.

The **isolation monitoring** determines whether the insulation resistance between active high-voltage components (e.g. high-voltage cables) and ground is above or below a required minimum value. If the insulation resistance falls below the minimum value, the danger exists that the vehicle parts will be energized with hazardous voltage. If a person were to touch a second active high-voltage component, he or she would be at risk of electric shock. There is therefore fully automatic isolation monitoring for the F10H high-voltage system. It is performed by the battery management electronics at regular intervals while the high-voltage system is active. Ground serves as the reference potential. Without additional measures only local isolation faults in the high-voltage battery unit could be determined in this way. However, it is equally important to identify isolation faults from the high-voltage cables in the vehicle to ground. For this reason all the electrically conductive housings of high-voltage components are connected to ground. This enables isolation faults in the entire high-voltage electrical system to be identified from a central point, the high-voltage battery unit.

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The proper electrical connection of all high-voltage component housings to ground is an important prerequisite for proper function of the isolation monitoring. Accordingly, this electrical connection must be restored carefully if it has been interrupted during repair work.

The isolation monitoring responds in two stages. When the insulation resistance drops below a first threshold value, there is still no direct danger to people. The high-voltage system therefore remains active; no Check Control message is output, but the fault status is naturally stored in the fault memory. In this way the Service technician is alerted the next time the car is in the workshop and can then check the high-voltage system. When the insulation resistance drops below a second, lower threshold value, this is accompanied not only by the storage of the fault in the fault memory, but also by the display of a Check Control message prompting the driver to visit a workshop.

However, the Service technician does not have to perform a fundamental measurement of the insulation resistance him/herself – this task is performed by the high-voltage system through the isolation monitoring. When an isolation fault is detected, the Service technician must run through a test schedule in the diagnosis system to find the actual location of the isolation fault.

In addition to the high-voltage interlock loop and isolation monitoring, there are further monitoring functions which are listed below:

- **12 V supply voltage from the safety battery terminal:**
  To be able to perform a quick shut-down of the high-voltage system in the event of an accident of corresponding severity, the solenoids of all electromechanical switch contactors are supplied with 12 V from the safety battery terminal. If the safety battery terminal is deployed in the event of an accident, this cuts the voltage supply and the contacts of the switch contactors open automatically.
  In addition, the battery management electronics control unit evaluates the voltage on this line electronically and also causes the high-voltage system to shut down including discharge of the link capacitors and the active short circuit of the electrical machine.
- **Contacts of the switch contactors:**
  After the battery management electronics control unit has requested the contacts of the switch contactors to open during the shut-down of the high-voltage system it checks whether they have actually been opened (via a voltage measurement check parallel to the contacts).
  In the highly unlikely case that the contact of a switch contactor does not open, there is still no direct danger to the customer or the Service Technician. However, for safety reasons a renewed start-up of the high-voltage system is prevented.
- **Pre-charge switch:**
  If for example during the start-up of the high-voltage system a fault is identified with the pre-charge switch, then the start-up is cancelled immediately and the high-voltage system is not put into operation.
- **Excess temperature:**
  The cooling system of the high-voltage battery ensures in all driving situations that the temperature of the battery cells remain in the optimal range. If due to a fault the temperature of one or several battery cells increases past the optimal range, the power is reduced initially to protect the battery cells. If the temperature continues to increase and thus threatens damage to the battery cells, the high-voltage system is then switched off.
- **Undervoltage:**

Undervoltage at a battery cell is avoided by the constant monitoring and adjustment of the cell voltage as required. The total voltage of the entire high-voltage battery unit is also monitored and used to determine the state of charge. If the total voltage has fallen to the extent that the high-voltage battery is discharged completely, another discharge is prevented.

5.3. Service information

5.3.1. Installation and removal

The complete high-voltage battery unit can only be replaced as a complete assembly in a BMW Dealer Service workshop, if this is indicated by the test plan of the ISTA BMW diagnosis system.

The return of the faulty high-voltage battery unit can only be done via the "Warranty Parts Process" and not via the normal "Replacement Parts Process". The high-voltage battery unit of the F10H belongs to hazard class 9 (UN 3090) and cannot be sent by air freight.

⚠

Because the high-voltage battery unit is a high-voltage component, it is essential to observe the electrical safety rules before starting work and to always follow the repair instructions during removal and installation.

In addition, prior to removal the refrigerant must be drawn off from the refrigerant circuit and after installation re-introduced into the system.

⚠

A precise description of the work sequence for removal and installation can be found in the repair instructions in ISTA BMW diagnosis system.

A special tool is used to install and remove the high-voltage battery unit. It helps in lifting the high-voltage battery unit out of and back into the vehicle. Using this tool will reduce the physical strain on the Service Technicians when removing and installing the unit.

⚠

In the event of replacement of the high-voltage battery unit, initialization is required with help of the ISTA BMW diagnosis system.

5.3.2. Charging the high-voltage battery unit

If the 12 V battery of a F10H is flat, it can be charged as in conventional vehicles.

The operating strategy will control the state of charge of the high-voltage battery unit so that a certain amount of energy always remains in the high-voltage battery unit.

⚠

As the high-voltage battery unit may suffer damage in the event of excessive discharging, it must be ensured before long immobilization periods (with help of the state of charge display) that the high-voltage battery unit is fully charged.

Quick charging

In exceptional cases, the high-voltage battery unit can also be charged by the engine when stationary.

The following steps must be performed for Quick charging:

- Start engine
- Engage “P” and engage automatic hold brake
- Press and hold down the brake pedal
- With help of the brake pedal keep the speed of the engine at about 2000 rpm.

After a few minutes the high-voltage battery is fully charged again. The state of charge can be controlled with help of the state of charge display in the instrument cluster.

⚠

The charging process can only be carried out successfully if the brake pedal is permanently applied. Otherwise, the required charging status of the high-voltage battery will not be reached.

⚠

The charging status can also take place at idle speed, however, this increases the charging time significantly.

5.3.3. Safe working practices for working on a high-voltage system

⚠

Work on live high-voltage components is strictly prohibited. Prior to every operation which involves a high-voltage component, it is essential to disconnect the high-voltage system from the voltage supply and to secure it against unauthorized reconnection.

1. Switch off terminal 15.
2. **De-energize** the high-voltage by opening the high-voltage safety switch (Safety Disconnect).
3. **Secure** the high-voltage safety connector from being reconnected.
4. Switch on terminal 15.
5. **Verify** that the "High-voltage system switched-off" Check Control message is displayed in the instrument cluster.
6. Switch off terminal 15 and terminal R.

F10H Complete Vehicle

5. High-voltage Battery Unit

The following chapters provide brief descriptions on how to implement the electrical safety rules in the F10H.

Preparations

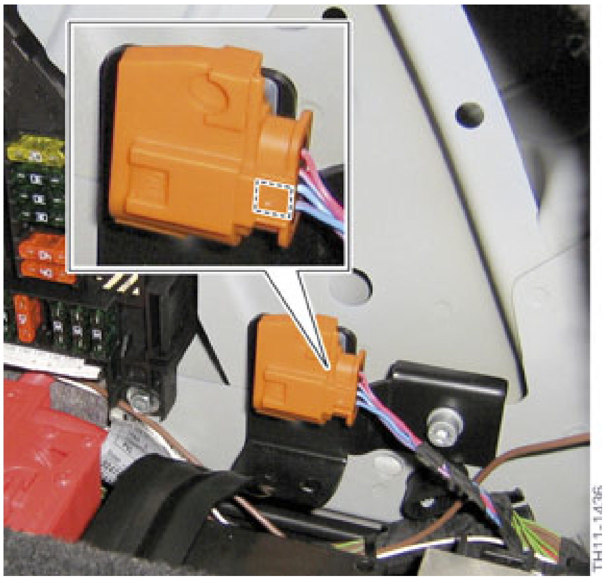
Prior to beginning any work, the vehicle must be secured against rolling (engage the parking lock of the transmission and activate the parking brake). Disconnect any connected charging cables. Terminal 15 and terminal R must be switched off.

From an electrical safety viewpoint, work can begin on the system when it has been switched off and put to a de-energized state. However, the high-voltage system may still be active and current may flow via the contacts of the switch contactors. During shut-down this may result in a load in the contacts which can be avoided in the following manner: Allow the vehicle or the control units to go to sleep. Then the high-voltage system is already shut down.

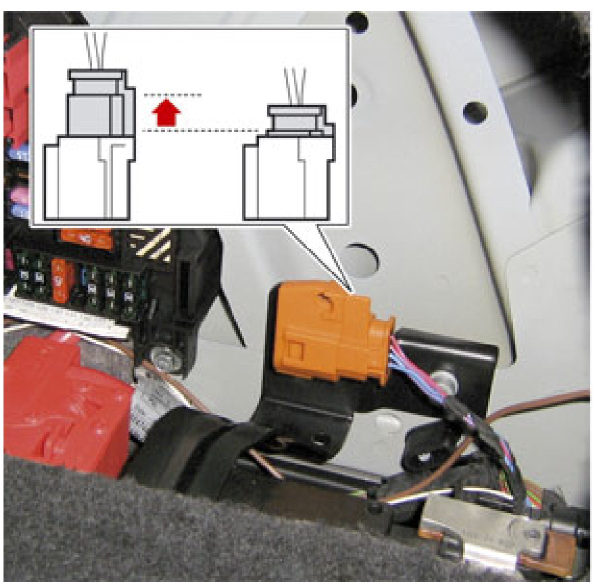
Disconnect the high-voltage system from the supply

The high-voltage system in the F10H is disconnected from the supply with the high-voltage safety connector. To disconnect from the supply, the connector must be pulled from the associated socket. This interrupts the circuit of the high-voltage interlock loop. As a result of the open circuit of the high-voltage interlock loop, the voltage supply of the switch contactors is interrupted and the high-voltage system put into a de-energized state.

This image shows the high-voltage safety connector in a connected state. The circuit of the high-voltage interlock loop is not interrupted.



To pull apart the socket and the connector, the mechanical lock shown in the image must be pressed.



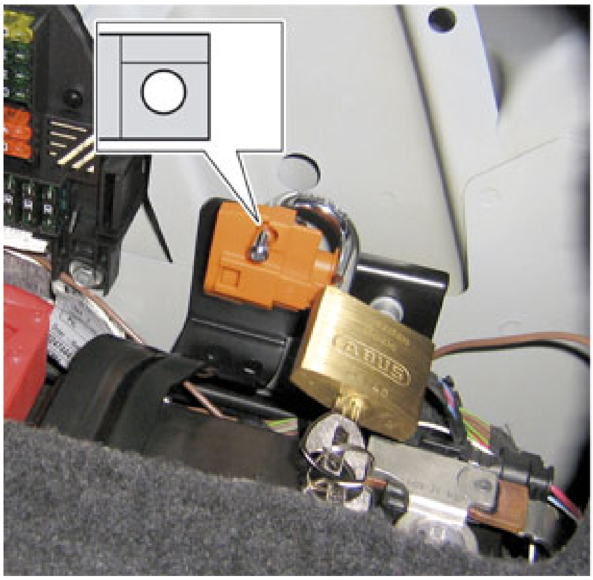
As soon as the mechanical lock has been removed, the connector can be pulled from the socket a few millimeters.

⚠

Do not pull any further or harder once resistance can be felt. The connector and socket of the high-voltage safety connector cannot be completely disconnected from each other.

Provide the high-voltage system with a safeguard against unintentional restarting

Securing against restart is also effected at the high-voltage safety connector. A commercially available padlock is required for this purpose.



By separating the socket and connector of the high-voltage safety connector, a hole appears through both parts. The loop of a typical pad-lock must be inserted in this hole.

TH11-1430

The pad lock can now be closed. The key must be stored in a safe place during work on the high-voltage system so that an unauthorized person cannot unlock the lock.
By inserting and closing the padlock at the high-voltage safety connector the connector can no longer be joined. This is an effective way of ensuring that the high-voltage system is not switched on again without the knowledge and consent of the Service Technician.

Verifying safe isolation from the power supply

The de-energized state is not verified using a measuring device or via the ISTA BMW diagnosis system in the BMW Dealer Service workshop. Instead, the high-voltage components measure the voltage themselves and transmit the measured result via bus signal to the instrument cluster.

The instrument cluster does not generate the Check Control message to display the de-energized state unless all involved high-voltage components consistently signal the de-energized state. This Check Control symbol in red shows a crossed-out flash symbol.

Check Control symbol "High-voltage system de-energized"

In order to verify the de-energized state, you must switch on terminal 15 and wait until you see the Check Control message with the symbol shown above on the instrument cluster. Then, and only then, you have ensured that the high-voltage system is de-energized. After the de-energized state has been verified, terminal 15 and terminal R must be switched off again before you can start the actual work.

If the Check Control message is not displayed, you must not carry out any work on high-voltage components!

5.3.4. Procedure after an accident

The safety concept of the high-voltage system ensures that, even during or after an accident, there is no danger to the customer or the Service Technician. The high-voltage system is automatically deactivated in the event of an accident in such a way that no dangerous voltages are applied at those points on the high-voltage components which are accessible from the outside. Deactivation of the high-voltage system is done as follows:

In normal operation the battery management electronics is supplied via terminal 30F. The coils of the electromechanical switch contactors are also supplied. Deactivation in the event of an accident is done by a deployed battery safety terminal. It contains an additional (normally closed) contact. This switch contact opens when the battery safety terminal is triggered at the same time as the positive battery cable is severed. Opening this switch contact causes the switch contactors in the high-voltage battery unit to open directly so that no more dangerous voltage can be fed into the high-voltage electrical system from the high-voltage battery unit. The electrical machine electronics receives a crash signal from the Crash Safety Module (ACSM). The EME then discharges the link capacitors immediately.

After an accident the battery safety terminal remains in the above-described state such that the high-voltage battery unit is not operational. Thus the high-voltage system remains inactive even if terminal 15 is switched on again.

Before working on the high-voltage components or on the battery safety terminal of an F10H which has been involved in an accident with a triggered battery safety terminal, contact the Technical Support (PUMA) of the BMW Group.

5.3.5. Transport mode

To protect the high-voltage battery unit, the following functions are not available in transport mode:

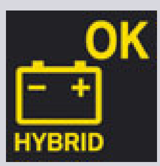
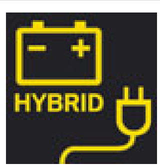
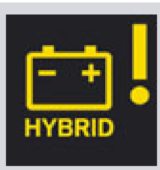
- Electrical driving
- Boost function
- Automatic engine start/stop function

The high-voltage battery unit is always charged in transport mode if the combustion engine is running.

Display of the state of charge

Like with other vehicles, in transport mode the state of charge of the 12 V battery is displayed as a Check Control message. In the F10H (also in transport mode) a Check Control message is displayed for the state of charge of the high-voltage battery unit.

The state of charge of the high-voltage battery unit is displayed in three stages:

Battery condition	Display in instrument cluster	Action
Battery state of charge of the high-voltage battery is OK		No further work necessary
High-voltage battery is flat		Charge high-voltage battery!
High-voltage battery is dead		Replace high-voltage battery unit!

If the high-voltage battery unit was dead and recharged, the display remains in the instrument cluster until the high-voltage battery unit is replaced. After resetting transport mode, there are no Check Control messages on the state of charge of the high-voltage battery unit displayed in the instrument cluster.

5. High-voltage Battery Unit

5.4. Technical data

In contrast to the F04, the high-voltage battery unit in the F10H is supplied by BMW.

The following is a summary of the important characteristics of the high-voltage battery unit:

Nominal voltage	316.8 V
Battery cells	96 at 3.3 V each
Storable amount of energy	1350 Wh
Used amount of energy	600 Wh
Maximum power	43.2 kW
Storage technology	Lithium-ion
Dimensions (width, height, depth)	690 mm x 237 mm x 210 mm
Weight	about 46 kg
Cooling system	Cooling with refrigerant (R134a)

6. Hybrid Brake System

6.1. Introduction

The function of the brake system of the ActiveHybrid 5 is to decelerate the car safely under stable conditions. Like the F04, the F10H can also be decelerated in two different ways:

- Conventional hydraulic braking
- Regenerative braking

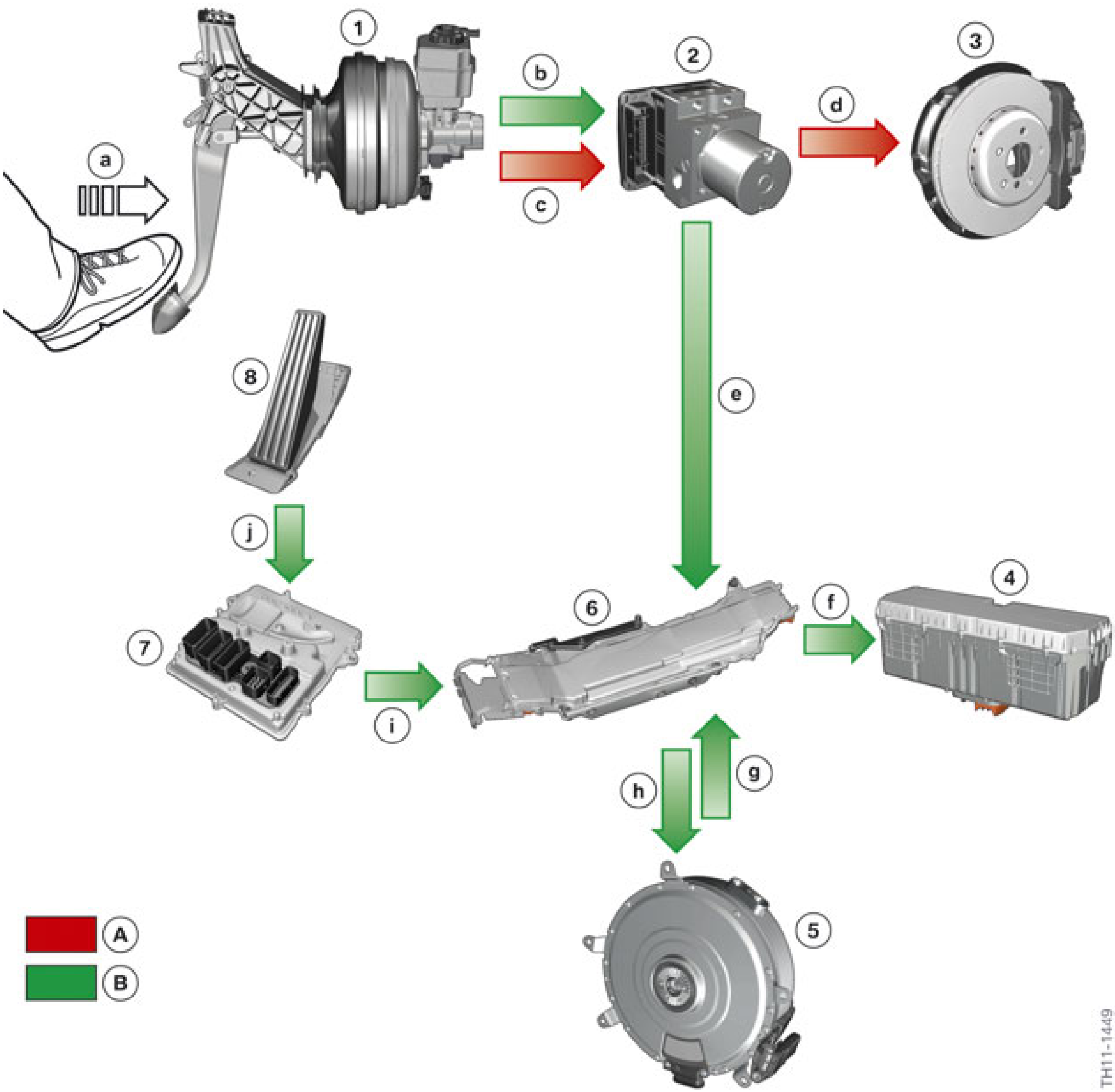
Thanks to energy recovery it is possible to convert the kinetic energy of the vehicle into electrical energy with help of the electrical machine and feed this to the high-voltage battery unit.

Although the service brake of the ActiveHybrid 5 is based on that of a conventional F10, this chapter only the hybrid-specific components and functions are described.

In comparison to the conventional F10, the following new or modified components are used:

- Electric vacuum pump
- Brake vacuum sensor
- Plunger tandem brake master cylinder with brake pedal travel sensor.

6.2. System overview



F10H system overview of hybrid brake system

Index	Explanation
A	Hydraulic braking
B	Regenerative braking
1	Brake pedal and brake booster with brake pedal travel sensor
2	Dynamic Stability Control (DSC)
3	Wheel brakes
4	High-voltage battery unit

Index	Explanation
5	Electrical machine
6	Electrical machine electronics (EME)
7	Digital Engine Electronics (DME)
8	Accelerator pedal module
a	Pressing of the brake pedal
b	Electrical signal "Brake pedal travel" from the brake pedal travel sensor to the Dynamic Stability Control
c	Hydraulic pressure from the brake booster to the DSC
d	Hydraulic pressure from the DSC to the four wheel brakes
e	Data bus message "Target braking torque" from the DSC to the electrical machine electronics
f	Electrical energy to be stored in the high-voltage battery (DC voltage)
g	Electrical energy generated by the electric motor (AC voltage)
h	Phase voltages for activating the electric motor
i	Data bus message "Accelerator angle" from the DME to the electrical machine electronics (energy recovery in coasting (overrun) mode)
j	Electrical signal "Accelerator pedal angle" from the accelerator pedal module to the DME (energy recovery in coasting (overrun) mode)

6.3. Hydraulic braking

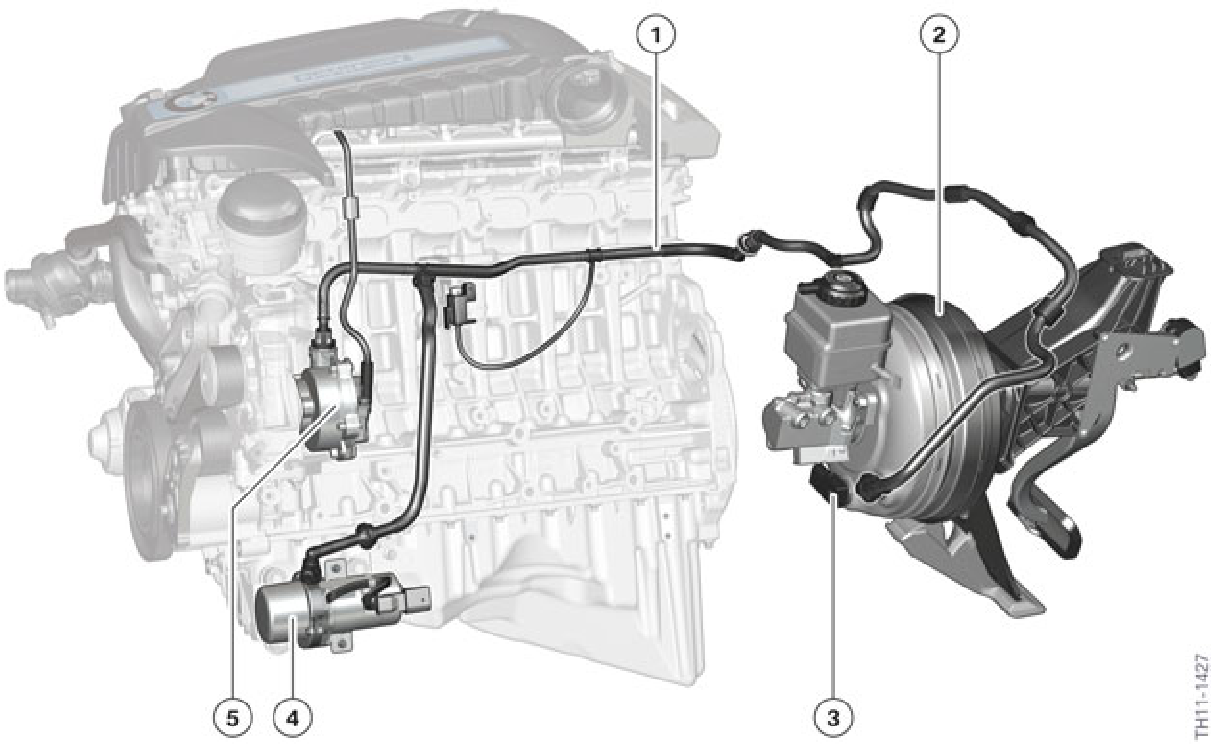
F10H Hydraulic braking

When the driver depresses the brake pedal, a direct mechanical connection is established to the brake servo and thus to the hydraulic brake system. Activation is therefore the same as in a conventional vehicle:

- Brake pedal – mechanical connection – brake servo
- Brake servo – pneumatic boosting – brake master cylinder
- Brake master cylinder – hydraulic boosting and distribution into two brake circuits – Dynamic Stability Control
- Dynamic Stability Control – driving dynamics control and electronic brake force distribution – wheel brakes

The main difference between the service brake of the F10H and the conventional F10 is in the vacuum supply.

F10H Complete Vehicle
6. Hybrid Brake System



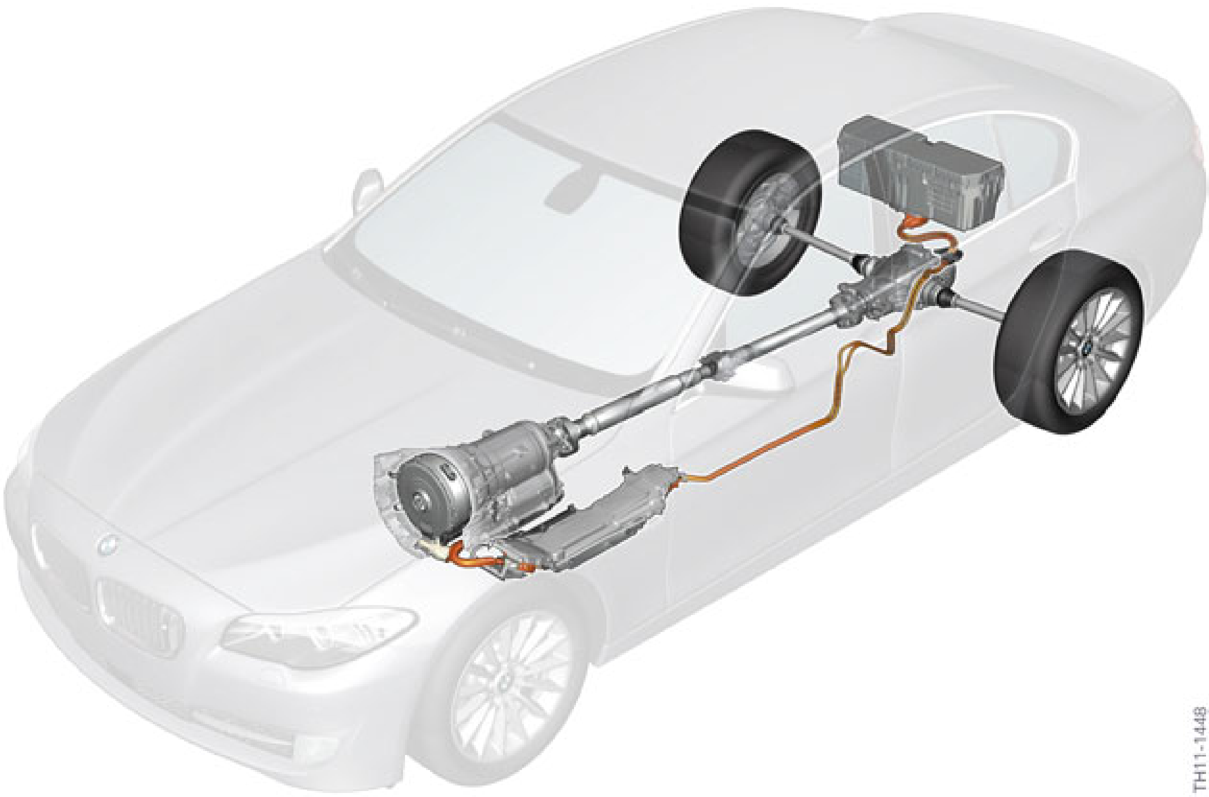
F10H vacuum supply

Index	Explanation
1	Vacuum line
2	Brake booster
3	Brake vacuum pressure sensor
4	Electric vacuum pump
5	Mechanical vacuum pump

In the fully electric driving phases, the combustion engine is off and thus can not activate the mechanical vacuum pump. To ensure the supply of a brake vacuum also in these phases, an auxiliary electric vacuum pump in installed.

The vacuum in the brake booster is measured by the brake vacuum sensor and read in by the DME. The activation and monitoring of the electric vacuum pump is done via the EME.

6.4. Regenerative braking

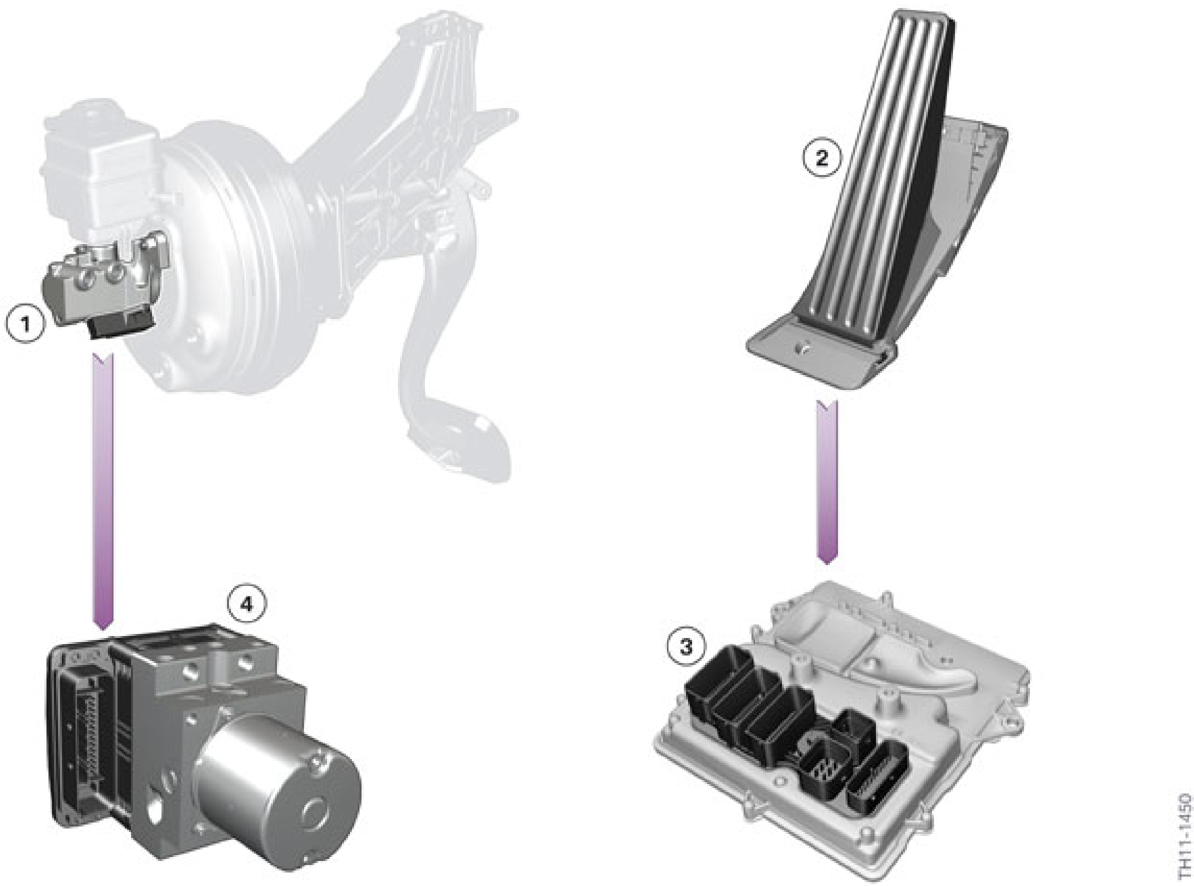


F10H Braking energy regeneration

During regenerative braking the electrical machine works as a generator and brakes with the use of kinetic energy collected from the drivetrain and wheels. The high-voltage battery is then charged by the electrical machine electronics using this energy.

Tandem brake master cylinders are used currently in BMW vehicles. As in the F04, the specially designed “plunger tandem brake master cylinder” is also used in the F10H. It offers advantages such as, for example, more compact dimensions. It was also chosen for the F10H because the brake pedal travel sensor could be easily integrated in this structure. The mechanical structure and the measurement principle of the brake pedal travel sensor of the tandem brake master cylinder in the F10H is the same as that in the F04.

In addition, the response path has been enlarged to extend the pedal travel for regenerative braking.



F10H components, input signals of brake energy regeneration

Index	Explanation
1	Tandem brake master cylinder with brake pedal travel sensor
2	Accelerator pedal module
3	Digital Engine Electronics (DME)
4	Dynamic Stability Control (DSC)

The accelerator pedal angle and the brake pedal travel are decisive input variables for regenerative braking.

- The brake pedal travel is measured by the brake pedal travel sensor and read in by the DSC.
- The accelerator pedal angle is measured by the accelerator pedal module and read in by the DME.

The electric motor is already operated as a generator when the brake pedal is not depressed but the accelerator pedal is already at an angle of zero degrees. The electric motor electronics activates the electric motor in such a way that a brake force is obtained for the complete vehicle which corresponds to a conventional vehicle in overrun mode. However the extent of energy regeneration (at this point) is still at a low level.

If now the brake pedal is depressed, initially (as in every conventional brake system) some free play (response travel) in the travel must be overcome during which still no hydraulic braking occurs. Already here, however, the brake pedal travel is evaluated and a greater brake force than that in pure overrun mode is generated with the aid of the electric motor.

If the brake pedal is depressed beyond the “response travel”, both brake modes are simultaneously active, because now hydraulic braking is added to brake energy recovery. The brake force generated by the electric motor is increased further as the brake pedal travel increases until it reaches the maximum.

6.4.1. Special states

Regenerative braking with help of the drive train only effects the rear axle of the F10H. The brake force on the rear axle must not exceed a specific value in proportion to that on the front axle. This would otherwise compromise driving stability. For this reason too the maximum deceleration which can be achieved through brake energy recovery is limited.

The maximum permissible brake force by brake energy recovery is subject to stability monitoring of slip, lateral accelerations and stability control processes. It is thus guaranteed that the vehicle constantly remains in a stable driving condition also during brake energy regeneration.

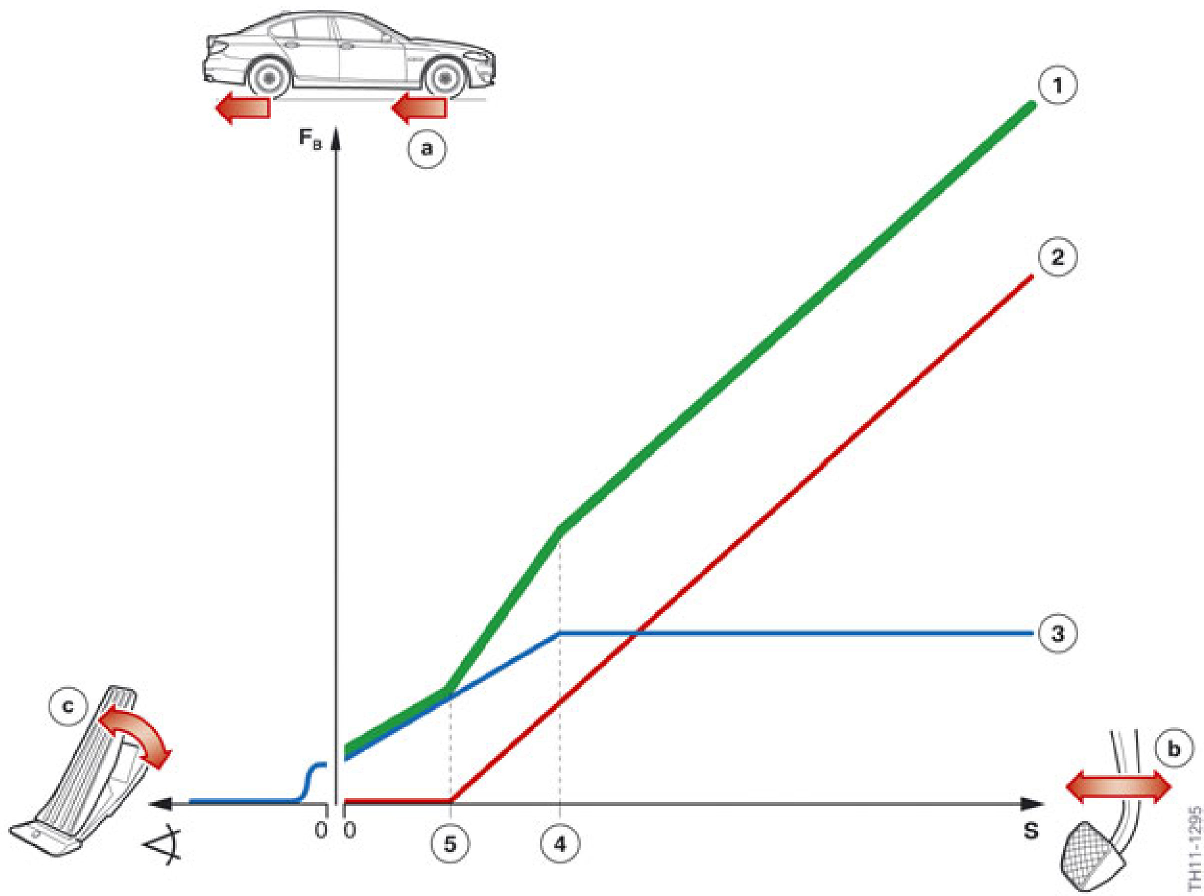
The DSC hereby contributes to fuel savings while taking active safety into consideration.

If the high-voltage battery is already fully charged, it cannot store any more electrical energy. This special state may be a reason why brake energy regeneration cannot be possible, but which arises rarely. A sufficiently large reserve in the state of charge of the high-voltage battery is maintained during normal driving by the operational strategy. In other words, intentionally energy is again and again drawn from the high-voltage battery and thereby the state of charge is maintained in a range which makes available (even during longer instances of downhill driving) capacity for the electrical energy generated during regenerative braking.

6.5. Distribution of hydraulically and regeneratively generated brake force

In generator mode the electric motor generates a braking torque in the drivetrain. This braking torque causes a brake force on the rear wheels.

The following diagram summarizes how the entire brake force is distributed into the hydraulic and regenerative modes. In the diagram it is presupposed that there is no unstable driving state and the high-voltage battery is able to store electrical energy.



F10H Diagram for distributing the brake force

Index	Explanation
a	Brake force on the wheels
b	Brake pedal travel
c	Accelerator pedal angle
1	Total brake force
2	Hydraulically generated brake force
3	Regeneratively generated brake force
4	Brake pedal travel at which the maximum possible regenerative brake force acts
5	Brake pedal travel at which the hydraulic brake force begins (end of response travel)

F10H Complete Vehicle

7. Low-voltage Electrical System

7.1. Voltage supply

The 12 V electrical system of the F10H is essentially the same as that of the F10. The main difference lies in the fact that the energy is no longer supplied by the generator, but by the high-voltage electrical system. The high voltage of the high-voltage battery is converted to a lower voltage (about 14 V) using a DC/DC converter. The electric voltage supply of the 12 V vehicle electrical system is thus no longer dependent on the engine speed of the combustion engine when driving.

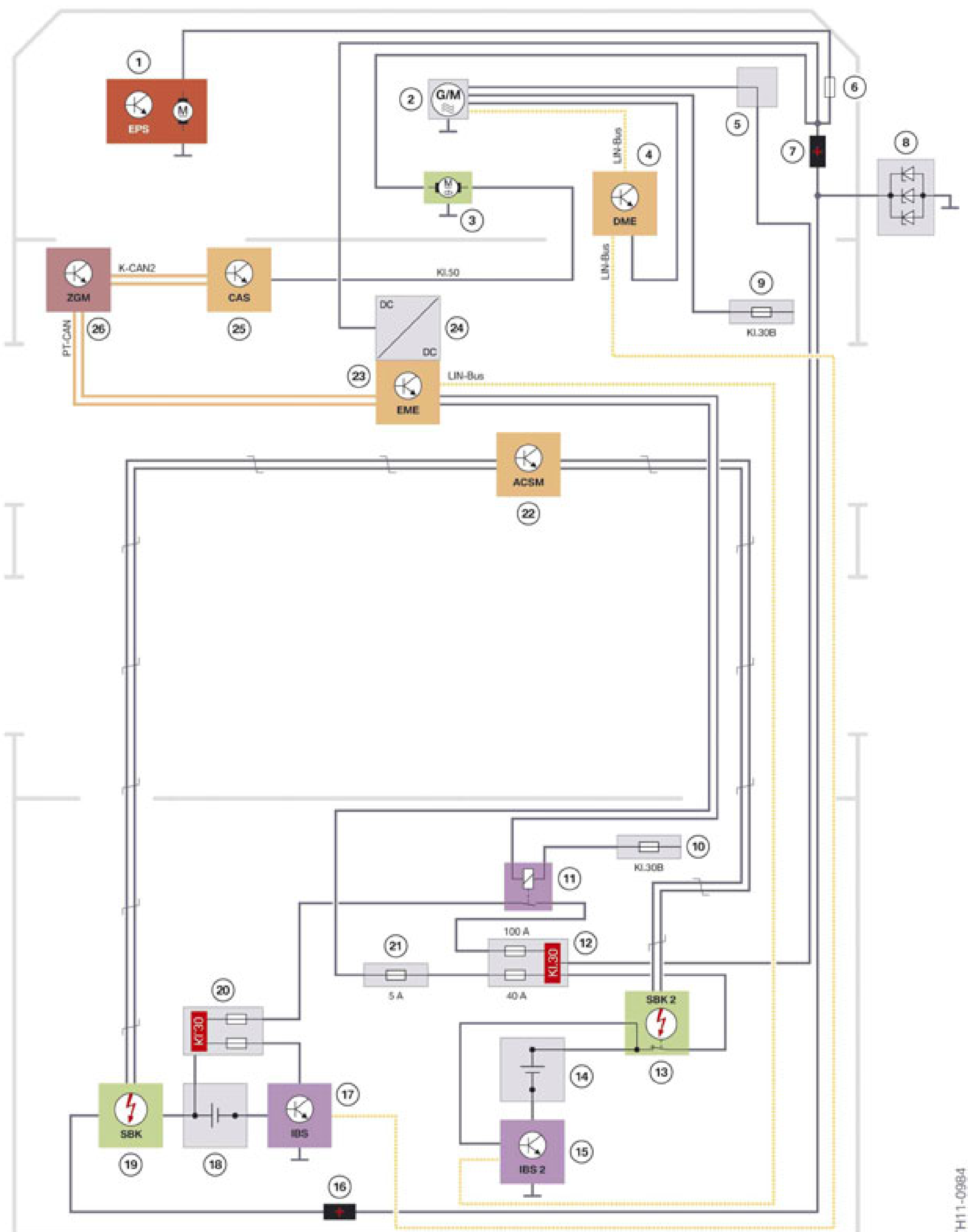
The second difference is that a new component with new functions (the starter motor generator) is being used. The starter motor generator can (as the name suggests) operate as both a starter motor or a generator. In motorized operation (as a starter motor), it allows us to start of the combustion engine (when at operating temperature) not only at standstill but also during electric driving. The energy required to start the combustion engine is taken from the auxiliary battery. The initial start of the combustion engine is effected as before via the conventional starter motor.

If the combustion engine is running, the starter motor generator works in generator mode like a traditional alternator. However, the energy generated here is stored in the auxiliary 12V battery.

The starter motor generator and the auxiliary battery form an independent 12 V vehicle electrical system. In certain situations such as in the event of a malfunction with the hybrid system for example, the starter motor generator can assume the energy supply of the standard 12 V vehicle electrical system. In this case the two 12 V vehicle electrical systems are connected via a cut-off relay.

7. Low-voltage Electrical System

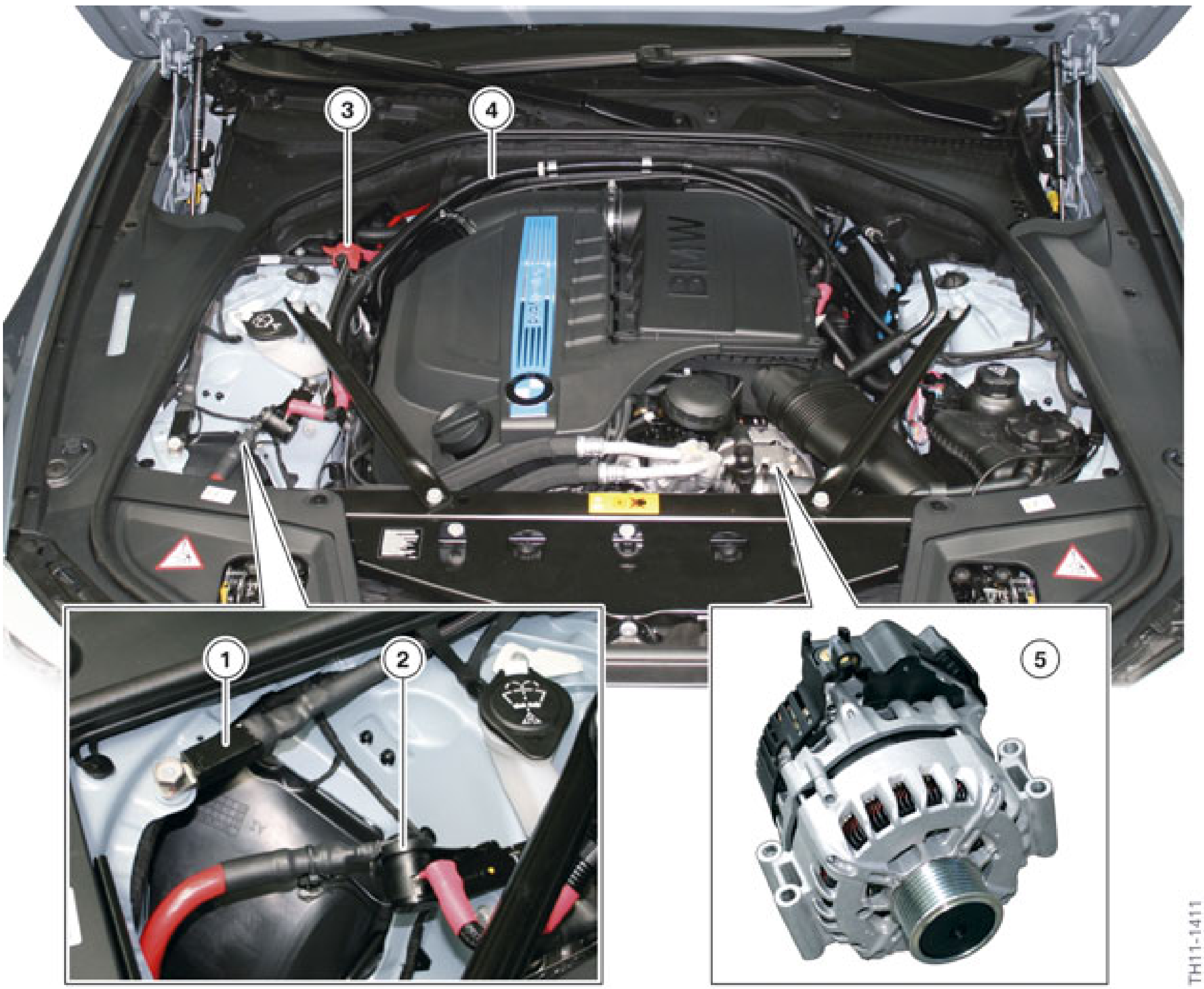
7.1.1. System overview



F10H 12 V voltage supply system wiring diagram

Index	Explanation
1	Electric Power Steering (EPS)
2	Starter generator
3	Starter motor
4	Digital Engine Electronics (DME)
5	Power distribution box, engine compartment
6	Fuse for Electric Power Steering
7	Jump start terminal point
8	Reverse polarity protection module
9	Fuse, terminal 30B in junction box
10	Fuse, terminal 30B in junction box, rear
11	Cut-off relay
12	Power distribution box of auxiliary battery
13	Safety battery terminal 2
14	Auxiliary battery
15	Intelligent battery sensor 2
16	Transfer support point (change of lines in the vehicle to lines at the vehicle underbody)
17	Intelligent battery sensor
18	12 V battery
19	Safety battery terminal
20	Power distribution box on the 12 V battery
21	Fuse of line to EME (5 A)
22	Crash Safety Module
23	Electrical machine electronics
24	DC/DC converter
25	Car Access System
26	Central gateway module

7.1.2. Reverse polarity protection module



Components of the voltage supply in the engine compartment

Index	Explanation
1	Reverse polarity protection module
2	Power distribution box, engine compartment
3	Jump start terminal point
4	Positive battery cable for the starter motor generator
5	Starter generator

The reverse polarity protection is used to avoid subsequent damage to the vehicle electrical system and connected electronic components for polarity reversal, for example during a charging procedure of the 12 V battery via an external charger. The diodes in the alternator are used to fulfill this task in conventional vehicles with a combustion engine.

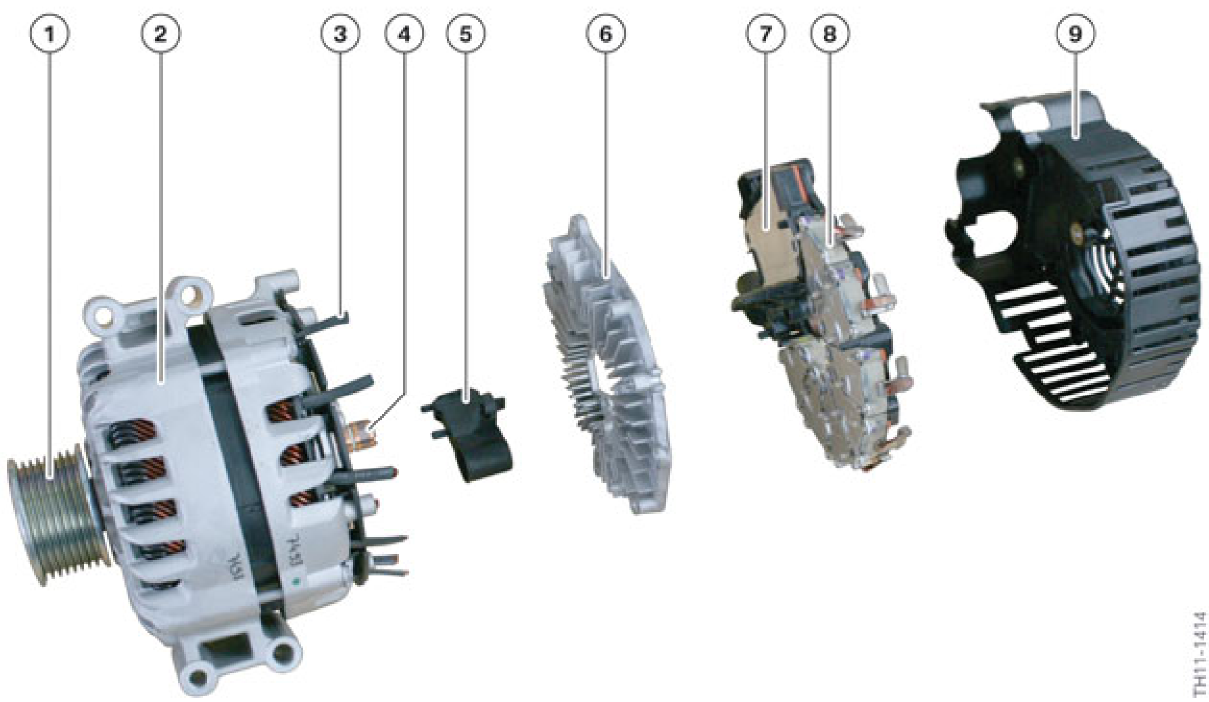
As there is no conventional alternator in the F10H, the reverse polarity must be carried out by a new component, the reverse polarity protection module. The reverse polarity protection module is installed in the engine compartment near the right headlight. It is connected on one side to the battery positive wire and on the other to ground. Inside the reverse polarity protection module are three Zener diodes which are able to restrict the reverse voltage for a certain period. In the event of longer polarity reversal, they may suffer damage without damaging neighboring components.

7.2. Start-up system

The start-up system comprises several components.

7.2.1. Starter generator

The starter motor generator is an electrical machine which can be operated as a motor or a generator. In generator mode the starter motor generator is driven by the combustion engine via a drive belt. The electrical energy generated here is stored in the auxiliary battery. This operation corresponds to the operation of the conventional alternator. The starter motor generator can start the combustion engine (at operating temperature) without causing a voltage dip in the 12 V vehicle electrical system. The energy required for starting the engine in this case is taken from the auxiliary battery.



Structure of the starter motor generator

Index	Explanation
1	Belt pulley
2	Housing of the starter motor generator with stator windings and rotor
3	Connection of one of seven stator windings
4	Slip rings
5	Carbon brushes for excitation current

Index	Explanation
6	Heat sink for performance electronics
7	Control module with Local Interconnect Network connection to the DME
8	One of seven modules of the power electronics
9	Protective cover with air openings

The design of the starter motor generator is based on the salient-pole machine (claw pole machine) with link claw magnets. The permanent magnets in the rotor ensure a higher starting torque. To optimize performance there is a direct current winding with separate excitation in the rotor. The copper wire of the stator windings has a square cross-section. A higher copper fill factor could be achieved with the square cross-section as opposed to the round one. The stator and the rotor are coordinated especially to the starter motor function (torque generation).

The control module has a similar function to the alternator controller in a traditional vehicle. In addition, the control module assumes control of the power electronics in the starter motor generator. During the start-up the control module checks the rotor magnetization, the angle of rotation and the rotational speed of the starter motor generator using a rotor position sensor in the housing. As soon as the control module identifies that the engine start is realized, the generator mode is used. The DME specifies the nominal voltage and the excitation current via local interconnect network bus to the control module. The three phase AC current generated in generator mode is converted by seven power electronics modules (MOSFET switches) to direct current.

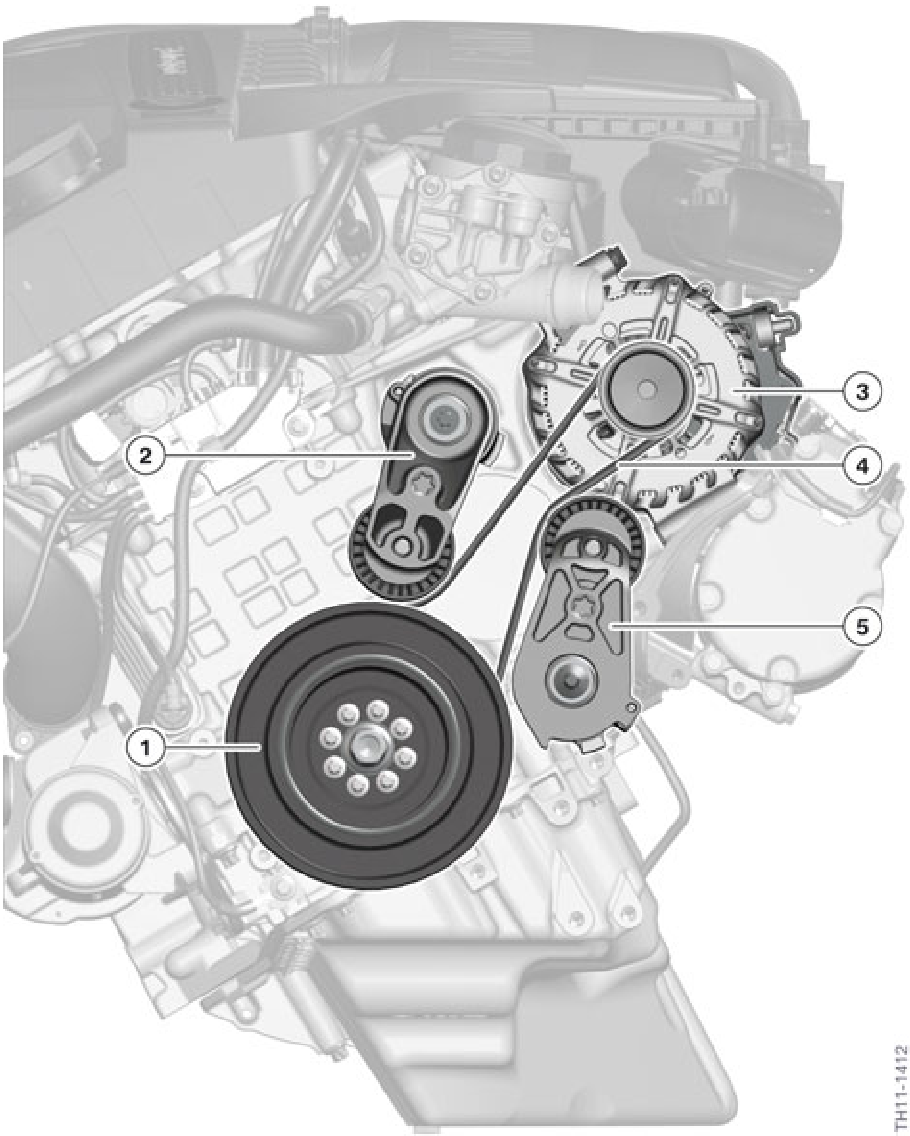
The starter motor generator can operate up to a temperature of 130 °C (266 °F). The temperature measurement is done using a temperature sensor located in the inside of the starter motor generator.

The starter motor generator is secured to the engine block using four bolts. The top bolts are 85 mm long and the rear bolts 105 mm.

The electrical power of the starter motor generator is 2.8 kW (14 V x 200 A). The manufacturer of the starter motor generator is Valeo.

Belt drive

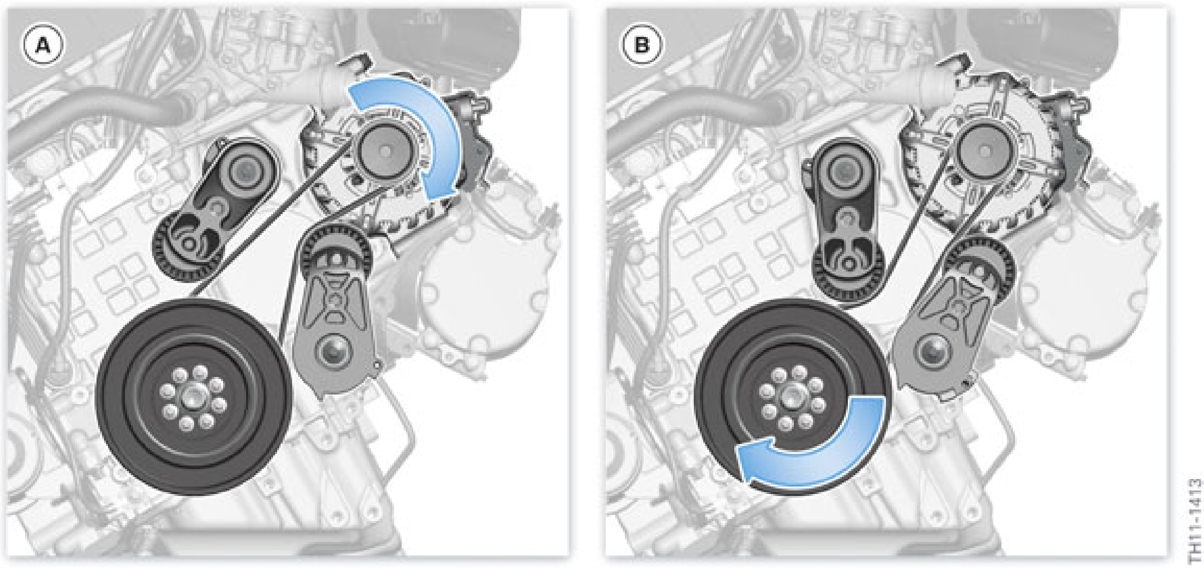
By using the electric A/C compressor and the electric power steering, the starter motor generator is the only ancillary component in the belt drive of the N55 engine in the F10H.



Belt drive of the N55 engine in the F10H

Index	Explanation
1	Torsional vibration damper with belt pulley
2	Top belt tensioner
3	Starter generator
4	Ribbed V-belt
5	Lower belt tensioner

As the starter motor generator can be operated as a both a generator and a motor, the belt must be prevented from slipping in each operating situation. For this reason a belt tensioner is fitted at the top and bottom of the drive belt.



Generator mode and starter motor operation of the starter motor generator

Index	Explanation
A	Motor operation (starting the combustion engine)
B	Generator mode (combustion engine is running and auxiliary battery is charging)

If the combustion engine is to start-up at operating temperature, the starter motor generator is operated as a motor where the auxiliary battery has been adequately charged. The DME provides the necessary information to the control module in the starter motor generator via a local interconnect network bus and the rotor in the starter motor generator begins to rotate. The belt pulley is also set in a rotational movement during the process whereby the direction of rotation is fixed by the design of the combustion engine. BMW engines are right-turning (clockwise) engines. If one looks at the engine from the front, the crankshaft and thus also the torsional vibration damper rotate clockwise. The top part of the drive belt is tensioned by the power transmission and the lower part of the drive belt is slackened. To prevent the drive belt slipping, it is tensioned using the lower belt tensioner. The belt tensioner is mechanically pre-tensioned by a torsion spring and presses the drive belt inwards and tensions it. The engine start via the starter motor generator is done very quickly, silently and with very few vibrations.

As soon as the control module identifies that the engine started, the generator mode of the starter motor generator is used. The torsional vibration damper rotates to the right and transfers the force over the drive belt to the belt pulley of the starter motor generator. The bottom part of the drive belt is tensioned in the process. Completely opposite to the motor operation, here the top part of the drive belt must be tensioned. This is done using the top belt tensioner. If the combustion engine is running, a choice can be made between generator and neutral mode. In generator mode the starter motor generator supplies current; in neutral mode it rotates load-free and does not supply any current.

Starting the combustion engine

The fact that the conventional starter motor, the starter motor generator and the electrical machine are installed in the F10H means the combustion engine has three different start-up systems.

The cold start of the engine is done via the conventional starter motor by pressing the START-STOP button. The energy required (in this case) is taken from 12 V battery of the standard vehicle electrical system. As the F10H is a hybrid vehicle, it should be noted here that there may possibly be no engine start after pressing the START-STOP button because for example the combustion engine is still at operating temperature and the state of charge of the high-voltage battery is sufficient. This means after the START-STOP button is pressed, the driving readiness is established and the vehicle can be driven off using the electrical machine.

The renewed engine start is done via the starter motor generator after the driving readiness has been established. Here the combustion engine is started both after an engine stop phase and also during electric driving with the corresponding torque requirement. The energy required to start the combustion engine is taken from the auxiliary battery of the starter motor generator.

In the event of a malfunction with the starter motor generator, the combustion engine is started via the electrical machine. The electrical machine takes the energy required for this from the high-voltage battery.

In emergency situation the system works as follows:

- If the combustion engine does not start via the conventional starter motor despite activation, a fault code entry is made and a switch is made to start the combustion engine using the starter motor generator.
- If the starter motor generator fails, a switch is made to start the combustion engine via the electrical machine.
- If an engine start is also not possible via the electrical machine, a Check Control message is displayed.

A new attempt to start the engine with all three start systems is possible again after the terminal change.

In the cold (at the latest less than -10 °C/ 14 °F), a successful engine start via the starter motor generator cannot be expected. The power of the auxiliary battery and the power transfer via the drive belts are not sufficient at these temperatures. An engine start via the electrical machine may not always work at these temperatures because the high-voltage battery may no longer be able to provide the necessary power.

Unlike the E72 and F04, the combustion engine in the F10H is no longer started with the assistance of the high-voltage battery, but instead with the 12 V battery, except in an emergency situation. This means the two 12 V batteries must have enough energy for the start-up of the combustion engine (sufficient state of charge). As the high-voltage battery does not have to provide any reserves (for weeks) for the engine start, a higher energy withdrawal from the high-voltage battery is possible.

Service information

A faulty starter motor generator cannot be repaired, it must be replaced as complete unit.

To establish a fault with the starter motor generator, the auxiliary battery (state of charge or state of health), cables (resistance), connections of cables (contact resistance), fit and tension of drive belt, grounding cable and the local interconnect network bus connection must first be checked. The starter motor generator should only be replaced if all these checks have failed.

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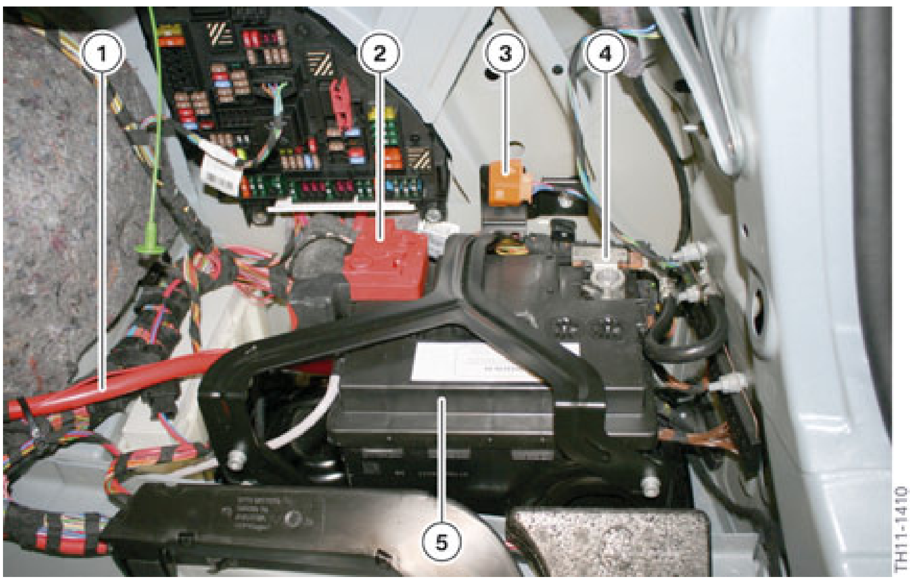
Always proceed according to the repair instructions for the removal and installation of the starter motor generator!

7.2.2. Auxiliary battery



Auxiliary battery

Index	Explanation
1	Positive battery cable for the power distribution box
2	Safety battery terminal of the auxiliary battery SBK2
3	High-voltage safety connector (Service Disconnect)
4	Intelligent battery sensor (auxiliary battery) IBS2
5	Auxiliary battery

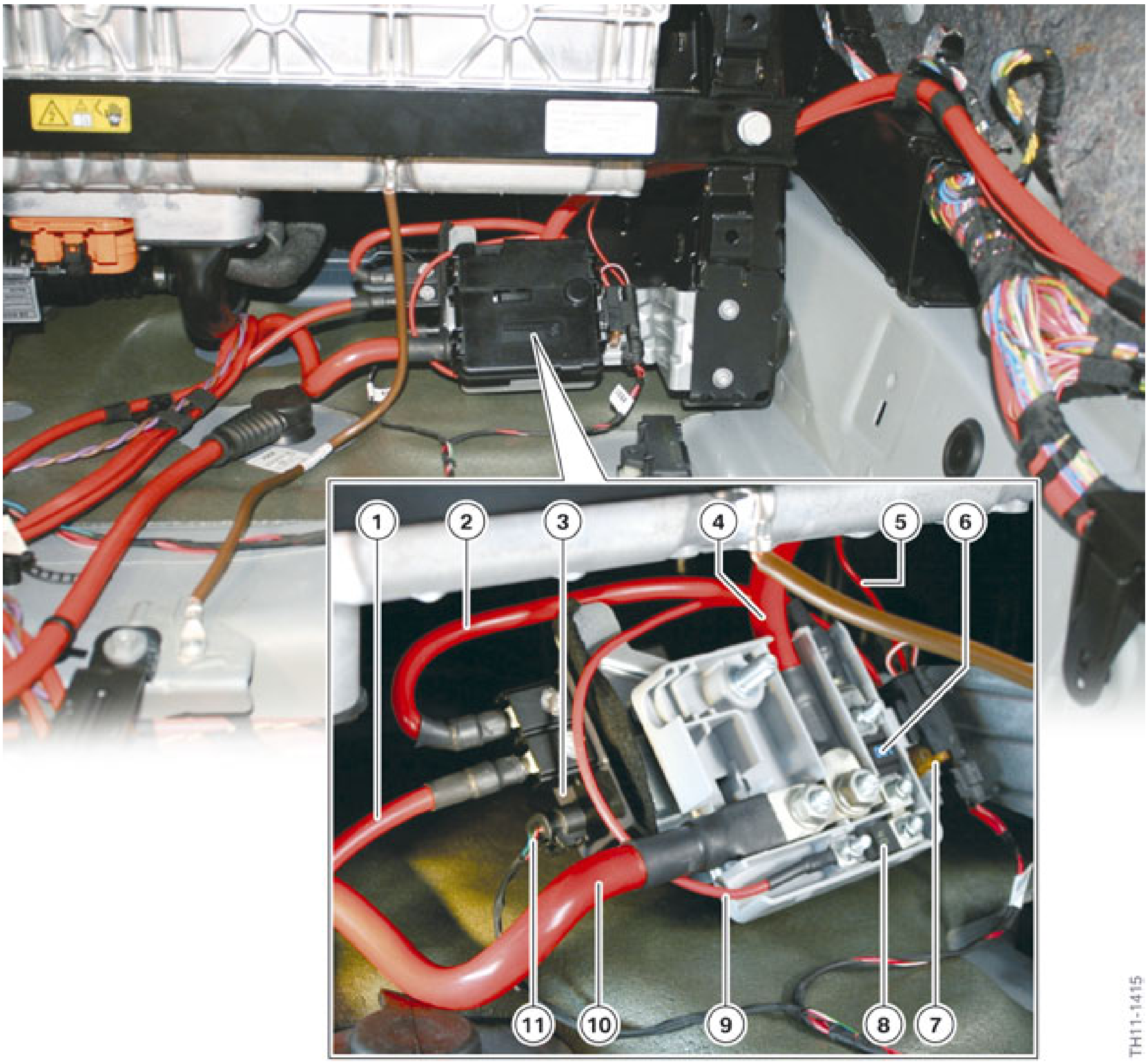
The electrical energy generated by the starter motor generator is mainly stored in the auxiliary battery. It is installed in the luggage compartment on the right. The auxiliary battery is a 50 Ah AGM battery.

Similar to the 12 V battery, the current, voltage and pole temperature of the auxiliary battery are measured by an intelligent battery sensor IBS2. The data are then forwarded via local interconnect network bus to the master control unit, the electrical machine electronics (EME).

In the event of an accident of sufficient severity, a safety battery terminal SBK2 ensures the disconnection of the positive battery cable between the auxiliary battery and starter motor generator. This safety battery terminal SBK2 is located directly at the positive terminal of the auxiliary battery. The pyrotechnic activation of the safety battery terminal for the auxiliary battery is done by the Crash Safety Module (ACSM).

A change of the auxiliary battery must always be registered.

7.2.3. Power distribution box and cut-off relay



Power distribution box and cut-off relay

Index	Explanation
1	Positive battery cable between power distribution box of the 12 V battery and cut-off relay
2	Positive battery cable between power distribution box of the auxiliary battery and cut-off relay
3	Cut-off relay
4	Positive battery cable between auxiliary battery and power distribution box
5	Line to the EME
6	100 A fuse

Index	Explanation
7	5 A fuse
8	40 A fuse
9	Line from the 40 A fuse to the 5 A fuse
10	Positive battery cable for the starter motor generator
11	Control line for cut-off relay

The intelligent battery sensor IBS2 receives the voltage supply via a line with a small cross-section from the safety battery terminal SBK2 of the auxiliary battery. The positive battery cable of the auxiliary battery is also fed from the safety battery terminal SBK2 of the auxiliary battery. Other lines also lead from this power distribution box to the starter motor generator, cut-off relay and EME. The power distribution box is not a new development, but has already been installed in some other models.

The positive battery cable runs from the power distribution box of the auxiliary battery to the starter motor generator. This positive battery cable has a cross-section of 110 mm^2 and is not secured against overload current.

A cut-off relay is installed on the bottom side of the power distribution box. The two 12 V vehicle electrical systems are connected via this cut-off relay. If for example the auxiliary battery is fully charged and the engine speed of the combustion engine increases to over 2500 revolutions per minute, the two 12 V vehicle electrical systems are connected. The activation of the cut-off relay is carried out by the EME. Before the cut-off relay is closed, the voltages in both 12 V vehicle electrical systems are adjusted. The DME defines the value of the nominal voltage and sends it to the starter motor generator. Any possible equalization currents are minimized putting less stress on the switch contacts of the cut-off relay.

As a result of the installed link claw magnets in the rotor, at engine speeds over 2500 rpm the starter motor generator induces (even in neutral mode) a voltage which leads to the undesired charging of the auxiliary battery. To avoid damage to the starter motor generator and the auxiliary battery, the cut-off relay is always closed from this speed on. In this case the excess electrical energy is fed to the conventional 12 V vehicle electrical system.

The cut-off relay is opened by a start-up process in order to avoid a voltage dip in the conventional 12 V vehicle electrical system.

In the event of a malfunction with the hybrid system the starter motor generator can assume the supply of the network consumers which is normally done by the DC/DC converter in the EME. Many functions such as seat heating for example are deactivated.

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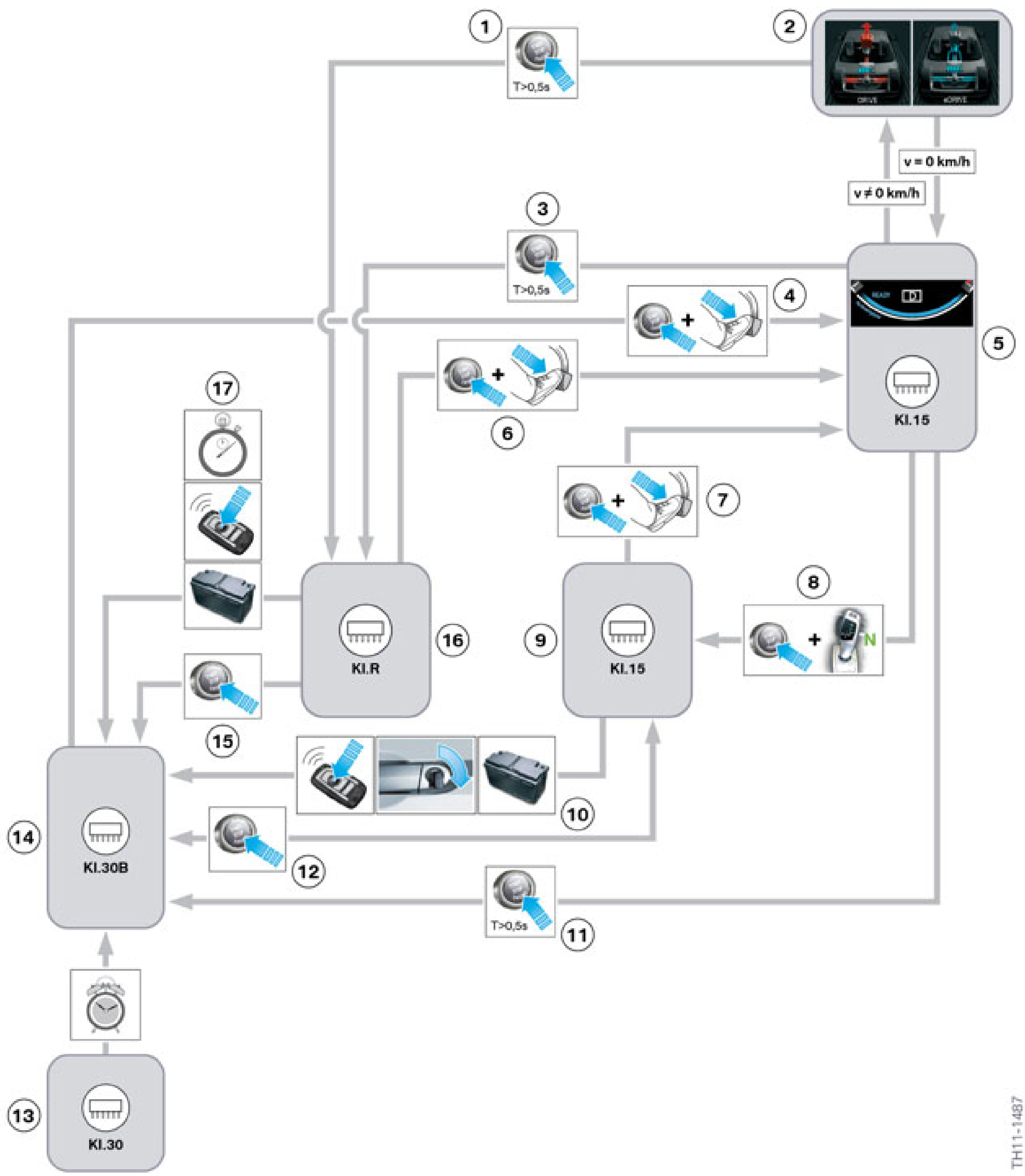
Note: Under no circumstances can the standard 12 V electrical system supply voltage to the auxiliary 12 V (50 Ah) battery.

There are two fuses in the power distribution box of the auxiliary battery. The 100 A fuse secures the positive battery cable from the power distribution box to the cut-off relay. The power distribution box is a component from the "module" which is why the 40 A fuse is always installed. The line to the EME has a smaller cross-section however, and this is why a 5 A fuse is connected in series.

7.3. Terminal control

7.3.1. "Ready to drive" mode



Terminal control in the F10H from driver's view

Index	Explanation
1	When the START-STOP button is pressed for a long time the terminal status changes from "Terminal 15 driving" to "Terminal R".
2	Driving with assistance of the combustion engine or the electrical machine
3	When the START-STOP button is pressed for a long time the terminal status changes from driving readiness to terminal R.
4	When the START-STOP button and the brake pedal are pressed simultaneously, the driving readiness is activated (with or without start-up of the combustion engine) **from terminal 30B**
5	Driving readiness with activated terminal 15
6	When the START-STOP button and the brake pedal are pressed simultaneously, the driving readiness is activated (with or without start-up of the combustion engine) **from terminal R**
7	When the START-STOP button and the brake pedal are pressed simultaneously, the driving readiness is activated (with or without start-up of the combustion engine) **from terminal 15**
8	When the selector lever is at "N" and driving readiness is ended with the START-STOP button, terminal 15 remains switched on for 15 minutes (car wash function)
9	Terminal 15
10	Terminal 15 is switched off if the vehicle has been locked or the state of charge of the battery is too low. (Vehicle then goes to terminal 30B)
11	When the START-STOP button is pressed for a long time the terminal status changes from driving readiness to terminal 30B.
12	When the START-STOP button is pressed the terminal status changes between terminal 15 and terminal 30B
13	Terminal 30
14	Terminal 30B
15	When the START-STOP button is pressed the terminal status changes from terminal R to terminal 30B.
16	Terminal R
17	Change from terminal R to terminal 30B if more than eight minutes have passed or the car has been locked or the state of charge of the vehicle battery is too low.

The first driving readiness is activated when the brake pedal and the START-STOP button are pressed simultaneously. Here the activation of the driving readiness is carried out by each terminal status (terminal 30B, terminal R and terminal 15). In subsequent driving the driving readiness is automatically reached at each vehicle stop. Always when the vehicle comes to a standstill, for example in traffic or at lights, and the state of charge of the high-voltage battery is sufficient, is the driving readiness activated. The activate driving readiness is indicated to the driver by the illumination of "READY" in the bottom part of the rev counter.

With the status "Driving readiness" the vehicle can be driven off using electrical or a combustion engine depending on the torque requirement. In comparison to the conventional vehicle with a single powertrain by the combustion engine, the driving readiness in a hybrid car cannot recognized at the running combustion engine. The requirements for non-start of the combustion engine, the so-called "silent start", are a sufficiently charged high-voltage battery and a combustion engine at operating temperature.

The driving readiness is deactivated by pressing the START-STOP button when the vehicle is stationary. The drive position "P" is automatically engaged in the process. The car wash function is an exception to this: If the driver engages the drive position "N" when the driving readiness is switched on and then presses the START-STOP button, the drive position "N" remains engaged and terminal 15 remains switched on.

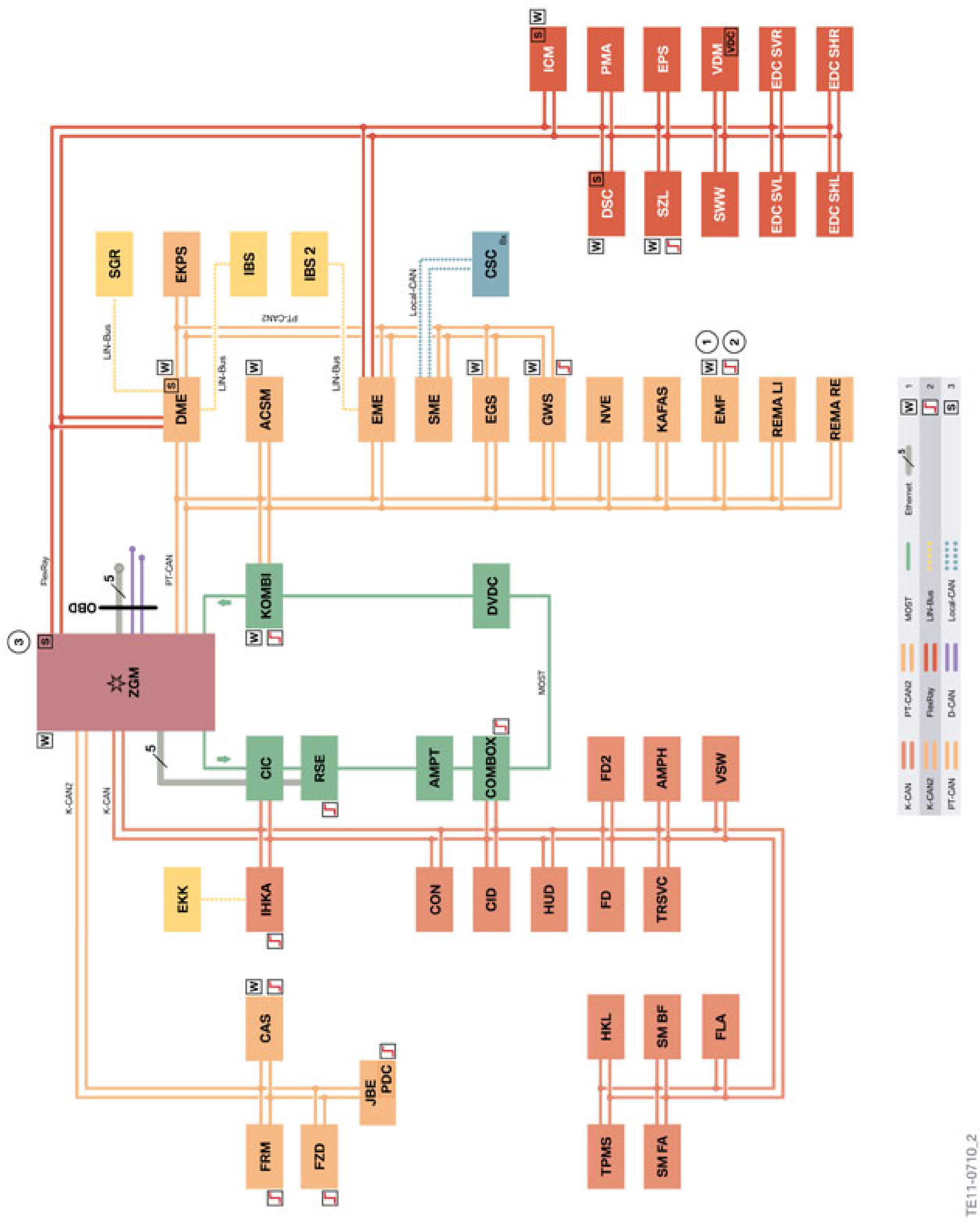
8. Bus Systems

The bus systems of the F10H are based on the bus systems of the F10. Thus all the main and sub-bus systems of the F10 are also used in the F10H. Compared with the bus systems of the F10 some new control units have been added, some have had to be adapted and some are not installed in the F10H at all. The resulting bus structure of the F10H is as follows.

F10H Complete Vehicle

8. Bus Systems

8.1. Bus overview



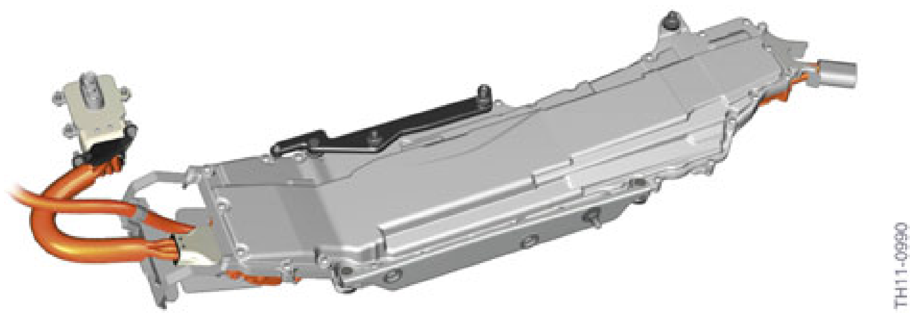
F10H bus overview

Index	Explanation
1	Wakeable control units
2	Control units authorized to perform wake-up function
3	Start-up nodes: Control units for start-up and synchronization of the FlexRay bus system
ACSM	Advanced Crash Safety Module
AMPH	Amplifier High (high fidelity amplifier)
AMPT	Amplifier Top (top high fidelity amplifier)
CAS	Car Access System
CID	Central information display
CIC	Car Information Computer
COMBOX	Combox telematics w/emergency call, Multimedia Combox
CON	Controller
CSC	Cell supervision circuit
D-CAN	Diagnosis-on-Controller Area Network
DME	Digital Engine Electronics
DSC	Dynamic Stability Control
DVDC	DVD changer
EDC SHL	Electronic Damper Control satellite, rear left
EDC SHR	Electronic Damper Control satellite, rear right
EDC SVL	Electronic Damper Control satellite, front left
EDC SVR	Electronic Damper Control satellite, front right
EGS	Electronic transmission control
EKK	Electrical refrigerant compressor
EKPS	Electronic fuel pump control
EME	Electrical machine electronics
EMF	Electromechanical parking brake
EPS	Electric Power Steering
Ethernet	Cable-based data network technology for local data networks
FD	Rear compartment display
FD2	Rear compartment display 2
FLA	High-beam assistant
FlexRay	Fast, real-time and fault-tolerant bus system for use in automotive sector
FRM	Footwell module
FZD	Roof function centre
GWS	Gear selector switch

Index	Explanation
HKL	Automatic operation of tailgate
HUD	Head-Up Display
ICM	Integrated Chassis Management
IBS	Intelligent battery sensor
IBS 2	Intelligent battery sensor 2
IHKA	Integrated automatic heating / air-conditioning system
JBE	Junction box electronics
KAFAS	Camera Assisted Driver Assistance Systems
K-CAN	Body controller area network
K-CAN2	Body controller area network 2
KOMBI	Instrument cluster
LIN-Bus	Local interconnect network bus
Local CAN	Local Controller Area Network
MOST	Media Oriented System Transport
NVE	Night Vision Electronics
OBD	Diagnostic socket
PDC	Park Distance Control
PMA	Parking Maneuvering Assistant
PT-CAN	Powertrain controller area network
PT-CAN2	Powertrain controller area network 2
REMAFA	Reversible electric-driven reel, left (Not for US)
REMABF	Reversible electric-driven reel, right (Not for US)
RSE	Rear Seat Entertainment
SGR	Starter generator
SMBF	Seat module, passenger
SME	Battery management electronics
SMFA	Driver's seat module
SWW	Blind Spot Detection
SZL	Steering column switch cluster
TPMS	Tire Pressure Monitoring System
TRSVC	Control unit for reversing camera, Top View and Side View
VDM	Vertical Dynamics Management
VSW	Video switch
ZGM	Central gateway module

8.2. New control units

8.2.1. Electrical machine electronics (EME)



Electrical machine electronics

The function of the electrical machine electronics is to activate and regulate the permanently excited synchronous machine in the high-voltage electrical system. This requires the use of a bidirectional inverter which converts the high-voltage DC voltage of the high-voltage battery into a three-phase alternating current for the electrical machine. When the electric motor is running in generator mode the high-voltage battery is recharged via the inverter.

The EME also incorporates the DC/DC converter which is responsible for the power supply to the low-voltage electrical system. The EME is connected to the PT-CAN, PT-CAN2 and FlexRay.

8.2.2. Battery management electronics (SME)

The battery management electronics control unit (SME) is integrated in the high-voltage battery and is not accessible from the outside. To maximize the service life of the high-voltage battery, the SME ensures that it is operated in a precisely defined range. Other tasks of this control unit include start-up and shut-down of the high-voltage system, safety functions (for example high-voltage interlock loop, isolation monitoring) and specification of the available power of the high-voltage battery. The battery management electronics communicates with other control units via PT-CAN2.

8.2.3. Electric A/C compressor (EKK)

As in the E72 and F04, the F10H is equipped with an electric A/C compressor. To be able to provide the necessary power, the electric A/C compressor (EKK) is operated with high voltage. The EKK makes it possible to operate the air-conditioning refrigerant circuit in all driving situations. In addition to cooling the passenger compartment, the refrigerant circuit is also used to cool the high-voltage battery. The electric A/C compressor control unit is mounted on the A/C compressor housing and is connected via the LIN-bus to the integrated automatic heating/air-conditioning (IHKA).

8.2.4. Starter motor generator (SGR)

Starter generator

The starter motor generator is responsible for starting of the combustion engine mainly when either the automatic engine start-stop function is active or while electric driving. In addition, the starter motor generator charges the auxiliary battery from which the energy is then supplied for the start-up. The starter motor generator is connected to the DME via LIN-bus.

8.2.5. Intelligent battery sensor 2

The intelligent battery sensor 2 monitors the current, voltage and the pole temperature of the auxiliary battery. The results are forwarded via local interconnect network bus to the EME.

8.2.6. Reversible electromotive automatic reel (REMAFA and REMABF)

Although not available for the US market at the time of the launch, the reversible electromotive automatic reels were introduced in all F10 (EU) vehicles from autumn 2011 and now are also in the F10H vehicles. They are used on both front seatbelts (REMAFA and REMABF) and are part of the Pre-Crash function. The reversible electromotive reel reduces the belt slack when fastening the seat belt with less retracting force as soon as the driver drives off past 10 km/h (6 mph). Removing the belt slack guarantees that the seat belt is on the driver or front passenger and thus in the event of an accident ensures a better restraining effect can be achieved.

Another advantage of the reversible electromotive reel is the tensioning of the belt before a possible accident with increased retracting force. This reduces the possibility of the passenger slipping out of the belt (also called "submarining").

8.2.7. Cell supervision circuits (CSC)

Certain conditions must be observed for fault-free operation of the lithium-ion cells in the F10H: The cell voltage and the cell temperature cannot exceed or drop below certain values as otherwise the battery cells may suffer long-term damage. For this reason each high-voltage battery unit has several **cell monitoring circuits**, which may also be described as "Cell Supervisory Circuits (CSC)".

Communication between the 8 CSCs is done via a Local Controller Area Network. The Local Controller Area Network combines all CSCs together and is used for the communication of the CSCs with the battery management electronics. The battery management electronics control unit assumes the master function here. It is a low-voltage wiring harness with a maximum of 12 V.

8.3. Adapted control units

The **IHKA** had to be adapted to make possible the activation of the electric A/C compressor (EKK) in all operating conditions. The electric A/C compressor communicates with the IHKA via the LIN bus.

To be able to show additional displays for driving readiness, electric driving, brake energy regeneration and state of charge of the high-voltage battery which are relevant to the driver, the **instrument cluster** was adapted. In addition, the Check Control messages were enhanced with hybrid-specific messages.

The Car Information Computer **(CIC)** was adapted to enable additional, hybrid-specific displays to be shown in the central information display (CID). By selecting "Hybrid" in the "Vehicle Info" menu it is possible to display the energy and power flows for each driving situation and the state of charge of the high-voltage battery.

The software of the Digital Engine Electronics **(DME)** was adapted due to the torque coordination of the electrical machine/combustion engine. In addition, the DME communicates with the starter motor generator via the local interconnect network bus.

Rollover detection is required for the hybrid cars on a worldwide scale so that the high-voltage system is deactivated in the event of the car rolling over. The rollover detection is realized with the help of the sensors integrated in the Integrated Chassis Management control unit (roll rate sensor and vertical acceleration sensor). The **ACSM** had to be adapted with regard to the evaluation of these sensor signals. The safety battery terminal at the auxiliary battery is activated by the ACSM if required.

The software of the Dynamic Stability Control **DSC** was also adapted for the regenerative braking. This includes reading-in the brake pedal travel sensor which is connected directly to the DSC control unit as a hardware interface.

The **EGS** control unit was adapted due to the modified transmission. For example the electric transmission oil pump is controlled by the EGS control unit.

Due to the modified terminal control (driving readiness) the software in the **Car Access System** control unit was also adapted.

8.4. Omitted control units

Compared to the F10, some optional equipment is not offered in the F10H. For this reason some control units are no longer shown in the bus overview. The following table lists the control units no longer used.

Discontinued control unit	Function	Reason for discontinuation
AL	Active steering	Time-consuming and expensive coordination, in particular for electric driving. (In the F10 active steering is only offered in the package with Integral Active Steering.)
HSR	Integral Active Steering	Because Integral Active Steering is always offered only together with Active Steering, the control unit for rear axle slip angle control is also omitted.
ACC	Active speed control	Because of the complex control and the low production numbers involved, ACC is not featured in the F10H either.
BCU	Battery Charge Unit	In the F10H the electromechanical power steering (Electric Power Steering EPS) is always installed with 12 V.

9. Displays and Controls

The hybrid-specific operating conditions and the state of charge of the high-voltage battery are displayed in the instrument cluster and if desired in the Central Information Display.

The following hybrid-specific operating conditions are shown:

- "Ready to drive" display
- Display for electric driving
- Display for boost function
- Energy recovery

They are permanently displayed in the lower part of the rev counter in the instrument cluster. The hybrid-specific displays are called up in the CID via the "Vehicle Info > Hybrid" menu. Both the displays in the CID and in the instrument cluster are activated when terminal 15 is switched on.

9.1. Displays in the instrument cluster

The displays for the state of charge of the high-voltage battery and the hybrid-specific operating conditions in the instrument cluster are based on the display concept of the E72 and the F04. The current state of charge of the high-voltage battery is displayed on a scale from 1 to 10 when terminal 15 is switched on. The filling degree of the arc provides information on the state of charge of the high-voltage battery. At a scale value of 1 the high-voltage battery is fully charged. For optimal use of the high-voltage system the high-voltage battery is only charged fully during slow downhill driving. In normal driving it is charged up to a state of charge of about 80%. Note that only the customer-relevant range of the state of charge of the high-voltage battery is displayed, which corresponds to an actual state of charge of roughly 25% to 70%. This means that the hybrid system is still live when a scale value of zero is displayed.

The displays for the different operating conditions of the hybrid vehicle are summarized in the following table.

Driving readiness:
The "Ready to drive" status is signalled to the driver when the rev counter needle points to zero and at the same time in the lower part the blue writing "READY" flashes. This means the vehicle is stationary and can be set in motion at any time by pressing the accelerator pedal.
Depending on the state of charge of the high-voltage battery, temperature of the combustion engine and the position of the accelerator pedal, the vehicle is either driven electrically or using the combustion engine. If the vehicle for example is stationary at a railway crossing or a red light, the driving readiness is switched on.
If the customer has stopped the vehicle and wants to drive again a short time later, the driving readiness is switched on by pressing the START-STOP button. As the combustion engine is still at operating temperature and the high-voltage battery is still sufficiently charged, the combustion engine does not start.

Electric driving:
The vehicle can be driven by purely electric means up to a driving speed of about 55 km/h (34 mph). The power output from the high-voltage battery is shown on the left side with a blue arrow. Depending on the power requirement up to four arrows light up in succession. The needle of the rev counter is at "0" (combustion engine is off).
If all four arrows are lit up, the combustion engine is switched on upon an additional power requirement, for example if acceleration is desired. The four arrows move the needle of the rev counter so to speak and the powertrain of the vehicle changes from electric to combustion engine.

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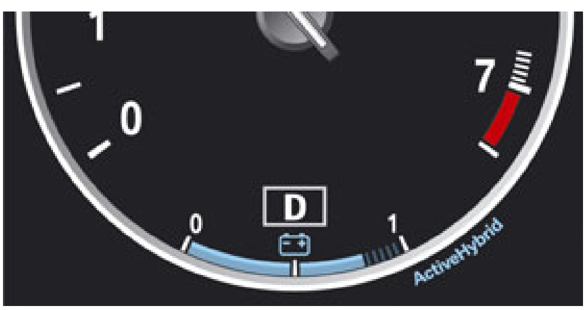
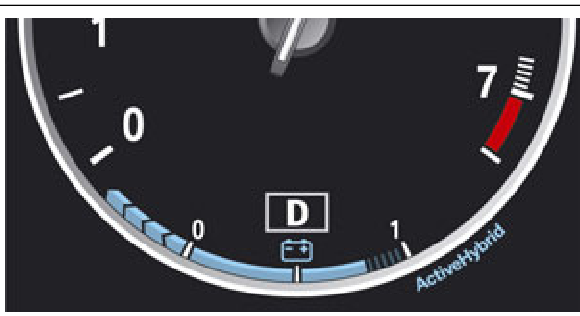
During electric driving please note that pedestrians and other road users cannot hear the usual sound of the conventional engine. For example pay particular attention when parking.



Operation using combustion engine:
From a driving speed of 55 km/h (34 mph) the drive of the vehicle is done using a combustion engine. The rev counter will display the current engine speed as usual. Only the display for the state of charge of the high-voltage battery will remain active on the hybrid-specific displays.



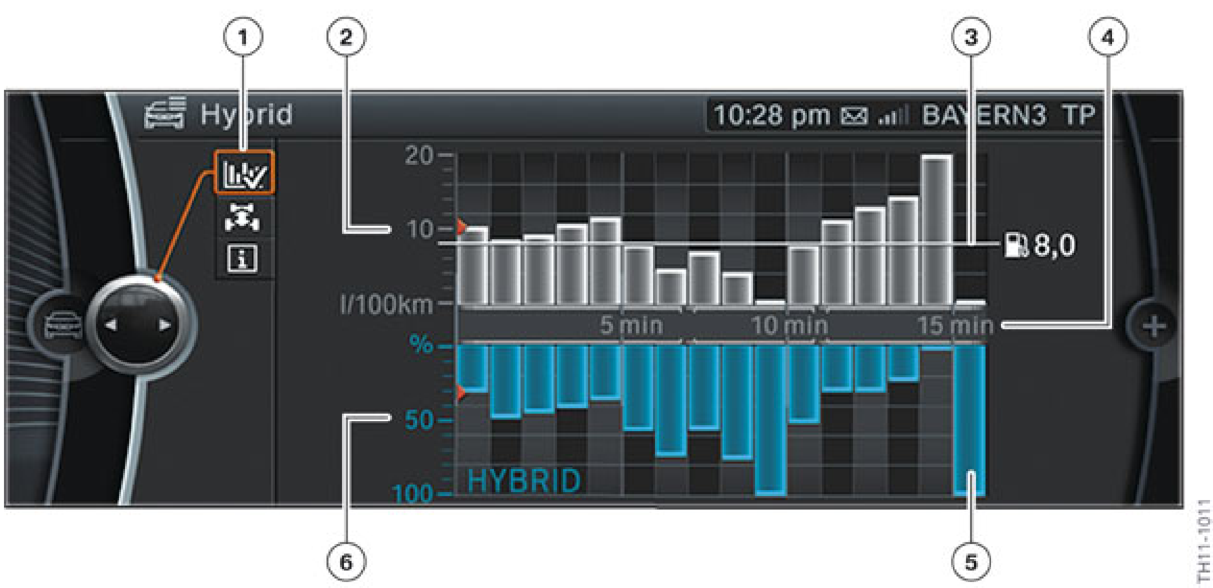
Boost function:
Under strong acceleration, for example when overtaking, power from the electrical machine is called upon in addition to the combustion engine. This provides the driver with maximum power. In addition, the accelerator pedal must be pressed down with force (kickdown). The rev counter shows the current engine speed and at the same time all four arrows on the left side of the display light up. The boost function can be used up to a driving speed of 160 km/h (100 mph).

Brake energy regeneration:
The hybrid system enables kinetic energy to be converted to electrical energy, for example during braking or in coasting (overrun) mode. The high-voltage battery is charged by this energy recovery.
The energy recovery is shown on the right side of the display via three blue arrows. The "+" symbol in the first arrow should rule out confusion with the arrows for electric driving. The blue arrow varies in length, depending on the deceleration or on the intensity of the brake pedal actuation.
Below a speed of roughly 15 km/h (10 mph) the display for brake energy regeneration does not light up although the vehicle is in coasting (overrun) mode or has just braked.

9.2. Displays in central information display

The energy/power flows and the charge state of the high-voltage battery can be displayed in the CID in all the vehicle's operating modes. In addition, the user can have the hybrid use from the last 16 minutes and the ECO PRO information displayed upon request. This provides the driver with an overview of the functioning of the hybrid system in different driving conditions, as well as optimal use of the hybrid vehicle.

9.2.1. Hybrid use



Display for utilization of the hybrid system

Index	Explanation
1	Selection of the display for hybrid use
2	Consumption scale of the combustion engine
3	Average consumption of the combustion engine
4	Time axis (16 minutes)
5	Bar representing minutes
6	Percentage scale for use of the electrical machine

The utilization of the hybrid system in the last 16 minutes of driving can be shown in the CID. One bar stands for a period of one minute. The time is also counted during engine stop phases. The higher the bar, the higher the fuel consumption or the use of the electrical machine.

The grey bars show the fuel consumption of the combustion engine. A line and a value to the right beside the diagram show the average consumption.

The blue bars show the percentage in which the electrical machine has been used. The electrical machine can be operated here as a generator (for brake recovery) or as an electric motor (for electric driving). The higher the bar, the greater the use of the hybrid system and thus higher fuel economy.

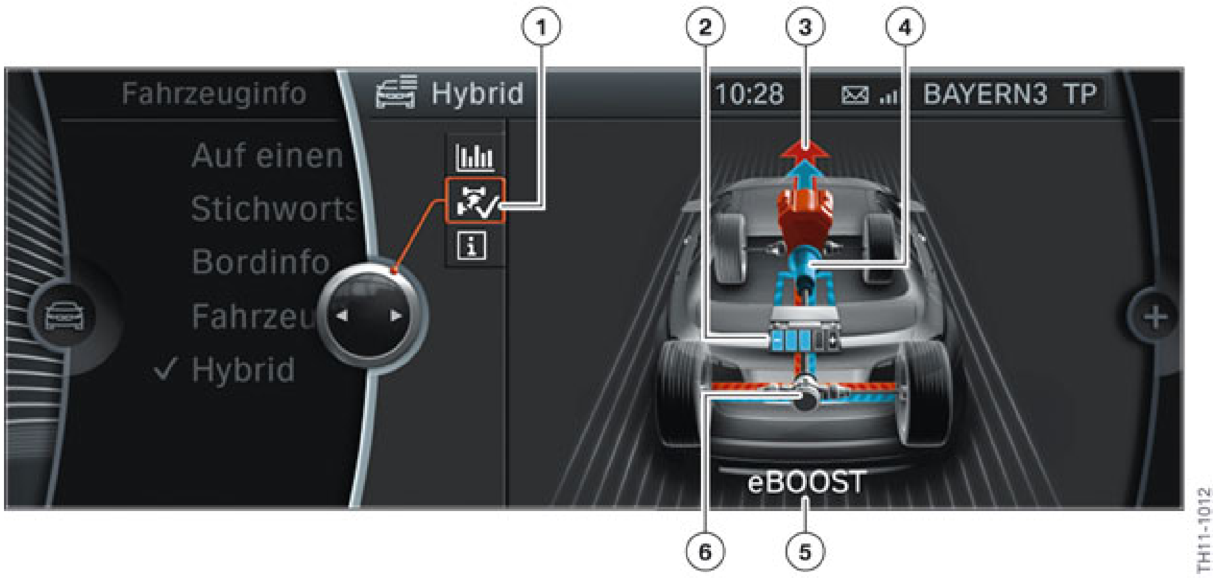
The two red marks on the vertical axis of the display indicate the bars from the last minute.

9.2.2. Energy/Power flows

The display of the energy/power flows in the CID functions according to the following principle:

- Blue: Electrical energy
- Red: Energy of the combustion engine
- Arrow: Direction of the energy/power flow

A driving situation is shown here as an example and the meaning of the symbols explained. The other driving situations can be deduced form there.



Hybrid display in the central information display during strong accelerate

Index	Explanation
1	Selection of display for energy/power flows
2	State of charge of the high-voltage battery
3	Drive arrow for combustion engine (red) and drive arrow for electric motor (blue)
4	Automatic transmission with electric motor
5	Text message for current driving situation "eBOOST"
6	Power flow to the rear axle

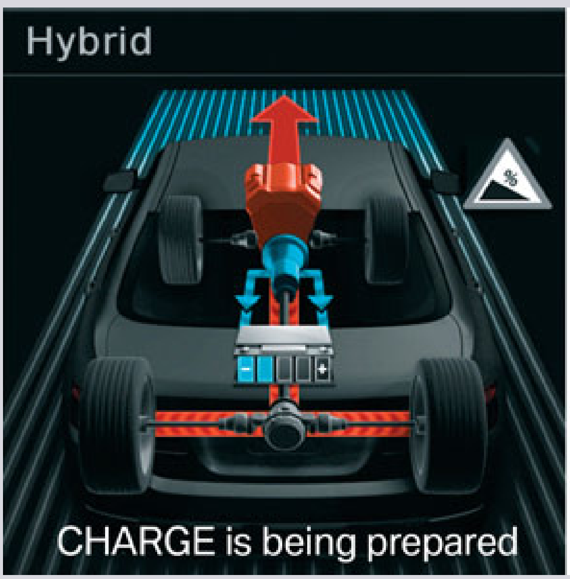
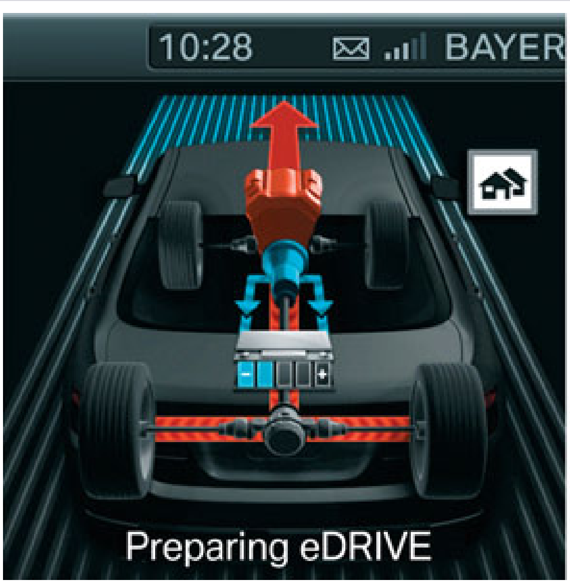
In the CID the boost function is represented by a red arrow (combustion engine contribution) and a slightly smaller blue arrow (electric motor contribution). The combustion engine is shown in red here. The activity of the electric motor in the automatic transmission is indicated by the blue color of the transmission. The five segments symbolize the state of charge of the high-voltage battery. In other words, one segment equals a state of charge of the high-voltage battery of 20%. The highest segment of the current display for the state of charge of the high-voltage battery is shown in various blue shades depending on the state of charge. This enables the intermediate stages in the range of 0% to 20% to be visualized. In the example above three segments are filled in. This equates to a usable state of charge of 60%. The power flow is depicted with two colors to show that the power flow to the rear wheels comes from both drive sources (combustion engine and electrical machine). The red arrow shows the combustion engine percentage and the blue arrow the electrical machine percentage. The current driving situation is also shown as a fade-in text message below the vehicle.

Adaptating to the road ahead

In the F10H for the first time the data from the navigation system can be used with the hybrid operating strategy. The navigation system is able to identify the stretches of road ahead and forward the data to the EME. For example before a road with reduced traffic is driven, the EME draws less energy from the high-voltage battery or tries to save as much electrical energy as possible in the high-voltage battery through energy recovery. This guarantees that the vehicle can drive through the reduced-traffic zone using purely electric means, thus providing for optimal use of the hybrid system.

A prerequisite for this intelligent intelligent energy management function is that the driver must have route guidance activated.

In the case of active route guidance and a selected display for energy/power flow, additional symbols and text messages are shown. The additional symbols in the energy/power flow display show that a situation has been identified ahead and the hybrid drive prepare itself for such a situation.

Display	Explanation
	Incline: "Preparing Charging" So that the high-voltage battery can charge during downhill driving through energy recovery, the high-voltage battery must be able to absorb the energy. A fully charged battery cannot absorb any more energy. For this reason the high-voltage battery is discharged in this case before downhill driving. This happens by increasing the percentage of electrical driving for the powertrain. The fuel consumption is also lowered as a result. **This feature is not yet available in the US.**
	Residential area or desired street: “Preparing eDrive" Before reaching a desired street or a residential area, the high-voltage battery is charged. Purely electric driving in the residential area or when approaching the destination is made possible with the use of the electric motor.

9.3. ECO PRO mode

The driver of a F10H can drive his vehicle even more efficiently by using the driving experience switch to activate ECO PRO mode. The ECO PRO mode consistently supports a driving style that reduces fuel consumption and ensures coordination of the hybrid system to maximize the range of the vehicle.

9.3.1. Activation and display

TH11-1368

Driving experience switch in the ActiveHybrid 5

ECO PRO mode is activated using the driving experience switch (previously referred to as "driving dynamics switch)". The "COMFORT" program is set as standard. To activate ECO PRO mode, the driving experience switch must be pressed in the "COMFORT" direction when terminal 15 is switched on until "ECO PRO" is displayed in the instrument cluster.

ECO PRO mode is deactivated again when terminal 15 is switched off.

TH11-1371

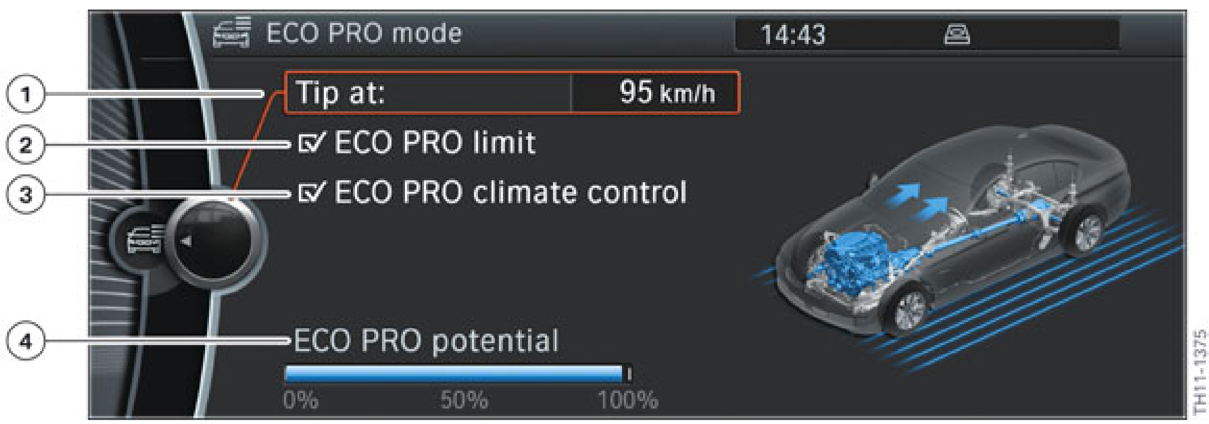
Displays in the instrument cluster when ECO PRO mode is activated.

Index	Explanation
1	ECO PRO bonus range
2	ECO PRO mode is active

The activated ECO PRO mode is indicated in the instrument cluster by the display of the "ECO PRO" writing in the rev counter beside the gear indicator. The so-called bonus range is shown below the rev counter.

The bonus range displays what range could be achieved by adapting the driving style. This display is called up automatically upon activation of ECO PRO mode. As long as the vehicle is driven efficiently, the bonus range is displayed in blue; otherwise this is shown in grey.

Upon activation of ECO PRO mode another window appears in the central information display to configure the ECO PRO mode.



Configure ECO PRO mode

Index	Explanation
1	Selection of speed with regard to ECO PRO tips
2	Activation/Deactivation of information when ECO PRO speed limit exceeded
3	Activation/Deactivation of restricted climate control in ECO PRO mode
4	The possible saving potential that can be achieved with the current configuration is displayed in percentage

If the driver is not driving his vehicle efficiently, for example accelerating too hard or incorrect gear selection, this is shown on the CID.

ECO PRO information

Index	Explanation
1	ECO PRO information is activated
2	Information in text form, for example "Moderate acceleration"
3	Information in graphic form
4	Name of the menu (ECO PRO information)

9.3.2. What is affected in ECO PRO mode?

The ECO PRO mode supports the driver in adopting an optimized-consumption driving style and reduces fuel consumption through intelligent control of energy and A/C management.

The following measures help to reduce fuel consumption:

- A modified accelerator pedal characteristic curve and shift program with automatic transmission helps the driver adopt a driving style that optimizes fuel consumption.
- Reduction of electric comfort consumers
- Power reduction of heating/air-conditioning
- Number and length of possible engine shutdown phases is maximized in ECO PRO mode

9.3.3. Reduction of electric comfort consumers

In ECO PRO mode a certain measure of reduction in comfort is tolerated. Thus for example the exterior mirror heating is switched off (this saves up to 100 W) and the maximum seat heating temperature is limited at about 37.5 °C (99.5 °F) [usually about 42 °C (107.6 °F)]. With the selection of ECO PRO mode for climate control, these functions are also reset in the operating condition which is active in the other programs of the driving experience switch.

9.3.4. Power reduction of heating/air-conditioning system

In ECO PRO mode an optimized operating strategy with less energy consumption (at acceptable restrictions to comfort) is used for climate control. The air-conditioning works in a more efficient manner with reduced air drying and cooling efficiency. This in turn requires less electrical energy.

The cooling of the high-voltage battery always has top priority and is not affected by the activation of ECO PRO mode.

If the required temperature can be achieved without cold production, the A/C compressor is switched off.

The ECO PRO mode climate control settings can be reset in ECO PRO mode by the user to the operating condition which is active in the other programs of the driving experience switch. The vehicle takes note of this setting for when ECO PRO mode is called up again.

9.4. Hybrid-specific Check Control messages

If faults occur in the F10H, the driver is informed about them by the Check Control messages. The following table summarizes the key hybrid-specific Check Control messages.

Check control message	Meaning	Cause
	Engine malfunctioning Restricted range and powertrain performance. Renewed engine start may not be possible. Please drive to next BMW Service workshop.	Malfunction with electrical machine, malfunction with electric vacuum pump.
	Engine malfunctioning Engine does not automatically switch on or off. If necessary start engine via Start/Stop button. Restricted powertrain functions. Have vehicle checked by your authorized BMW Service workshop.	EME cooling fault, electric vacuum pump fault, malfunction with PT-CAN or FlexRay, for example short circuit.
	HV battery Power of the HV battery is restricted. Possible to continue journey, if necessary (only using combustion engine). Have vehicle checked by your authorized BMW Service workshop.	Power of high-voltage battery too low.

Check control message	Meaning	Cause
	Recharge HV battery	Current state of charge High-voltage battery on transport path (state of charge too low -> recharge)
	HV battery OK	Current state of charge High-voltage battery on transport path OK
	Replace HV battery	Current state of charge High-voltage battery on transport path, battery damaged
	HV battery High-voltage system faulty. After stopping the engine, it may be no longer possible to continue journey. Please look for next authorized BMW Service workshop without delay.	isolation fault
	High-voltage system High-voltage system shut down. Engine start is no longer possible. Maintenance of high-voltage components is only by trained BMW Service personnel.	High-voltage system shut-down and in de-energized state for maintenance, service and repairs. High-voltage safety connector (Service Disconnect) removed, circuit of high-voltage interlock loop interrupted.

9.5. Operation

Despite two different drives and two different energy storage units, there are no differences in the operation of an ActiveHybrid 5 in comparison to a conventional vehicle. Subject to various parameters the hybrid system automatically performs the best settings in order to drive the vehicle efficiently and dynamically.

There is not a separate button for deactivation/activation of the automatic engine start-stop function in the F10H. The automatic engine start-stop function is automatically controlled by the hybrid system. Engine start-stop function can however be deactivated by placing selector lever into position "DS" or "M/S".

If the power of the combustion engine in the F10H is to be determined at a chassis dynamometer, it must be ensured that the vehicle is not driven "electrically".

The following procedure assures that the combustion engine is always running during a performance test.

- Vehicle is in driving readiness
- Deactivate driving readiness (terminal R is switched on and terminal 15 off)
- Press brake pedal
- Switch on hazard warning flashers
- Press DSC button until the message "Test mode" appears in the CID.

This test function is automatically deactivated upon the next terminal change or driving off.

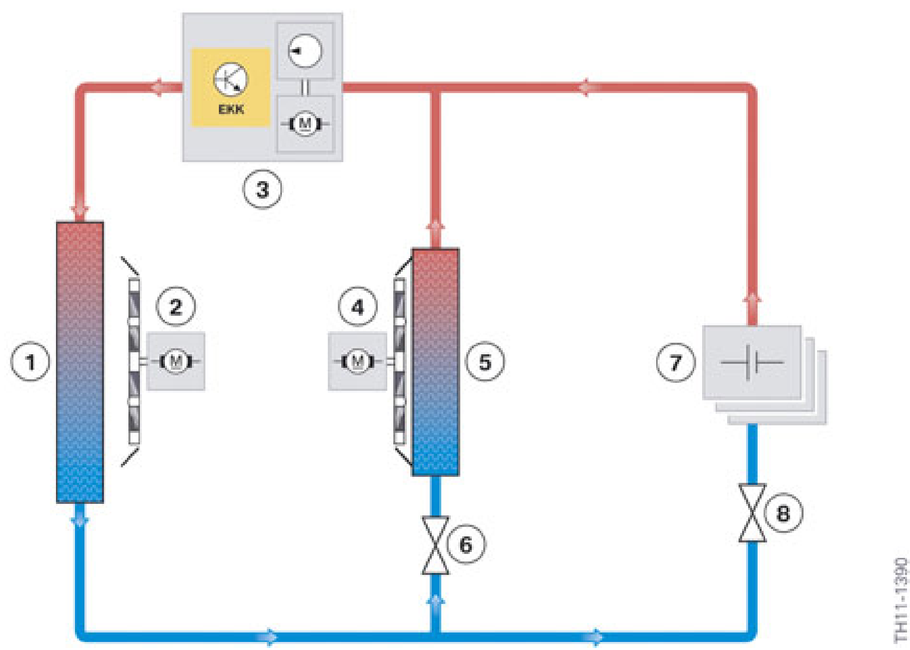
F10H Complete Vehicle

10. Climate Control

As with previous BMW active hybrid vehicles the F10H also uses an electric A/C compressor. Therefore it is possible to operate the air conditioning independently of the combustion engine. Thus the customer can enjoy the cooling effect of the air conditioning even while driving in pure electric mode and while stopped. Special silencing measures provides for quiet comfort. For example, even when the car is stationary and the combustion engine is off the air conditioning can barely be heard.

The stationary cooling and conservation cooling functions known from the F04 are also offered in the F10H.

10.1. System overview

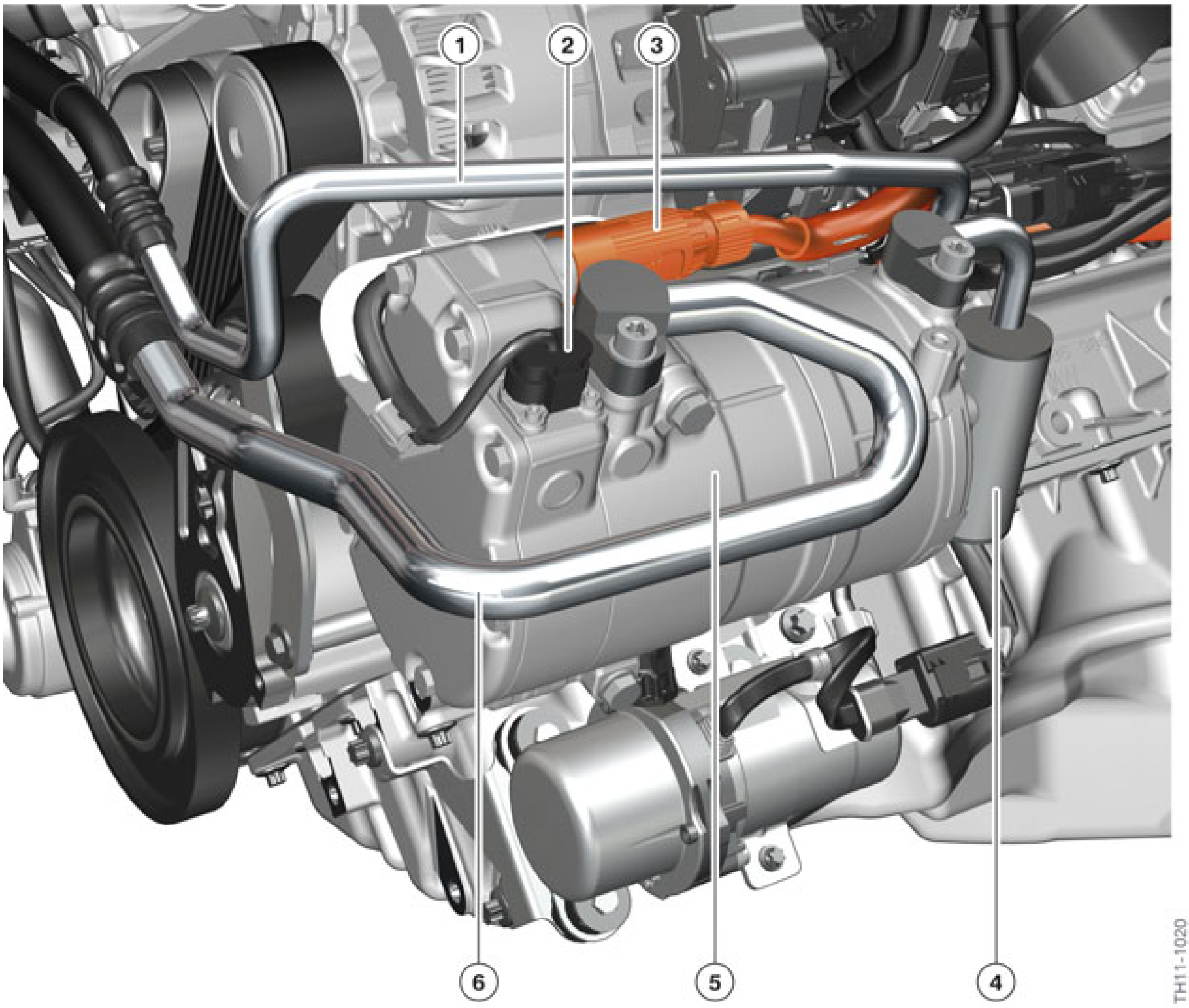


F10H climate control system overview

Index	Explanation
1	Condenser
2	Electric fan
3	Electrical A/C compressor
4	Blower motor
5	Evaporator
6	Shutoff valve in refrigerant circuit
7	High-voltage battery
8	Combined expansion and shut-off valve in the refrigerant circuit for cooling high-voltage battery

The top graphic shows the refrigerant circuits in the F10H. The refrigerant circuit for cooling the high-voltage battery is connected in parallel to the refrigerant circuit for cooling the passenger compartment.

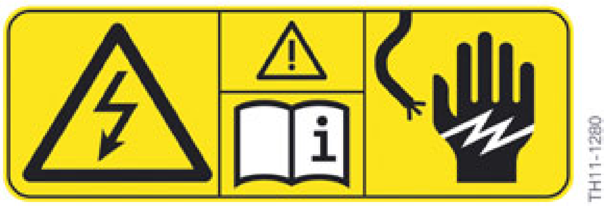
10.2. Electrical A/C compressor



F10H electric A/C compressor

Index	Explanation
1	Connection for gaseous refrigerant with high temperature and high pressure (pressure line)
2	Signal connector
3	High-voltage connector for compressor
4	Silencer
5	Electrical A/C compressor
6	Connection for gaseous refrigerant with low temperature and low pressure (intake pipe)

The electric A/C compressor is a high-voltage component!



Warning for high-voltage components

Each high-voltage component has on its housing an identifying label that enables Service Technicians and vehicle users to clearly identify the possible hazards that can result from working with high voltages.

Only BMW hybrid certified Service technicians are permitted to work on the designated high-voltage components. Certified technicians must: have suitable qualifications, comply with the safety rules and procedures and always follow the repair instructions to the letter.

Before working on a high-voltage component, you must apply the safety rules to shut down the high-voltage system. Once this has been accomplished according to procedure, all high-voltage components are no longer live and work can proceed in safety. There is, of course, a remote possibility that the correct shutdown procedure might be omitted, so an extra safety precaution has been implemented as a means of imposing an automatic shutdown of the high-voltage system.

The bridge for the high-voltage interlock loop is integrated in the high-voltage connector beside the contacts for the high voltage. The contacts of the bridge in the high-voltage connector are designed as leading contacts. This means when removing the high-voltage connector the contacts of the high-voltage bridge are separated first. The voltage supply of the EKK control unit is thus interrupted which in turn leads to an interruption of the high-voltage supply, even before the high-voltage connector is removed completely. Thus no electric arc is generated on the high-voltage contacts. The high-voltage contacts are protected from contact. The high-voltage connector of the electric A/C compressor is **not** part of the circuit of the high-voltage interlock loop.

The operating principle of the compressor corresponds to that of the E72 and the F04. To compress the refrigerant, the spiral compressor (also known as the scroll compressor) is used. The electric power of the electric A/C compressor is 4.5 kW.

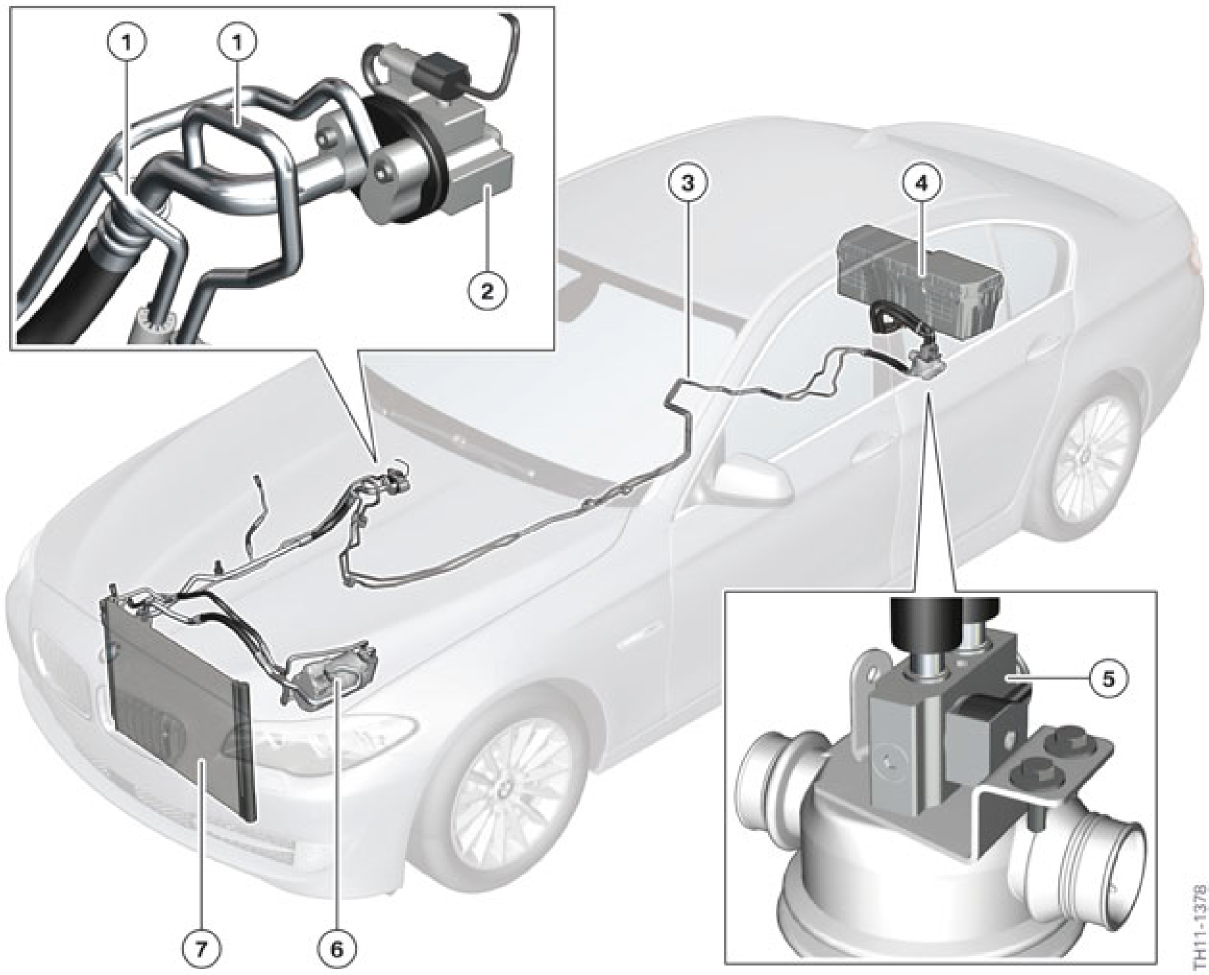
The supply voltage for EKK has a voltage range of about 200 V to 450 V. The power is reduced above and blow this voltage range or the EKK is switched off.

The manufacturer of the electric A/C compressor is Sanden.

10.3. Refrigerant circuit

The refrigerant circuit of the F10H has a similar design to that of the F04.

The following image shows the key components of the refrigerant circuit and their installation locations.



Refrigerant circuit in the F10H

Index	Explanation
1	Branching of refrigerant lines to high-voltage battery
2	Combined expansion and shutoff valve in the refrigerant circuit for cooling the passenger compartment
3	Refrigerant lines to high-voltage battery
4	High-voltage battery
5	Combined expansion and shut-off valve in the refrigerant circuit for cooling high-voltage battery
6	Electrical A/C compressor
7	Condenser in A/C system

The temperature has a decisive influence on the service life of the high-voltage battery. The cells of the high-voltage battery should not deliver or absorb electrical power at too high or too low a temperature. The optimal cell temperature is about 20 °C (68 °F); the battery cells should not exceed a maximum temperature of 40 °C (104 °F).

R134a is used as the refrigerant (which circulates in a circuit) absorbing heat at one point in the system and releasing it at another point. The heat absorbed from the passenger compartment and high-voltage battery is released to the ambient air in a heat exchanger (condenser) on the front of the vehicle. When the air-conditioning is activated for the passenger compartment or when cooling power is requested for the high-voltage battery, the electric A/C compressor is switched on and the system supplies the corresponding point with cooling. The passenger compartment cooling and the cooling of the high-voltage battery can be operated independently of each other. The energy required for this is drawn by the electric A/C compressor from the high-voltage battery. The BMW-approved PAG oil is used as the lubricant. So that the battery cooling and the passenger compartment cooling can be operated independently of each other, special solenoid valves are integrated into the refrigerant circuit. These open only the portion of the circuit that is actually required. This ensures high efficiency and proper control characteristics of the system.

The following table shows how the valves and the electric A/C compressor are controlled.

Cooling of	Combined expansion and shutoff valve for evaporator (passenger compartment)	Combined expansion and shutoff valve for high-voltage battery	Electrical A/C compressor
High-voltage battery	Closed	Open	switched on
Passenger compartment	Open	Closed	switched on
High-voltage battery and passenger compartment	Open	Open	switched on
No cooling	Closed	Closed	Switched off

⚠

The combined expansion and shutoff valve for the high-voltage battery is NOT required to be activated (OPEN) for the evacuation of the refrigerant.

The request of how much cooling power is required is measured and determined by the IHKA control unit. On the one hand, the request can come directly from the customer to cool the passenger compartment. On the other hand, the battery management electronics control unit (SME) can send a request for the high-voltage battery to be cooled as a data bus message to the IHKA control unit. The IHKA control unit coordinates these cooling requirements and activates the electric A/C compressor via a local interconnect network bus. The cooling requirements are prioritized depending on the temperature. For example at a high ambient temperature and very warm passenger compartment, a higher cooling power is demanded with higher priority. If the desired temperature is reached, the cooling power is reduced to maintain the temperature and set to a lower priority.

It is similar for the temperature of the battery cells. If the battery cells heat up to temperature of about 30 °C (86 °F), the cooling of the high-voltage battery has already begun. The cooling requested by the SME has an even lower priority here. It can for instance be declined by the high-voltage power management. At a higher cell temperature, the cooling requests receives top priority for the high-voltage battery and is always carried out.

10.4. Stationary cooling and conservation cooling

10.4.1. Stationary cooling

Due to the fact that the A/C compressor is electrically operated and that the high-voltage battery has high energy and power densities, a stationary cooling function is offered in the F10H.

Here the heated passenger compartment can be cooled prior to driving for about two minutes. The function can be conveniently activated by pressing the fourth button on the car's infrared remote control at outside temperatures > 15 °C (59 °F). If the customer opens a car door shortly before these two minutes have elapsed, the cooling period is extended by about 30 seconds. Stationary cooling can be activated again only after driving readiness is switched off (repeat interlock). The prerequisite for activating stationary cooling is a sufficient state of charge of the high-voltage battery (SoC > 42%).

10.4.2. Conservation climate control

F10H center console

Index	Explanation
1	Button for increasing air volume

The conservation cooling function is a further climate control function which has been made possible by the use of the electrically operated A/C compressor. The customer can activate this function when he/she leaves the car for a short time and then wishes to resume driving, for example to refuel the car, and would like to retain the pleasant interior temperature. Conservation cooling can be activated with the ignition switched off by pressing the button for increasing the air volume. The temperature in the passenger compartment is maintained for up to 6 minutes by activating the electric A/C compressor and the ventilation according to requirements.

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